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Finite-Difference Solution of Steady Flow in a Channel with a Quarter-Cylinder Obstruction: Gauss-Seidel, SOR, SLOR, ADI and Multigrid

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Abstract

A steady, laminar, incompressible two-dimensional flow through a 10 cm × 10 cm channel containing a quarter-cylinder of radius 2 cm at the lower-right corner is solved using the stream function-vorticity formulation. The inlet velocity is 1 m/s and the kinematic viscosity is 10^{-2} m²/s. The stream function and vorticity transport equations are discretized with finite differences on a Cartesian grid. The stream function, vorticity and velocity components are solved together until all four fields converge to a tolerance of 10^{-7} , after which the pressure field is obtained from a post-processing step. Five iterative schemes are compared: Gauss-Seidel, SOR, SLOR, SLOR combined with ADI, and a two-level Multigrid method. The first four are tested on coarse (61 × 61), intermediate (101 × 101) and fine (141 × 141) meshes, while Multigrid uses the fine mesh (141 × 141) with a 71 × 71 correction level. SOR converged fastest on every mesh, requiring 8,041 iterations and 9.85 s on the fine mesh, against 54,149 iterations and 33.47 s for Gauss-Seidel. The line-implicit schemes needed fewer iterations than Gauss-Seidel but were slower in wall-clock time because of their higher cost per iteration.

Keywords: computational fluid dynamics; stream function-vorticity formulation; finite difference method; iterative solvers; successive over-relaxation; multigrid

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1 Introduction

Flow around bodies placed inside channels is a common problem in fluid dynamics. It appears in many engineering systems, such as heat exchangers and cooling passages. When a body is placed near a wall, the flow near it becomes more complex: the velocity changes quickly in that region, and vorticity is generated along the body's surface. This makes the problem a good test case for comparing different numerical methods.

In this project, we study a steady, incompressible, two-dimensional flow through a square channel with a quarter-cylinder placed in the lower-right corner. The channel is 10 cm $\times$ 10 cm, and the cylinder has a radius of $R = 2$ cm, as shown in Figure 1.

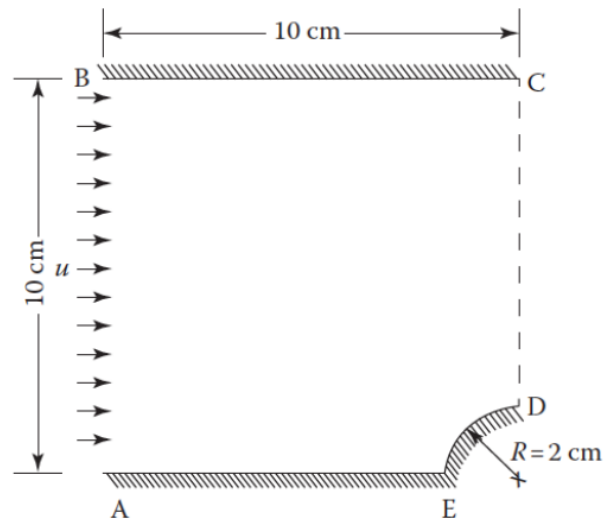


Figure 1. Physical domain

Flow enters from the left side (A-B) with a uniform speed of 1 m/s. The top of the channel (B-C) is a solid wall. The right side (C-D) acts as the outlet. The bottom of the channel, together with the cylinder surface (D-E-A), is also a solid wall.

We solve the flow using the stream function-vorticity (ψ - ξ) method. This method removes pressure from the momentum equations and automatically satisfies continuity. Once ψ and ξ are known, pressure is found separately by solving an additional equation.

The boundary conditions stated in the project are:

- **A-B (inlet):** $u = 1$ m/s, $v = 0$
- **B-C (top wall):** $\psi = \text{constant}$

- **C–D (outlet):** $\partial\psi/\partial x = 0$
- **D–E–A (bottom wall and cylinder):** $\psi = 0$
- **C–D (pressure):** $P = 0$

The goal of this project is to find the stream function, vorticity, velocity field (u , v), and pressure field (P) using a non-uniform finite-difference method. We also compare five iterative methods for solving the equations:

1. Gauss–Seidel
2. Successive Over-Relaxation (SOR)
3. Successive Line Over-Relaxation (SLOR)
4. Two-level Multigrid
5. SLOR combined with ADI

For the first three methods, we test three mesh sizes: coarse (61×61), intermediate (101×101), and fine (141×141). For Multigrid, we use only the fine mesh (141×141) and a coarse mesh (71×71). For each method and mesh, we record the wall-clock time and number of iterations needed to converge, so we can compare their speed and efficiency. Results are shown as line plots of ψ , ξ , u , v , and P at three locations ($x = 5$ cm, $x = 9$ cm, $x = 10$ cm), and as contour plots over the full domain.

The rest of this report is organized as follows. Section 2 gives the governing equations. Section 3 describes the grid and discretization. Section 4 describes the computational mesh. Sections 5–9 present each solver with its results. Section 10 compares all schemes.

2 Governing Equations and Mathematical Formulation

2.1 Assumptions

The flow is steady, laminar, incompressible, two-dimensional, with constant properties and no body forces.

2.2 Continuity and Momentum Equations

The flow is governed by the continuity equation and the two momentum equations:

$$\text{Continuity: } \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} = 0$$

$$\text{x-momentum: } u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} = -\frac{1}{\rho} \frac{\partial p}{\partial x} + \nu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right)$$

$$\text{y-momentum: } u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} = -\frac{1}{\rho} \frac{\partial p}{\partial y} + \nu \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} \right)$$

where u , v are the velocity components, p is pressure, ρ is density, and ν is kinematic viscosity.

2.3 Stream Function and Vorticity Definitions

$$u = \frac{\partial \psi}{\partial y}, \quad v = -\frac{\partial \psi}{\partial x}$$

$$\xi = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}$$

2.4 Core Equations Solved

$$\text{Stream function (Poisson equation): } \nabla^2 \psi = -\xi$$

$$\text{Vorticity transport: } u \frac{\partial \xi}{\partial x} + v \frac{\partial \xi}{\partial y} = \nu \left(\frac{\partial^2 \xi}{\partial x^2} + \frac{\partial^2 \xi}{\partial y^2} \right)$$

$$\text{Pressure (Poisson equation): } \nabla^2 p = 2(\psi_{xx}\psi_{yy} - \psi_{xy}^2)$$

where $\psi_{xx} = \partial^2 \psi / \partial x^2$, $\psi_{yy} = \partial^2 \psi / \partial y^2$, and $\psi_{xy} = \partial^2 \psi / \partial x \partial y$. The stream function and vorticity equations are solved together at every iteration: ψ is updated from ξ using the stream

function equation, u and v are then updated from the new ψ , and ξ is updated using the vorticity equation. This loop repeats until ψ , ξ , u , and v converge. Once these have converged, pressure is computed separately, as a post-processing step, using the pressure equation.

2.5 Equations at Interior and Boundary Nodes

The boundary segments A, B, C, D, E refer to Figure 1. AB is the inlet, BC is the top wall, CD is the right boundary, DE is the cylinder surface, and EA is the bottom wall. Nodes adjacent to the cylinder are treated with the same wall relations as DE, rather than with the interior equations below.

Stream function, ψ

Interior: $\nabla^2 \psi = -\xi$

Boundaries:

- AB: $\psi = U_\infty y$
- BC: $\psi = U_\infty h$, where h is the channel height
- CD: $\partial\psi/\partial x = 0$
- DE, EA: $\psi = 0$

Velocity, u and v

Interior and CD: $u = \partial\psi/\partial y$, $v = -\partial\psi/\partial x$

Boundaries:

- AB: $u = U_\infty$, $v = 0$
- BC: $u = 0$, $v = 0$
- DE, EA: $u = 0$, $v = 0$

Vorticity, ξ

Interior: $u \xi_x + v \xi_y = \nu \nabla^2 \xi$

Boundaries:

AB: $\xi = 0$ (because the flow is uniform here)

CD: $\xi = -\nabla^2\psi$

BC, DE, EA (solid, no-slip walls): $\partial^2\psi/\partial n^2|_{\text{wall}} = -\xi_{\text{wall}}$

- **BC:** $n = y$
- **EA:** $n = y$
- **DE (cylinder surface):** $n = x$ where the wall face is oriented vertically, $n = y$ where the wall face is oriented horizontally, and both directions are combined at diagonal nodes

Pressure, p

Interior: pressure equation, $\nabla^2 p = 2(\psi_{xx}\psi_{yy} - \psi_{xy}^2)$

Boundaries:

CD: $p = 0$

AB: $\partial p/\partial y = 0$ (from uniform inlet)

BC, EA, and DE (solid, no-slip walls):

$(\partial p/\partial x)_{\text{wall}} = -v(\partial \xi/\partial y)_{\text{wall}}$ - used where the wall is horizontal

$(\partial p/\partial y)_{\text{wall}} = v(\partial \xi/\partial x)_{\text{wall}}$ - used where the wall is vertical

Applied to each boundary:

- **BC:** $\partial p/\partial x = -v \partial \xi/\partial y$
- **EA:** $\partial p/\partial x = -v \partial \xi/\partial y$
- **DE (cylinder surface):** $\partial p/\partial x = -v \partial \xi/\partial y$ where the wall face is horizontal, $\partial p/\partial y = v \partial \xi/\partial x$ where the wall face is vertical, and both formulas are averaged at diagonal nodes.

3 Numerical Method and Discretization

3.1 Grid Generation

The domain is first covered by a uniform grid, spaced evenly in x and evenly in y. A node is marked solid if it falls inside the quarter-circle at the lower-right corner (center (L,0), radius R); all other nodes are fluid.

Near the cylinder, the uniform grid does not line up exactly with the circular boundary. Where the boundary passes between two grid nodes, the spacing to that node is shortened to match the true distance to the circle. Elsewhere, spacing stays equal to the uniform grid spacing.

This gives four local spacings at each node: $\Delta x^- (x_{i,j} - x_{i-1,j})$, $\Delta x^+ (x_{i+1,j} - x_{i,j})$, $\Delta y^- (x_{i,j} - x_{i,j-1})$, $\Delta y^+ (x_{i,j+1} - x_{i,j})$. Two spacing ratios are defined from these:

$$\beta = \frac{\Delta x^+}{\Delta x^-}, \alpha = \frac{\Delta y^+}{\Delta y^-}$$

Fluid nodes next to a solid node are flagged separately, since they use wall formulas instead of the interior formulas.

3.2 General Finite-Difference Formulas

Using the local spacing ratios α and β defined in Section 3.1, together with the backward spacings Δx^- and Δy^- , the first and second derivatives of any field variable u are approximated as follows.

(Proofs are included in the abstract.)

y-direction:

$$\left. \frac{\partial u}{\partial y} \right|_{i,j} = \frac{u_{i,j+1} + (\alpha^2 - 1)u_{i,j} - \alpha^2 u_{i,j-1}}{\alpha(\alpha + 1)\Delta y^-} + O((\Delta y^-)^2)$$

$$\left. \frac{\partial^2 u}{\partial y^2} \right|_{i,j} = \frac{2[u_{i,j+1} - (1 + \alpha)u_{i,j} + \alpha u_{i,j-1}]}{\alpha(\alpha + 1)(\Delta y_-)^2} + O(\Delta y_-)$$

x-direction:

$$\left. \frac{\partial u}{\partial x} \right|_{i,j} = \frac{u_{i+1,j} + (\beta^2 - 1)u_{i,j} - \beta^2 u_{i-1,j}}{\beta(\beta + 1)\Delta x^-} + O((\Delta x^-)^2)$$

$$\left. \frac{\partial^2 u}{\partial x^2} \right|_{i,j} = \frac{2[u_{i+1,j} - (1 + \beta)u_{i,j} + \beta u_{i-1,j}]}{\beta(\beta + 1)(\Delta x^-)^2} + O(\Delta x^-)$$

These four formulas are the building blocks used to discretize the governing equations at interior nodes. The variable u above stands for any field being discretized ψ , u , v , ξ , or p with the same formulas applied in each case.

3.3 Discretized Equations at Interior Nodes

Stream function:

$$\Psi_{i,j} = \frac{\xi_{i,j} + A2 \cdot \Psi_{i+1,j} + \beta \cdot A2 \cdot \Psi_{i-1,j} + B2 \cdot \Psi_{i,j+1} + \alpha \cdot B2 \cdot \Psi_{i,j-1}}{(1 + \beta)A2 + (1 + \alpha)B2}$$

$$\text{with } A2 = \frac{2}{\beta(\beta + 1)(\Delta x^-)^2} \text{ and } B2 = \frac{2}{\alpha(\alpha + 1)(\Delta y^-)^2}$$

Velocity:

$$u_{i,j} = \frac{\Psi_{i,j+1} + (\alpha^2 - 1)\Psi_{i,j} - \alpha^2 \Psi_{i,j-1}}{\alpha(\alpha + 1)\Delta y^-}$$

$$v_{i,j} = -\frac{\Psi_{i+1,j} + (\beta^2 - 1)\Psi_{i,j} - \beta^2 \Psi_{i-1,j}}{\beta(\beta + 1)\Delta x^-}$$

Vorticity equation:

$$A1 = \frac{u_{i,j}}{\beta(\beta + 1)\Delta x^-}, \quad B1 = \frac{v_{i,j}}{\alpha(\alpha + 1)\Delta y^-}$$

$$C1 = \frac{2v}{\beta(\beta + 1)(\Delta x^-)^2}, \quad D1 = \frac{2v}{\alpha(\alpha + 1)(\Delta y^-)^2}$$

$$A_P = A1(\beta^2 - 1) + B1(\alpha^2 - 1) + C1(1 + \beta) + D1(1 + \alpha)$$

$$\xi_{i,j} = \frac{(C1 - A1) \cdot \xi_{i+1,j} + (\beta^2 A1 + \beta C1) \cdot \xi_{i-1,j} + (D1 - B1) \cdot \xi_{i,j+1} + (\alpha^2 B1 + \alpha D1) \cdot \xi_{i,j-1}}{A_P}$$

Pressure:

First, the second derivatives and mixed derivative of ψ are evaluated at the node:

$$\Psi_{xx,i,j} = \frac{2[\Psi_{i+1,j} - (1 + \beta)\Psi_{i,j} + \beta\Psi_{i-1,j}]}{\beta(\beta + 1)(\Delta x^-)^2}$$

$$\Psi_{yy,i,j} = \frac{2[\Psi_{i,j+1} - (1 + \alpha)\Psi_{i,j} + \alpha\Psi_{i,j-1}]}{\alpha(\alpha + 1)(\Delta y^-)^2}$$

$\Psi_{xy}(i,j)$ is found in two steps: first, the first-derivative formula in x is applied to ψ then, the first-derivative formula in y is applied to the resulting values.

$$\Psi_{xy,i,j} = \frac{[\Psi_{i+1,j+1} + (\beta^2 - 1)\Psi_{i,j+1} - \beta^2\Psi_{i-1,j+1}] + (\alpha^2 - 1)[\Psi_{i+1,j} + (\beta^2 - 1)\Psi_{i,j} - \beta^2\Psi_{i-1,j}] - \alpha^2[\Psi_{i+1,j-1} + (\beta^2 - 1)\Psi_{i,j-1} - \beta^2\Psi_{i-1,j-1}]}{\alpha(\alpha + 1)\beta(\beta + 1)\Delta x^- \Delta y^-}$$

Then $S_{i,j} = 2[\Psi_{xx}(i,j) \cdot \Psi_{yy}(i,j) - \Psi_{xy}(i,j)^2]$, and pressure is updated using:

$$C2 = \frac{2}{\beta(\beta + 1)(\Delta x^-)^2}, \quad D2 = \frac{2}{\alpha(\alpha + 1)(\Delta y^-)^2}$$

$$p_{i,j} = \frac{C2 \cdot p_{i+1,j} + \beta \cdot C2 \cdot p_{i-1,j} + D2 \cdot p_{i,j+1} + \alpha \cdot D2 \cdot p_{i,j-1} - S_{i,j}}{(1 + \beta)C2 + (1 + \alpha)D2}$$

3.4 Discretized Boundary Conditions

Stream function, ψ :

$$AB: \psi_{1,j} = U_{\infty} \cdot y_{1,j}$$

$$BC: \psi_{i,N_y} = U_{\infty} \cdot h$$

$$CD: \psi_{N_x,j} = \psi_{N_x-1,j}$$

$$DE, EA: \psi_{i,j} = 0$$

Velocity, u and v:

AB:

$$u = U_\infty, v = 0$$

BC:

$$u = 0, v = 0$$

CD:

$$u_{N_x,j} = \frac{\psi_{N_x,j+1} + (\alpha^2 - 1)\psi_{N_x,j} - \alpha^2\psi_{N_x,j-1}}{\alpha(\alpha + 1)\Delta y_{N_x,j}^-}$$

$$v_{N_x,j} = -\frac{\psi_{N_x,j} - \psi_{N_x-1,j}}{\Delta x_{N_x,j}^-}$$

DE, EA:

$$u = 0, v = 0$$

Vorticity, ξ :

AB:

$$\xi_{1,j} = 0$$

BC:

$$\xi_{i,N_y} = \frac{2[\psi_{i,N_y} - \psi_{i,N_y-1}]}{(\Delta y_{i,N_y}^-)^2}$$

CD:

$$\beta_b = \frac{\Delta x_{i-1,j}^-}{\Delta x_{i,j}^-}:$$

$$\xi_{N_x,j} = -\left\{ \frac{2[\psi_{N_x-2,j} - (1 + \beta_b)\psi_{N_x-1,j} + \beta_b \cdot \psi_{N_x,j}]}{\beta_b(1 + \beta_b)(\Delta x_{N_x,j}^-)^2} + \frac{2[\psi_{N_x,j+1} - (1 + \alpha)\psi_{N_x,j} + \alpha \cdot \psi_{N_x,j-1}]}{\alpha(1 + \alpha)(\Delta y_{N_x,j}^-)^2} \right\}$$

DE, by wall orientation:

$$\text{vertical-facing: } \xi_{i,j} = -2\psi_{i-1,j}/(\Delta x_{i,j}^-)^2$$

horizontal-facing: $\xi_{i,j} = -2\psi_{i,j+1}/(\Delta y_{i,j}^+)^2$

diagonal: $\xi_{i,j} = -2\psi_{i-1,j+1}/[(\Delta x_{i,j}^-)^2 + (\Delta y_{i,j}^+)^2]$

EA:

$$\text{EA: } \xi_{i,1} = \frac{-2\psi_{i,2}}{(\Delta y_{i,1}^+)^2}$$

Pressure, p:

CD:

$$p_{N_x,j} = 0$$

BC:

$$p_{i,j} = p_{i+1,j} + \frac{v \cdot \Delta x_{i,j}^+ \cdot [\xi_{i,j} - \xi_{i,j-1}]}{\Delta y_{i,j}^-}$$

AB:

$$p_{1,j} = p_{1,j+1}$$

DE:

$$\text{vertical-facing: } p_{i,j} = p_{i,j+1} - \frac{v \cdot \Delta y_{i,j}^+ \cdot [\xi_{i,j} - \xi_{i-1,j}]}{\Delta x_{i,j}^-}$$

$$\text{horizontal-facing: } p_{i,j} = p_{i-1,j} - \frac{v \cdot \Delta x_{i,j}^- \cdot [\xi_{i,j+1} - \xi_{i,j}]}{\Delta y_{i,j}^+}$$

$$\text{outer corner: } p_{i,j} = \frac{1}{2} \left[\left(p_{i,j+1} - \frac{v \cdot \Delta y_{i,j}^+ \cdot [\xi_{i,j} - \xi_{i-1,j}]}{\Delta x_{i,j}^-} \right) + \left(p_{i-1,j} - \frac{v \cdot \Delta x_{i,j}^- \cdot [\xi_{i,j+1} - \xi_{i,j}]}{\Delta y_{i,j}^+} \right) \right]$$

EA:

$$p_{i,j} = p_{i-1,j} - \frac{v \cdot \Delta x_{i,j}^- \cdot [\xi_{i,j+1} - \xi_{i,j}]}{\Delta y_{i,j}^+}$$

4 Computational Mesh

All three meshes are uniform Cartesian grids with the same number of nodes in each direction, so the spacing is constant and identical in x and y , $\Delta x = \Delta y = L/(N - 1)$. The three resolutions used are listed below.

Table 1. Node counts and grid spacing of the three meshes.

Mesh	Nodes ($N \times N$)	$\Delta x = \Delta y$ (mm)
Coarse	61×61	1.667
Intermediate	101×101	1.000
Fine	141×141	0.714

Each node is classified as either a solid node or a fluid node. A node is marked solid when its distance from the cylinder center is smaller than R , and fluid otherwise. Among the fluid nodes, those having at least one immediate neighbor to the right or below inside the solid region are marked as circle-adjacent nodes. These circle-adjacent nodes form the discrete representation of the curved boundary D–E and are shown by green circles in the mesh figures. A small number of nodes fall exactly on the arc.

The boundary D–E is treated directly on the circle-adjacent nodes. On these nodes the stream function is set to zero, consistent with the wall condition along D–E–A, and both velocity components are set to zero. The circle-adjacent nodes are excluded from the interior update of ψ and ξ . The wall vorticity on these nodes is evaluated from a one-sided second derivative of ψ , and the form used depends on the local orientation of the surface. Three cases are distinguished: nodes where the solid lies to the right and the surface is locally vertical, nodes where the solid lies below and the surface is locally horizontal, and diagonal nodes where the solid lies both to the right and below. The expressions used in each case are given with the boundary conditions in the previous section.

The accuracy of this representation improves directly with mesh refinement, as can be seen by comparing the three figures below: the chain of circle-adjacent nodes follows the true arc more closely as the spacing is reduced.

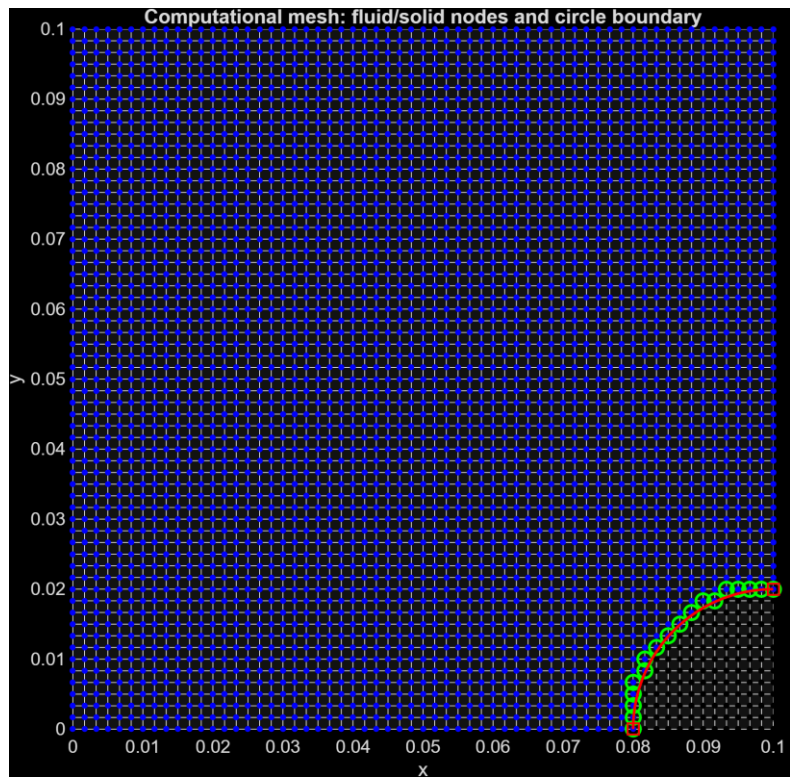


Figure 2. Coarse mesh (61×61)

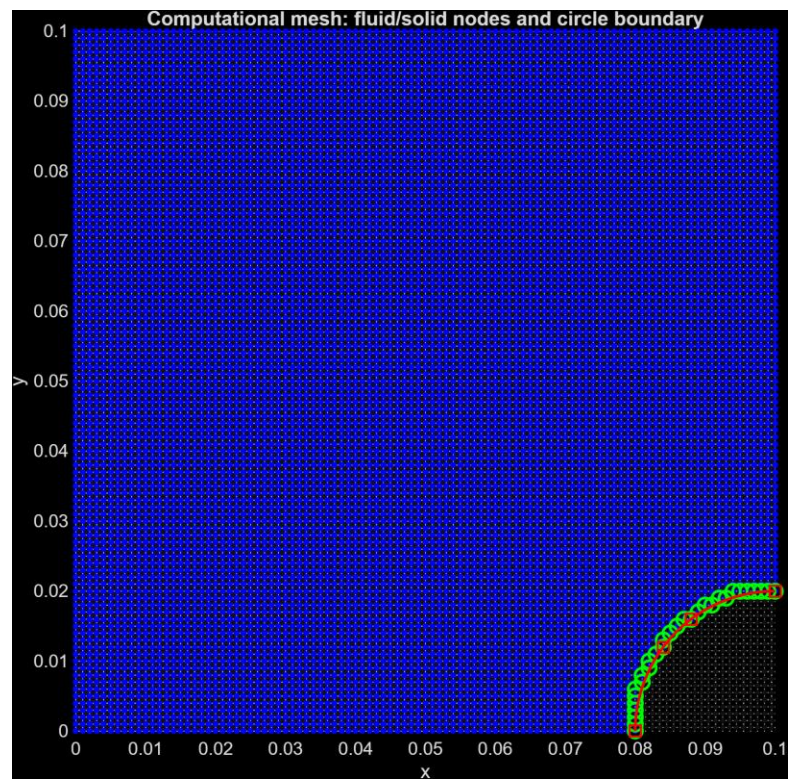


Figure 3. Intermediate mesh (101×101)

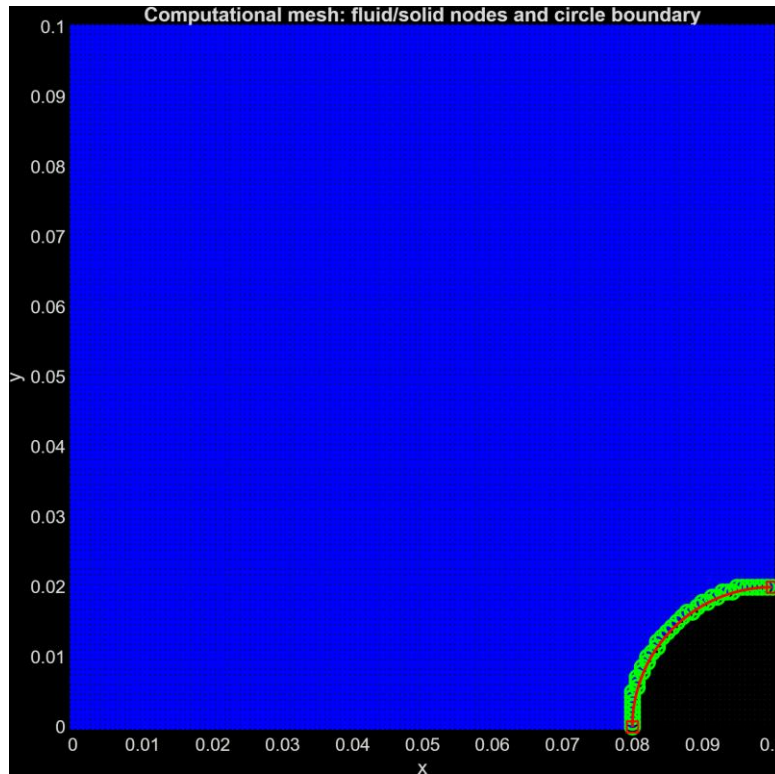


Figure 4. Fine mesh (141×141)

5 Gauss-Seidel Method

The Gauss-Seidel scheme is used here as the baseline solver. It is a point-iterative method: the nodes are swept in order of increasing i and j , and every updated value is used immediately in the same sweep rather than being stored until the end of the iteration.

Each outer iteration follows a fixed sequence:

1. The stream function is updated at all interior fluid nodes from the discretized Poisson equation, with the vorticity treated as known coefficient from the previous step.
2. The stream function on the outlet segment C-D is re-imposed, since this boundary condition depends on the interior solution and therefore changes at every iteration.
3. The velocity components are calculated from the updated stream function at the interior nodes and on the boundary C-D.

4. The vorticity boundary values on A-B, B-C, C-D, D-E and E-A are recomputed from the new stream function field.
5. The vorticity is updated at all interior fluid nodes from the discretized vorticity equation, with the velocity and psi components treated as known coefficients from the previous step.

Convergence is declared when the maximum absolute change between two successive iterations falls below the tolerance simultaneously for the stream function, the vorticity and both velocity components.

The pressure field is then obtained as a post-processing step, as described earlier.

Table 2. Iteration count and wall-clock time of the Gauss–Seidel scheme on the three meshes.

Mesh	Iterations	Wall-clock time (s)
Coarse (61×61)	6,504	0.91
Intermediate (101×101)	23,186	7.61
Fine (141×141)	54,149	33.47

5.1 contours (coarse mesh)

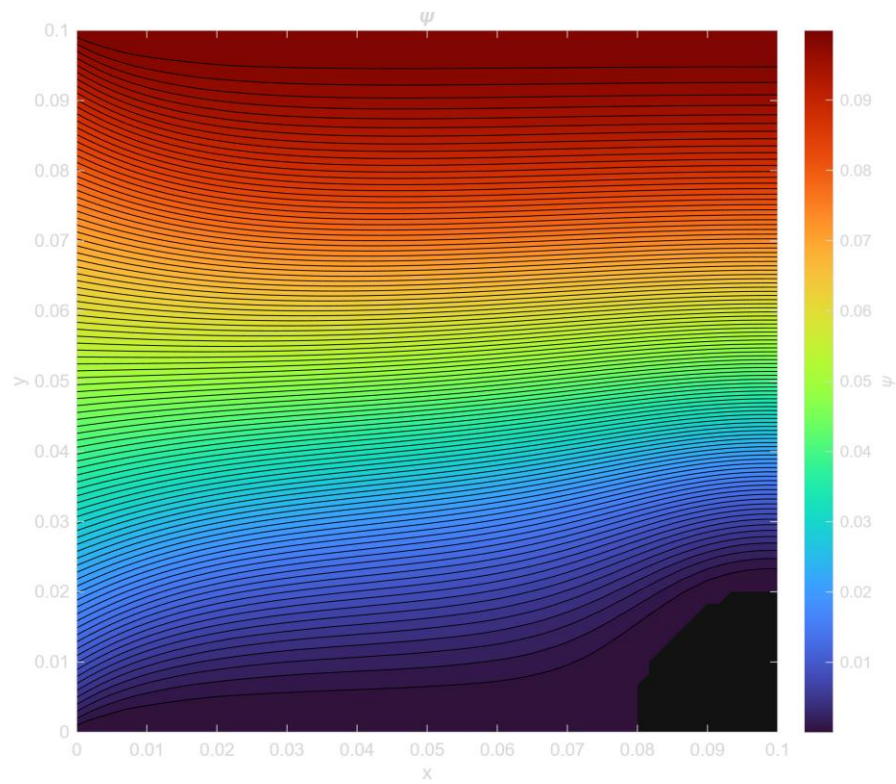


Figure 5. Stream function contour on the coarse mesh.

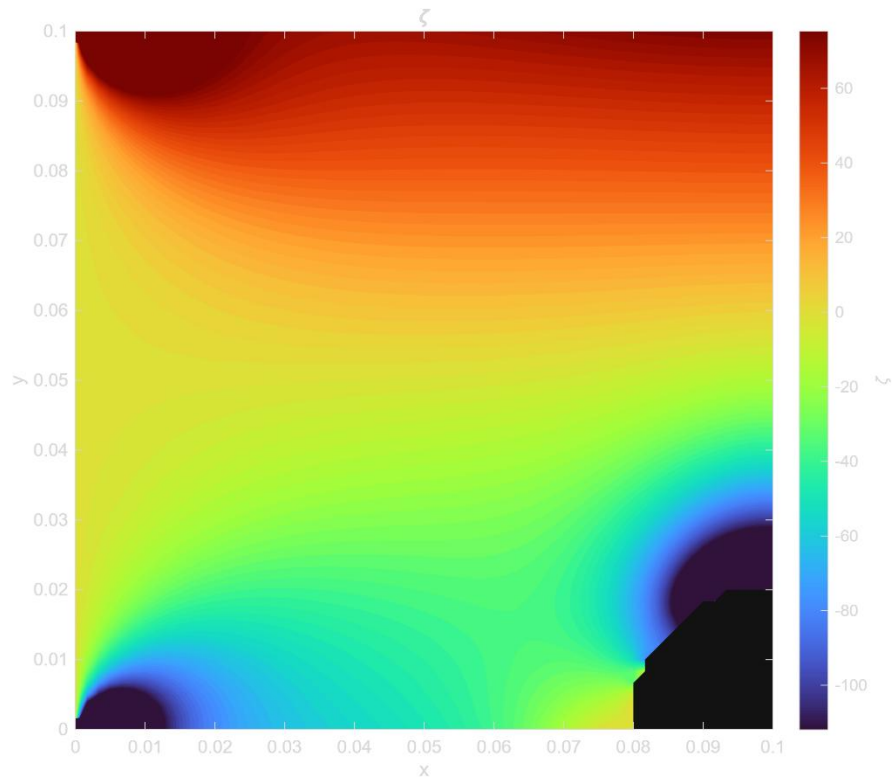


Figure 6. Vorticity contour on the coarse mesh.

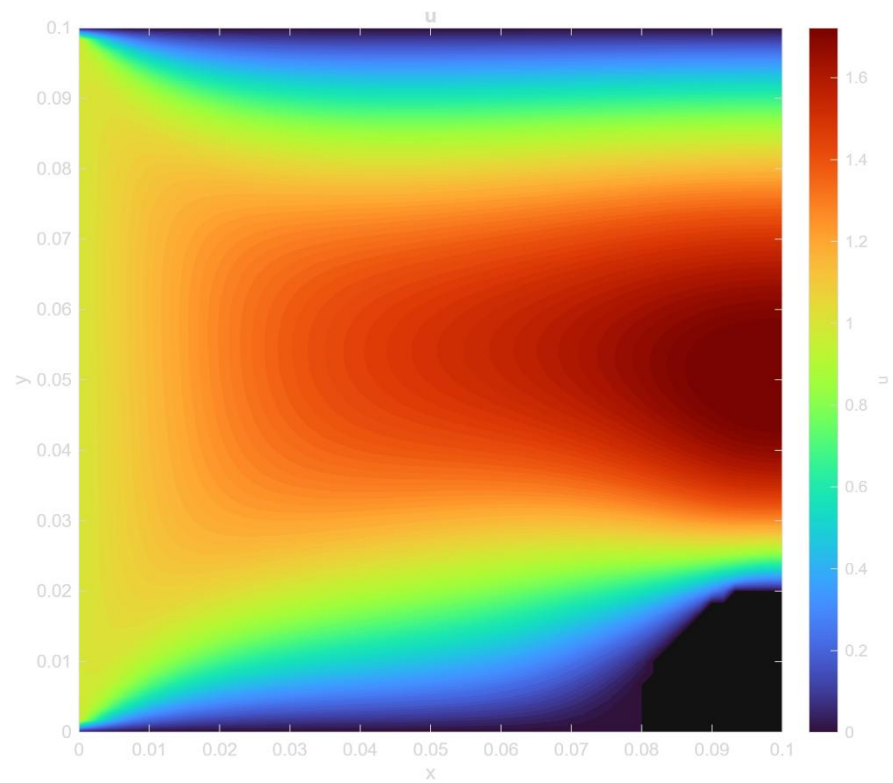


Figure 7. Horizontal velocity contour on the coarse mesh.

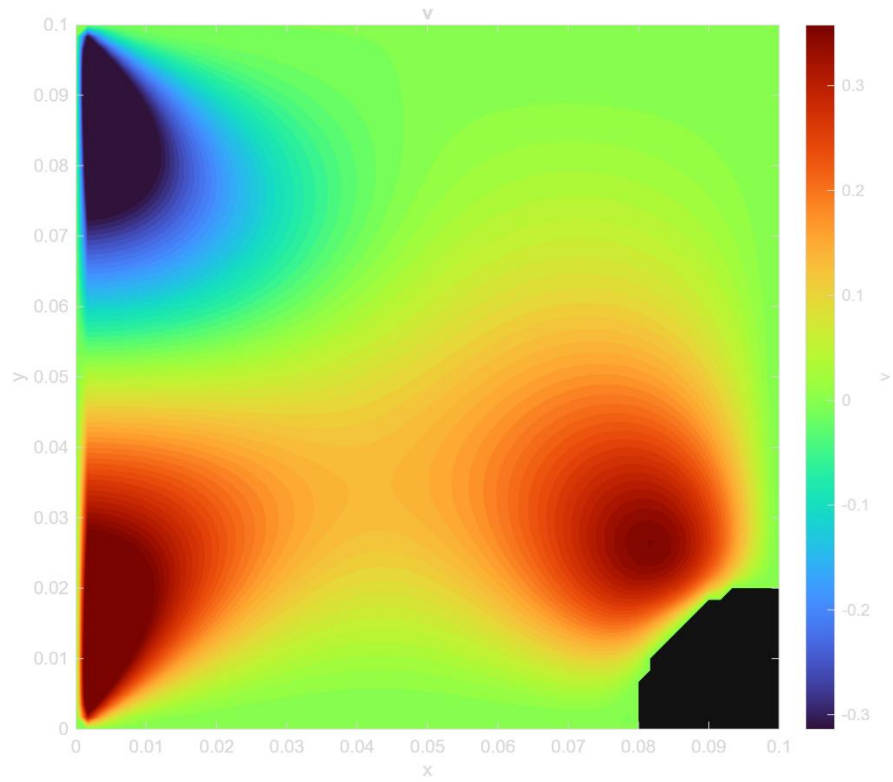


Figure 8. Vertical velocity contour on the coarse mesh.

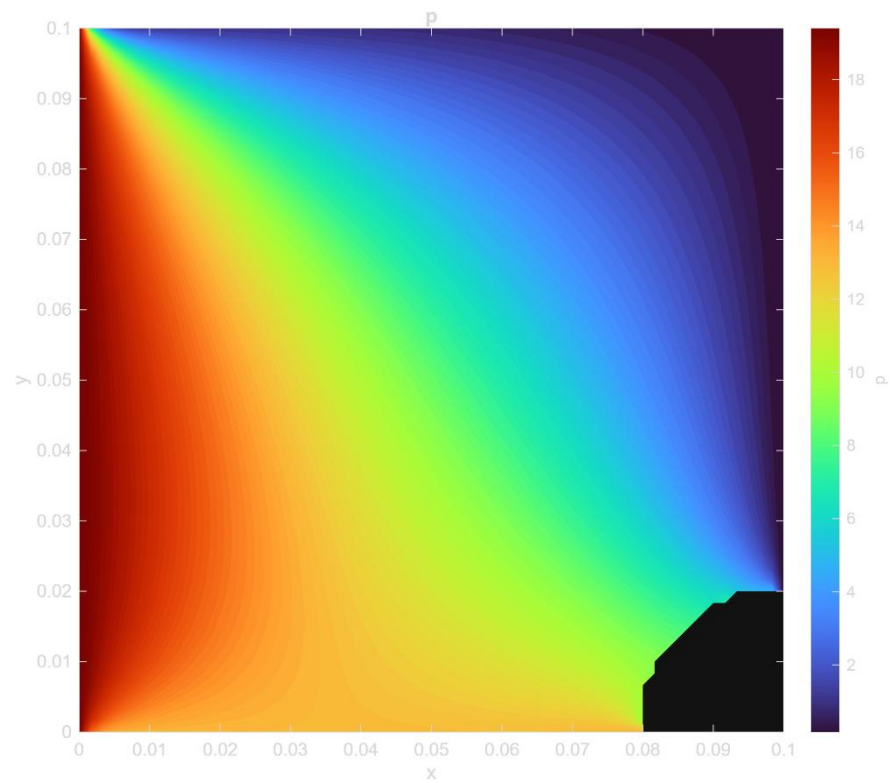


Figure 9. Pressure contour on the coarse mesh.

5.2 line plots (coarse mesh)

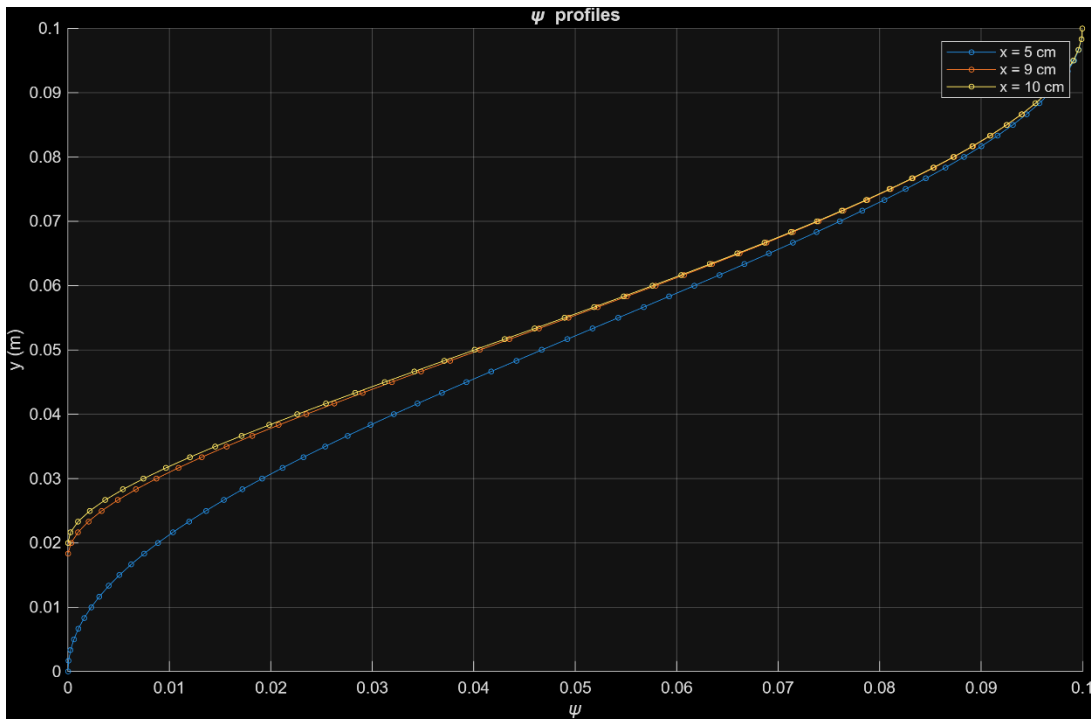


Figure 10. Stream function profiles at $x = 5, 9$ and 10 cm on the coarse mesh.

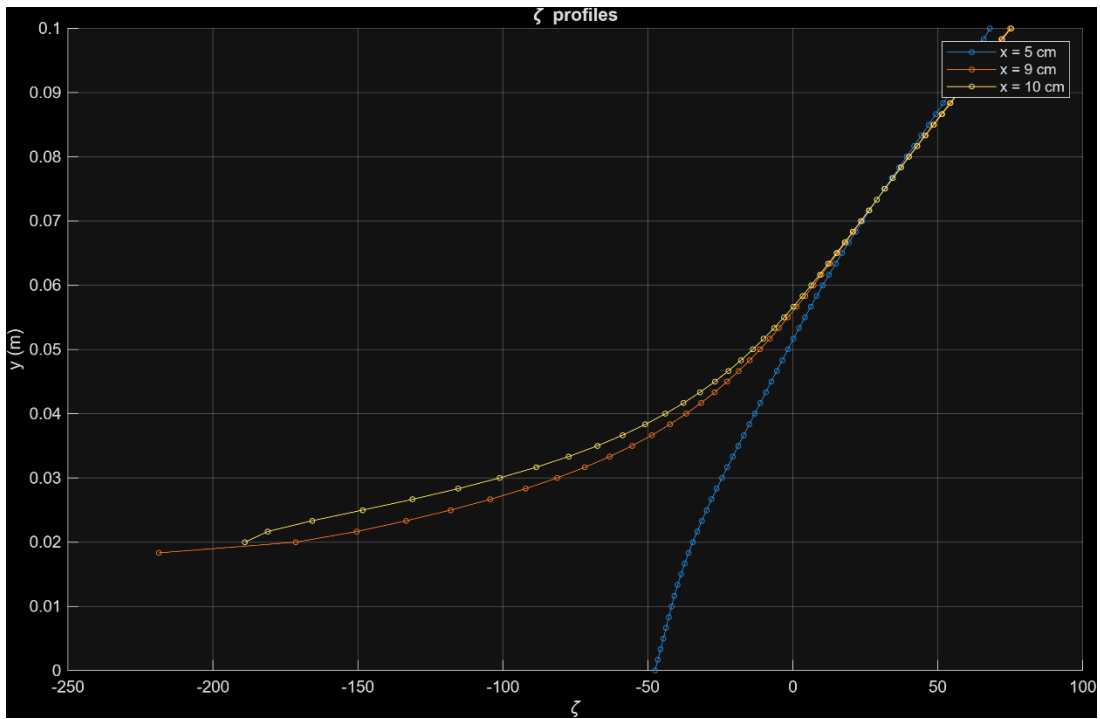


Figure 11. Vorticity profiles at $x = 5, 9$ and 10 cm on the coarse mesh

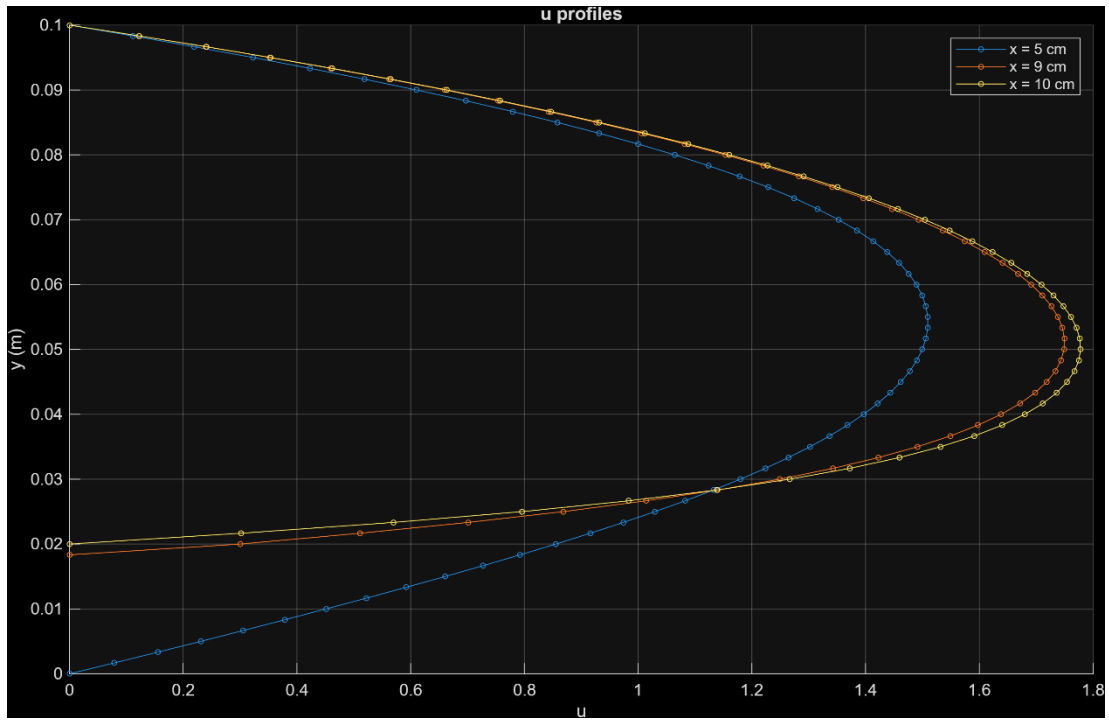


Figure 12. Horizontal velocity profiles at $x = 5, 9$ and 10 cm on the coarse mesh.

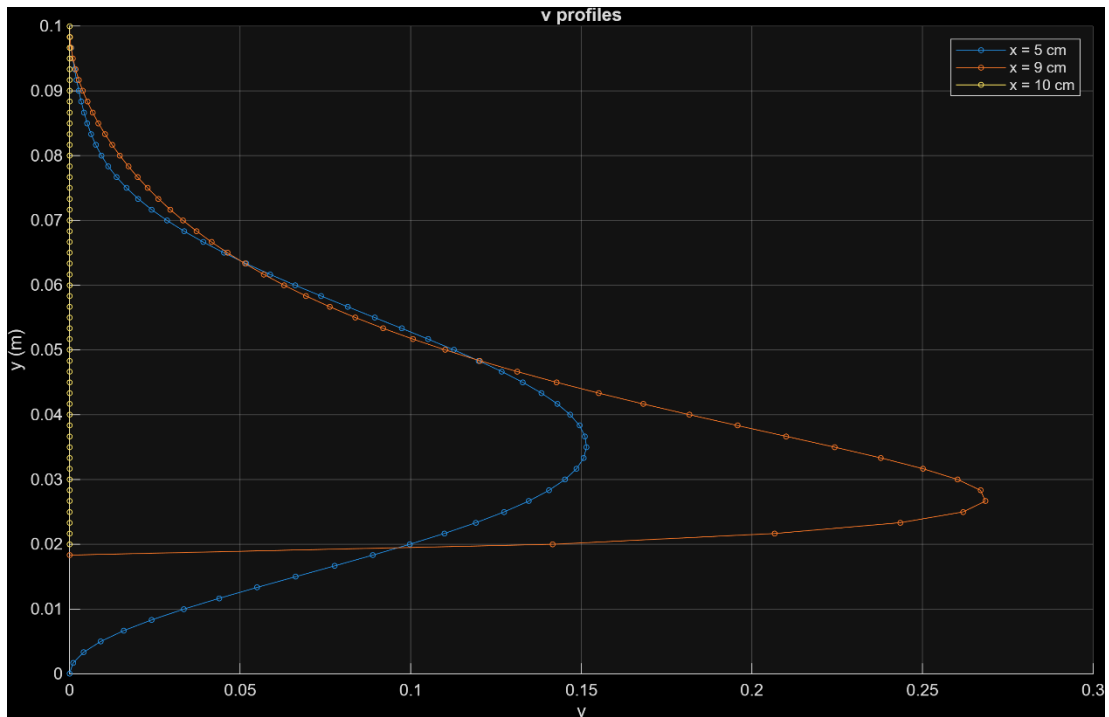


Figure 13. Vertical velocity profiles at $x = 5, 9$ and 10 cm on the coarse mesh.

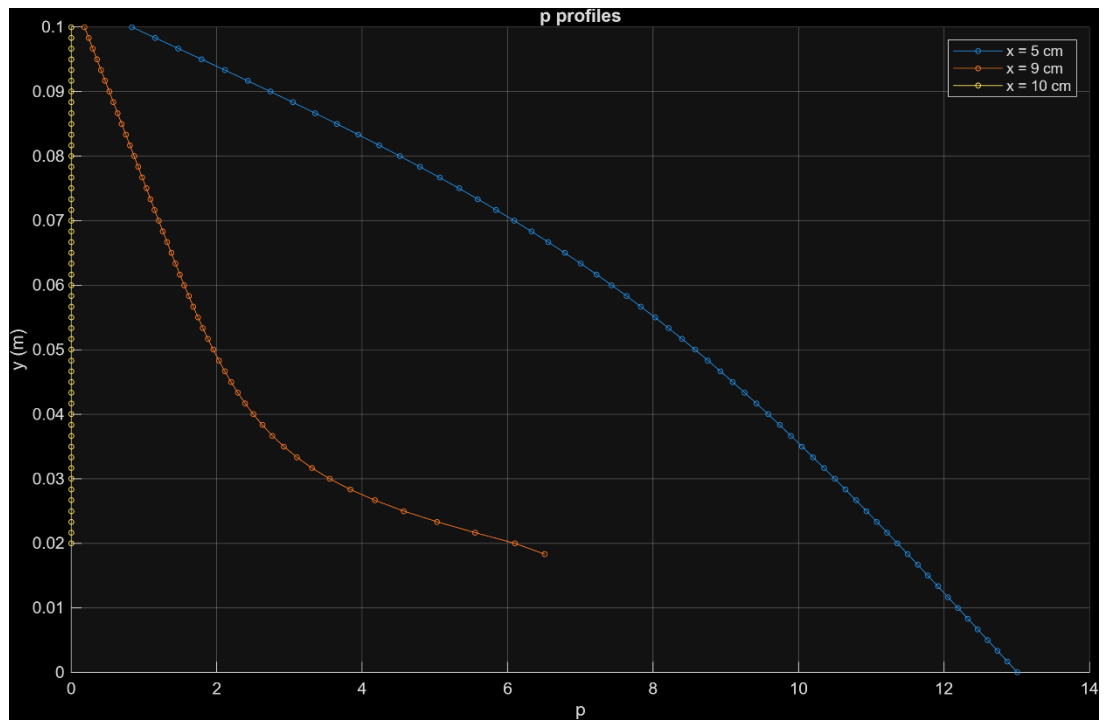


Figure 14. Pressure profiles at $x = 5, 9$ and 10 cm on the coarse mesh.

5.3 contours (intermediate mesh)

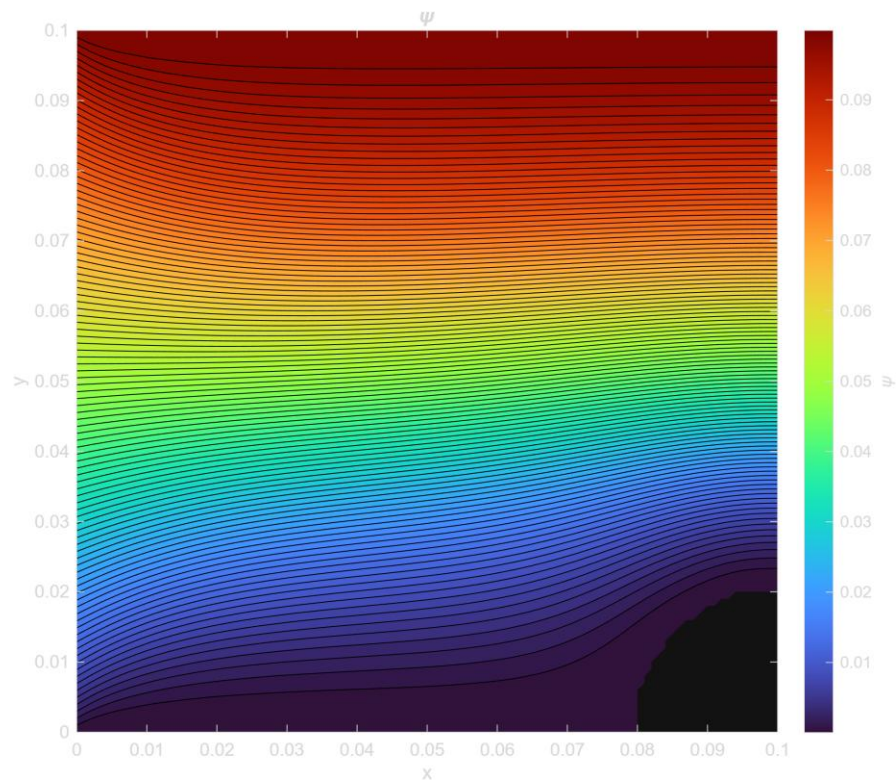


Figure 15. Stream function contour on the intermediate mesh.

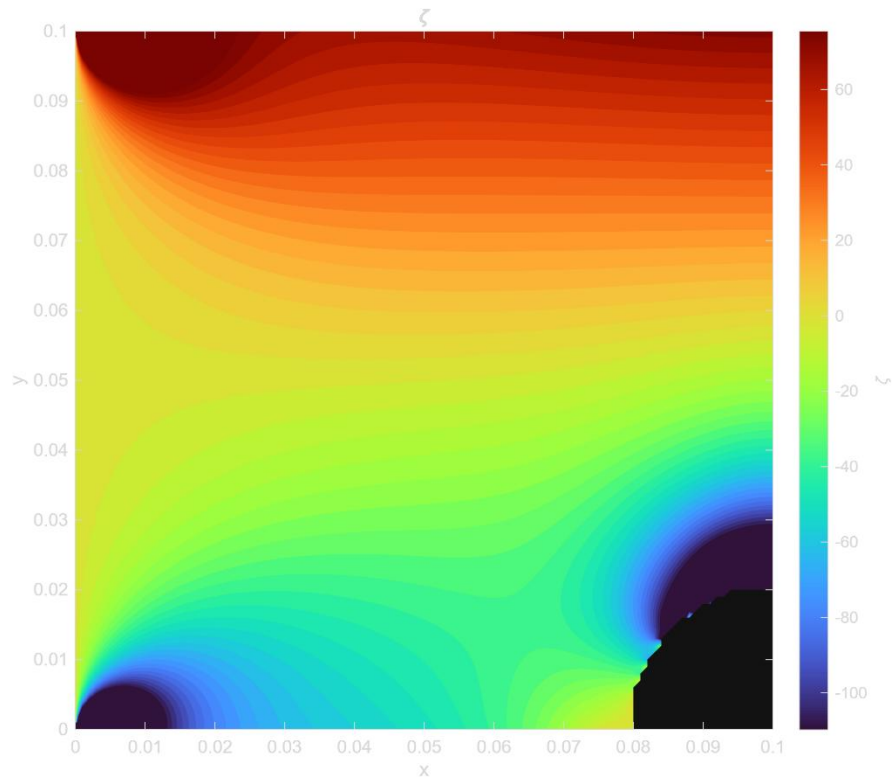


Figure 16. Vorticity contour on the intermediate mesh.

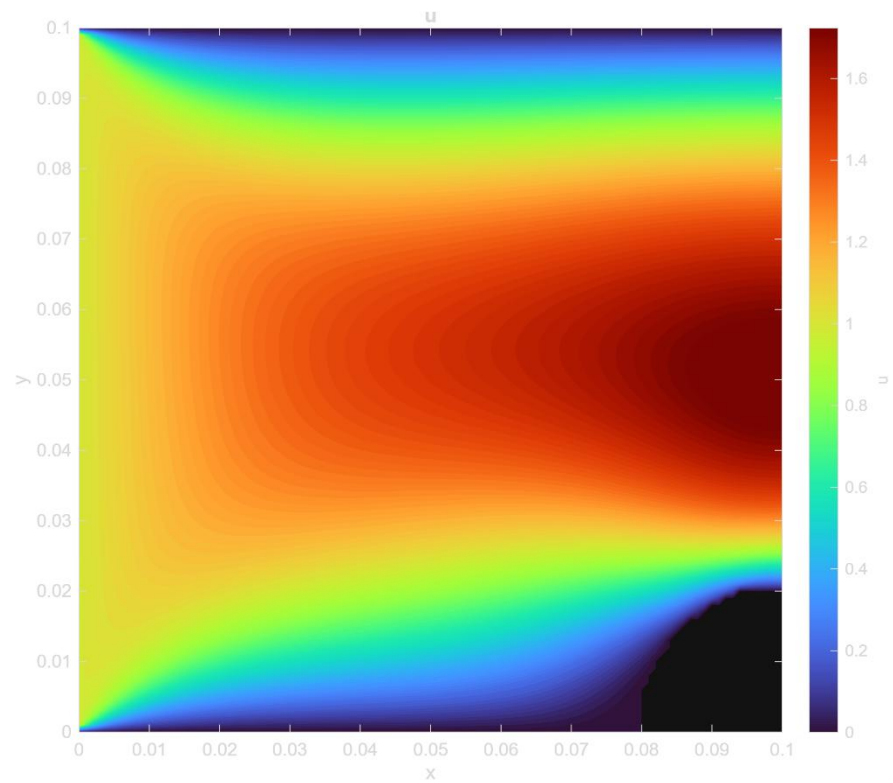


Figure 17. Horizontal velocity contour on the intermediate mesh.

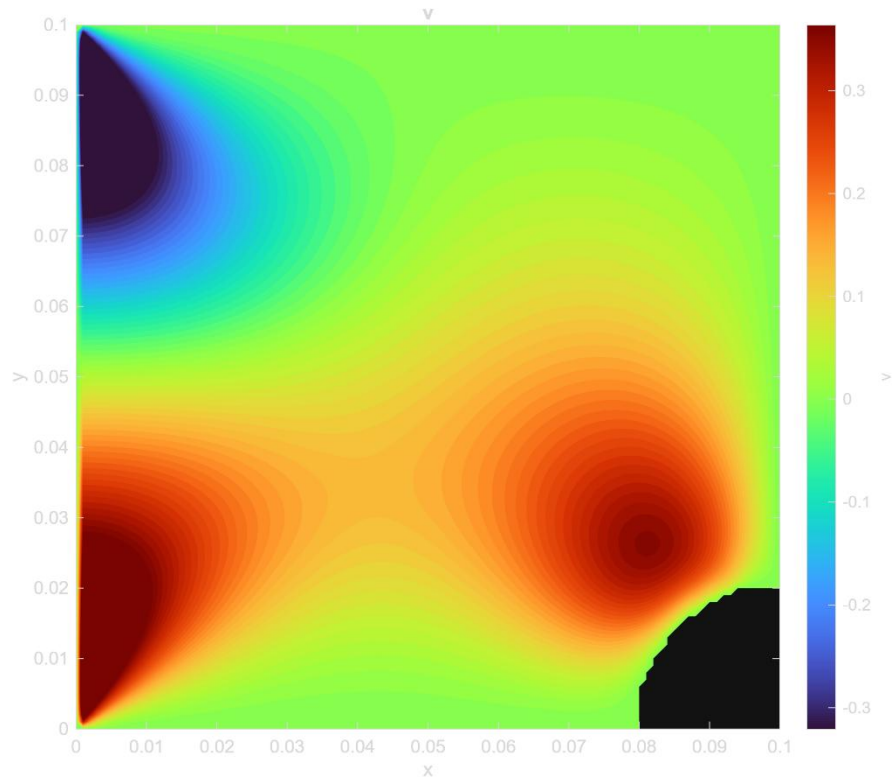


Figure 18. Vertical velocity contour on the intermediate mesh.

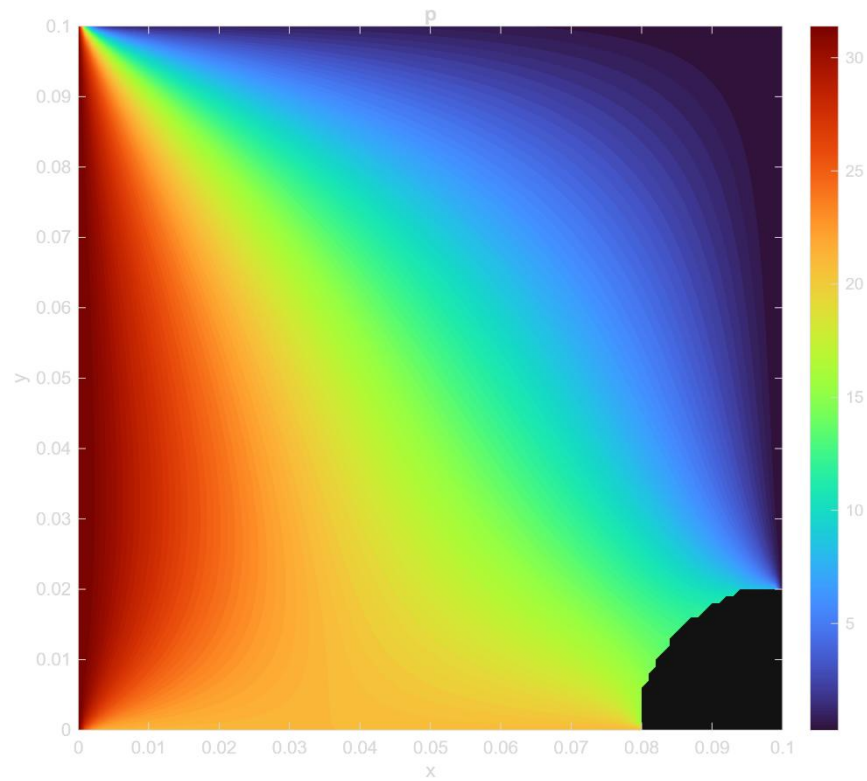


Figure 19. Pressure contour on the intermediate mesh.

5.4 line plots (intermediate mesh)

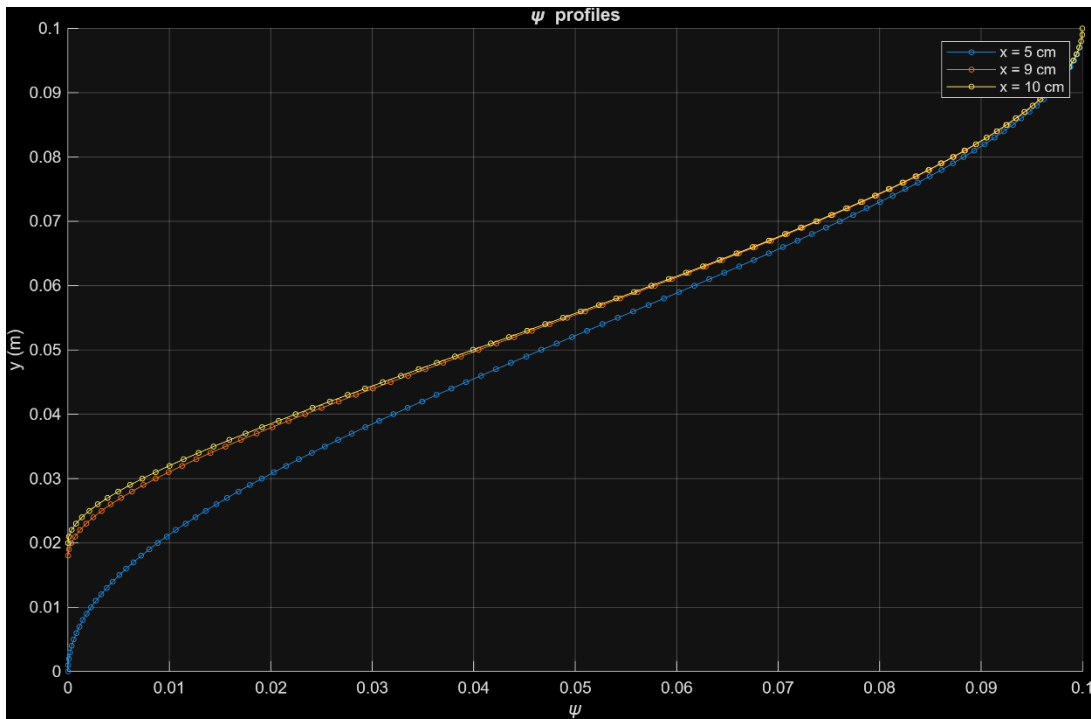


Figure 20. Stream function profiles at $x = 5, 9$ and 10 cm on the intermediate mesh.

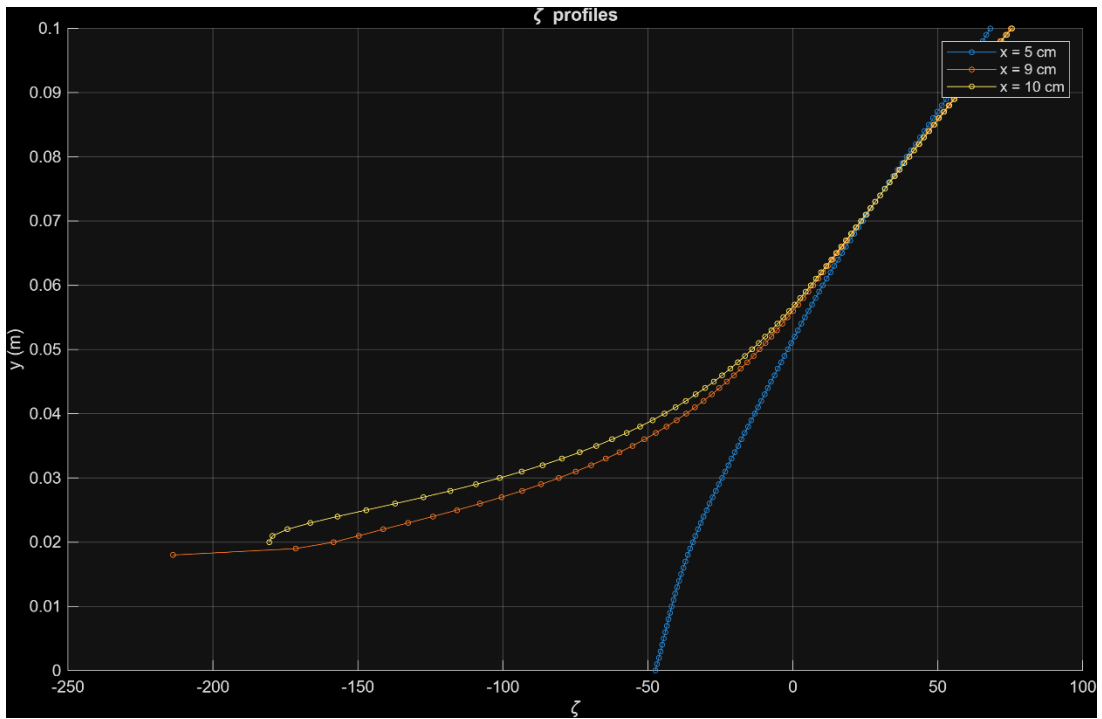


Figure 21. Vorticity profiles at $x = 5, 9$ and 10 cm on the intermediate mesh.

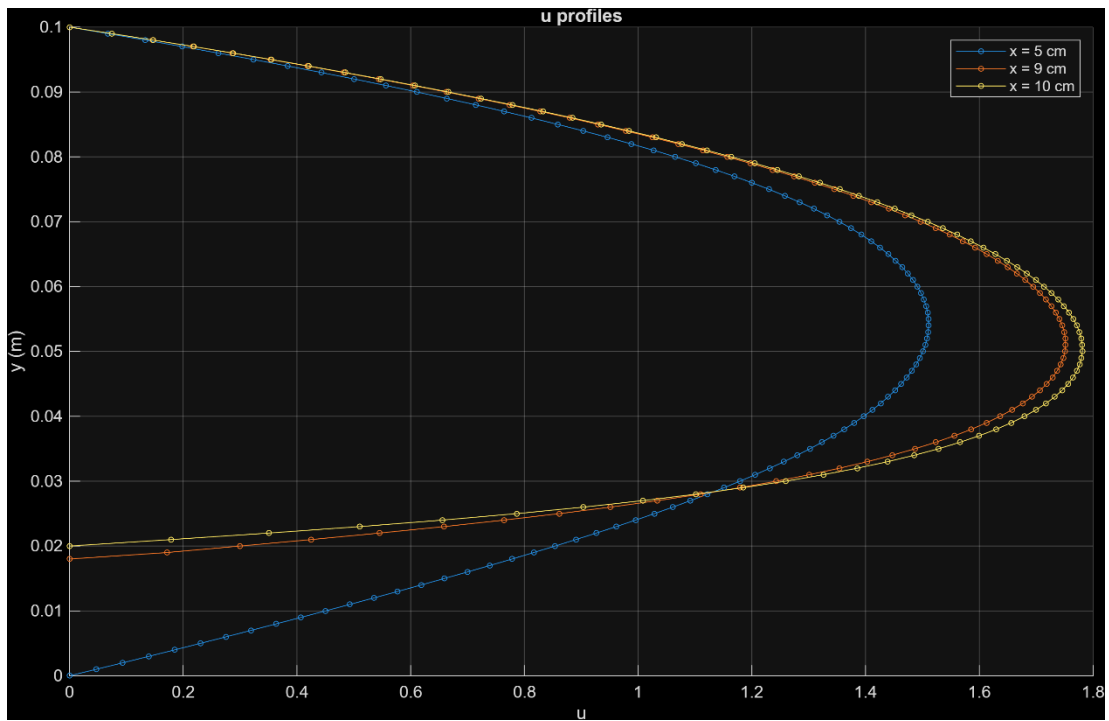


Figure 22. Horizontal velocity profiles at $x = 5, 9$ and 10 cm on the intermediate mesh.

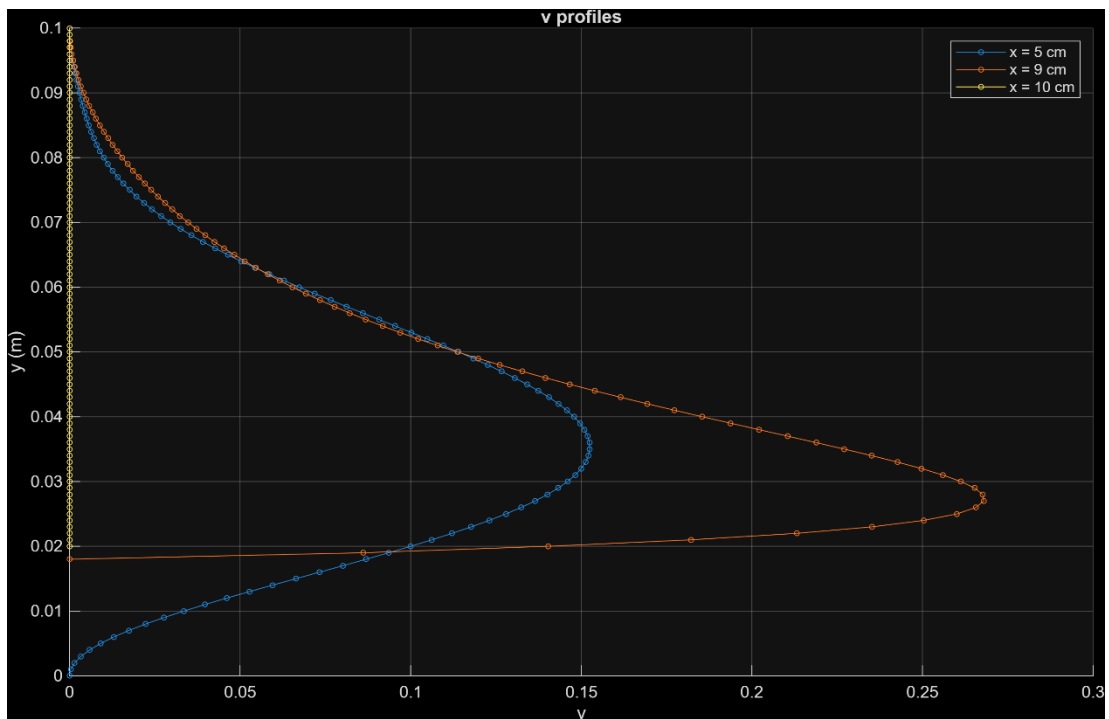


Figure 23. Vertical velocity profiles at $x = 5, 9$ and 10 cm on the intermediate mesh.

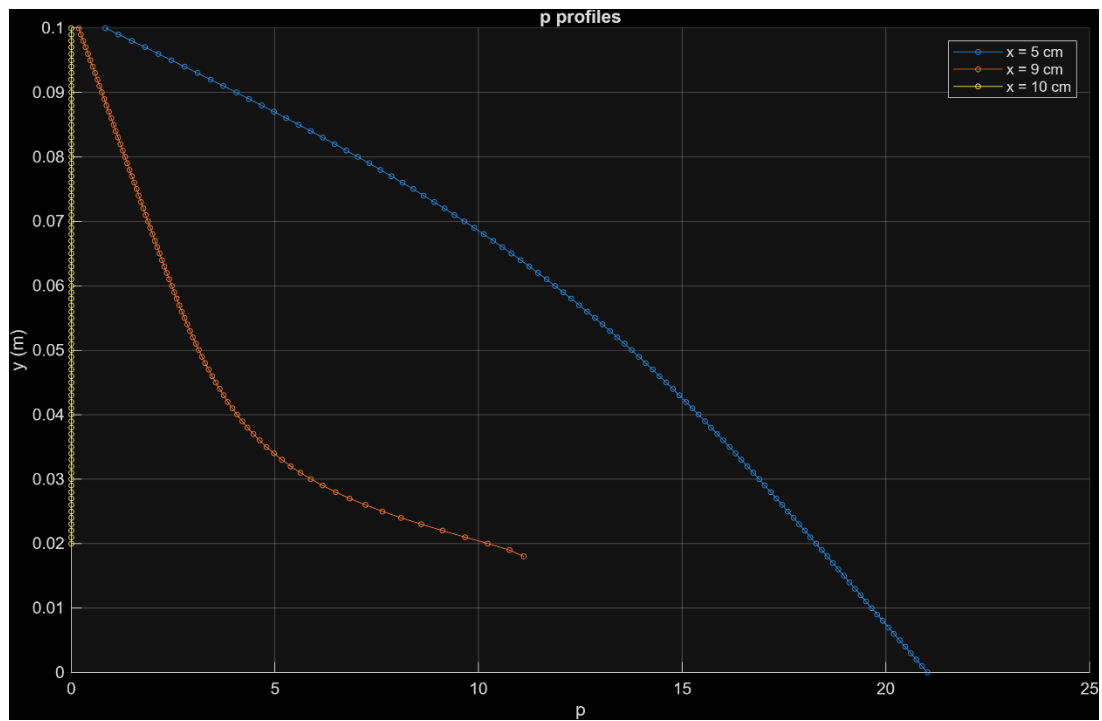


Figure 24. Pressure profiles at $x = 5, 9$ and 10 cm obtained with the Gauss–Seidel scheme on the intermediate mesh.

5.5 contours (fine mesh)

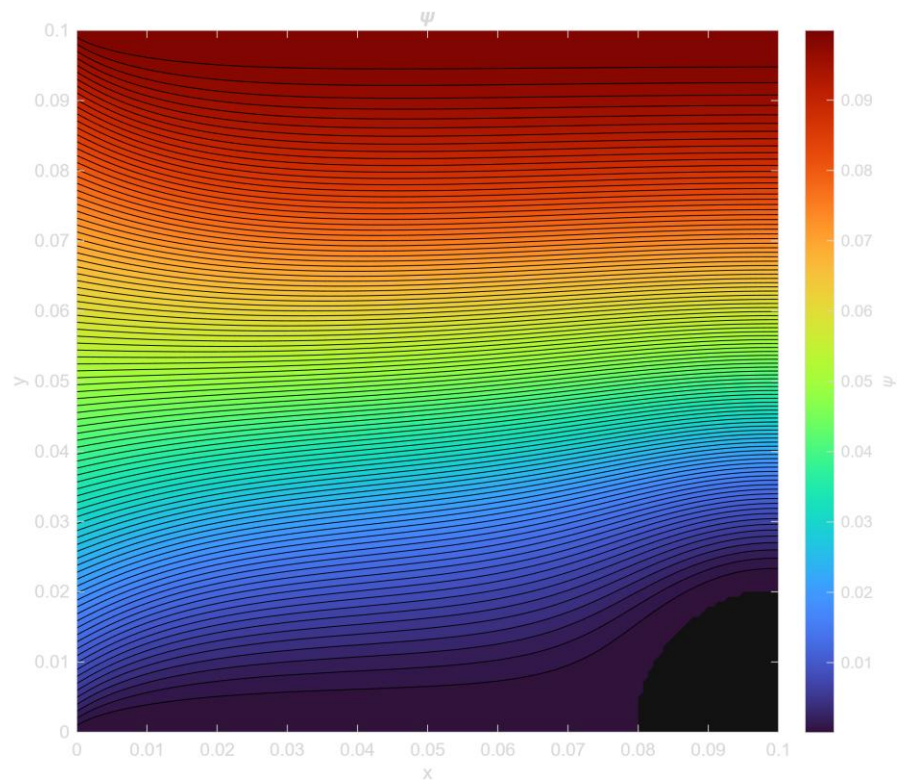


Figure 25. Stream function contour on the fine mesh.

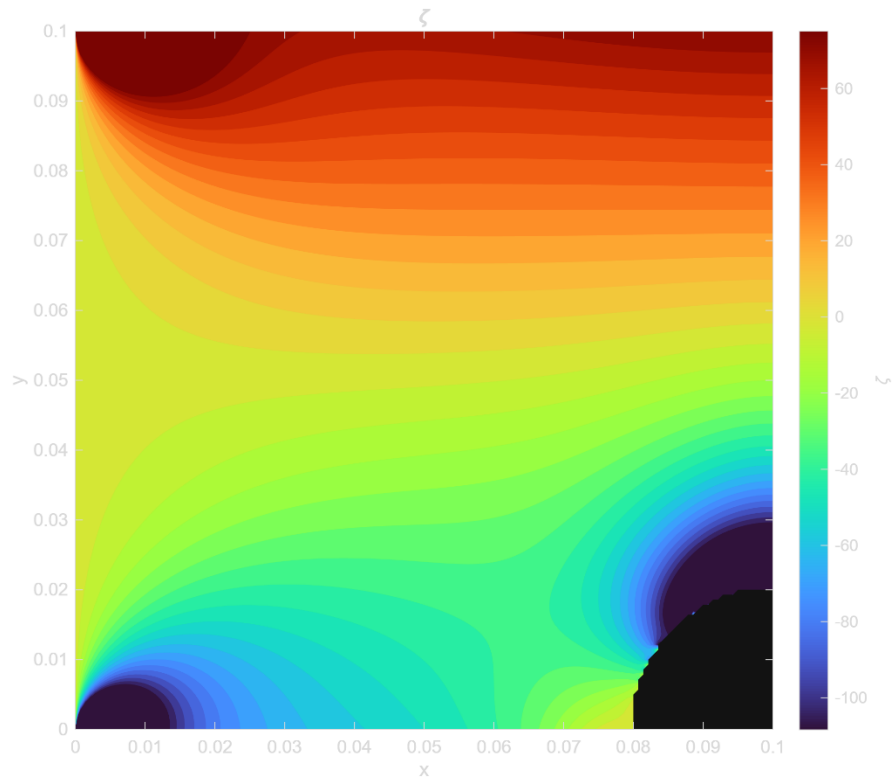


Figure 26. Vorticity contour on the fine mesh.

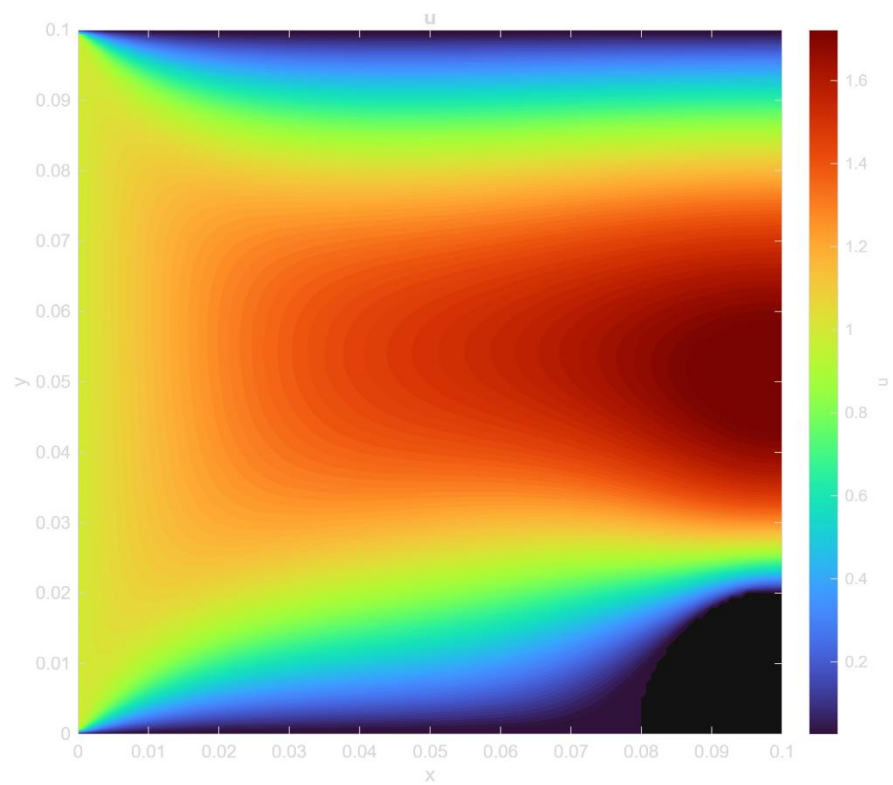


Figure 27. Horizontal velocity contour on the fine mesh.

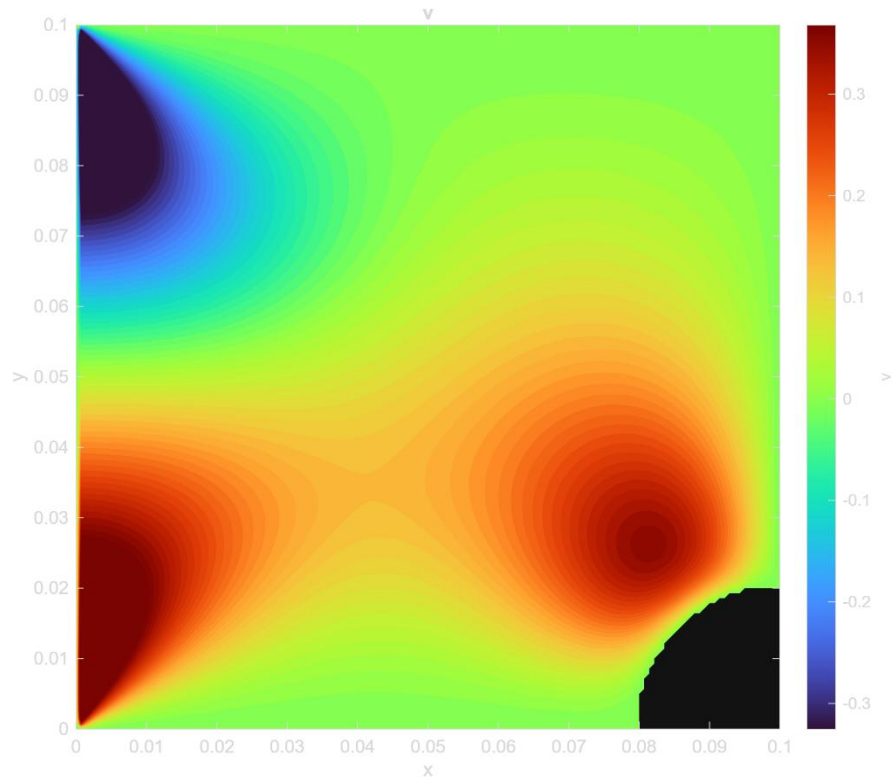


Figure 28. Vertical velocity contour on the fine mesh.

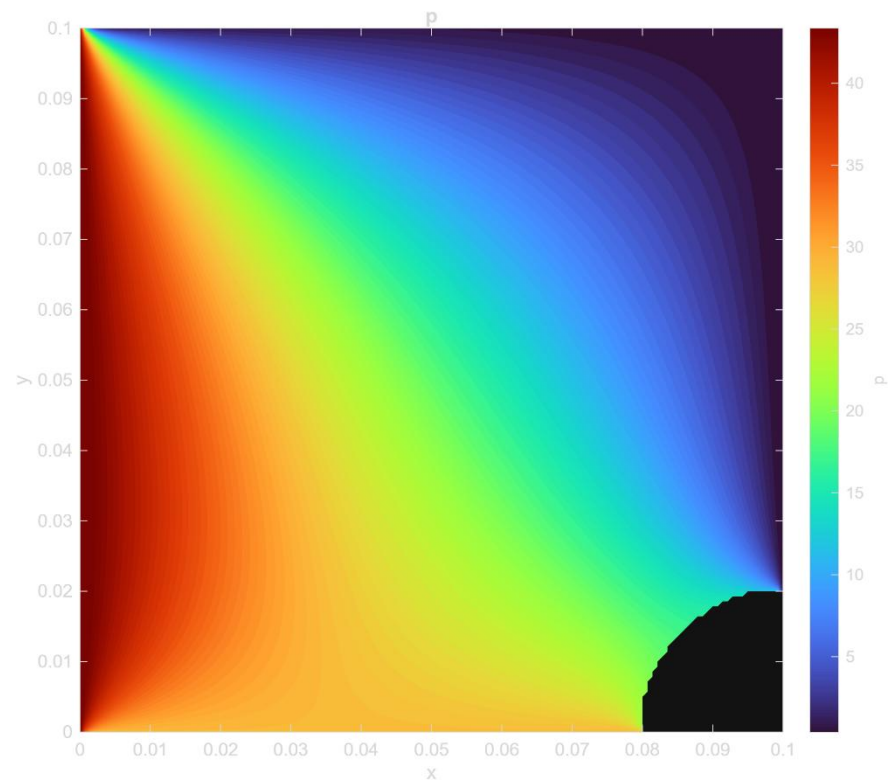


Figure 29. Pressure contour on the fine mesh.

5.6 line plots (fine mesh)

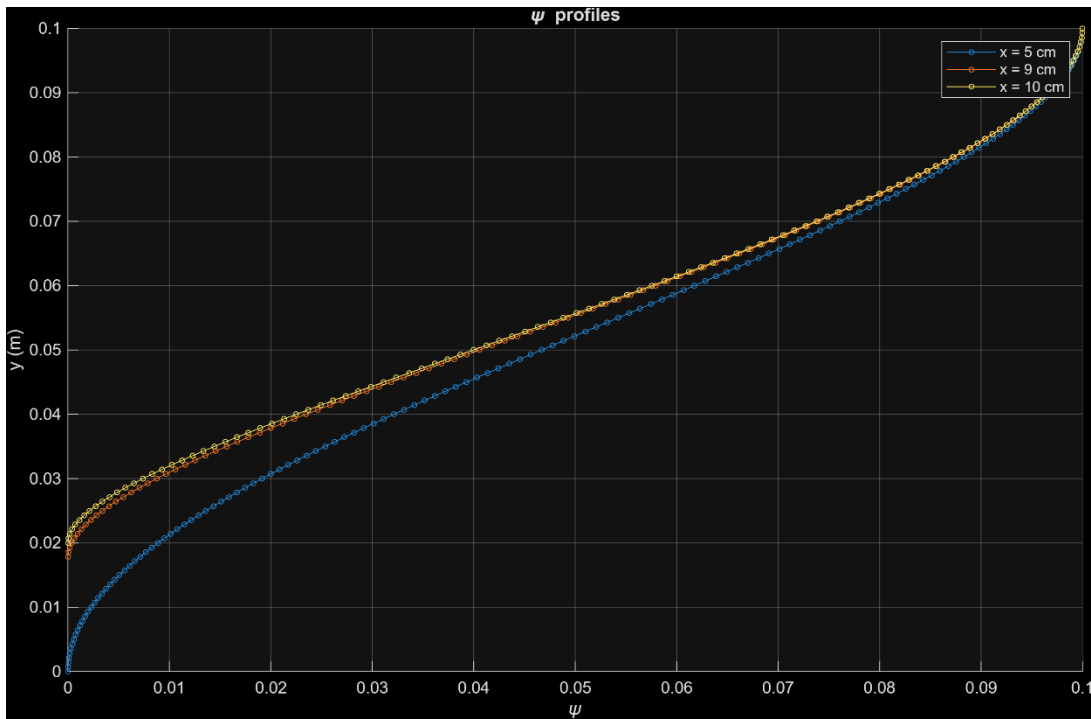


Figure 30. Stream function profiles at $x = 5, 9$ and 10 cm on the fine mesh.

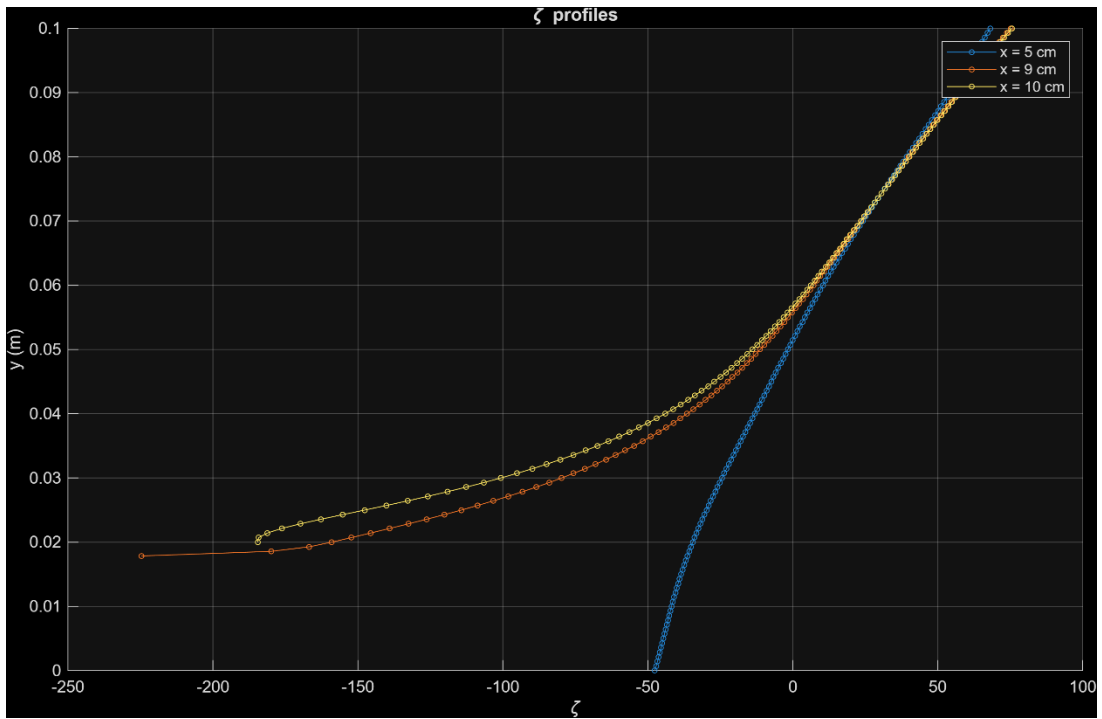


Figure 31. Vorticity profiles at $x = 5, 9$ and 10 cm on the fine mesh.

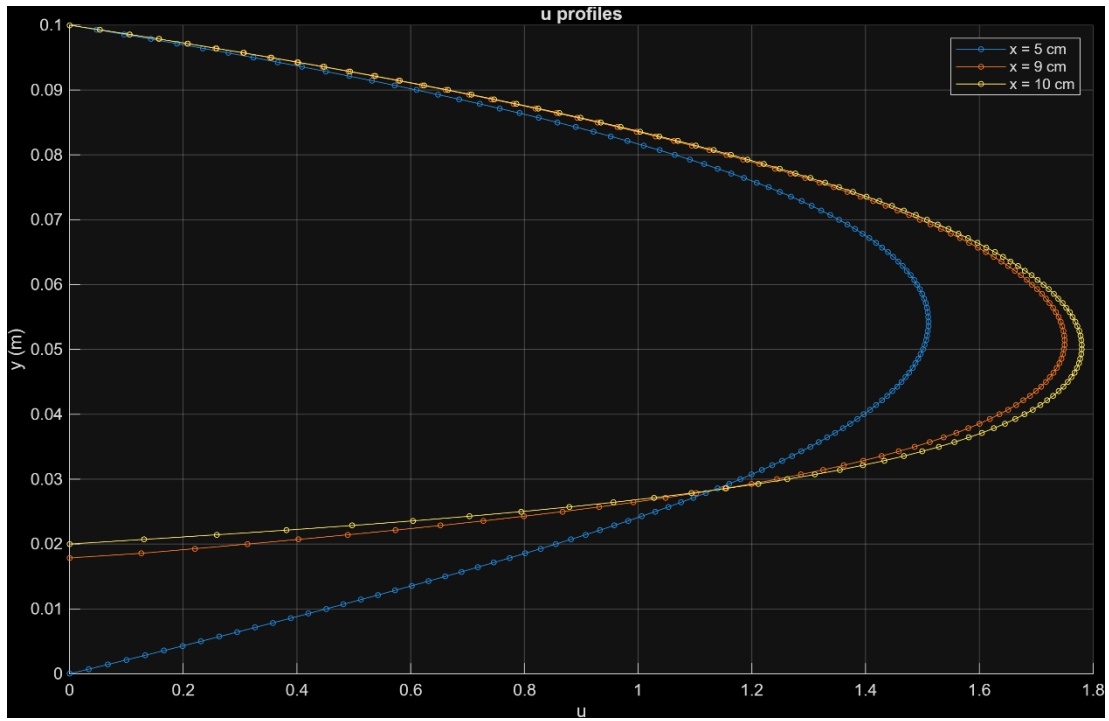


Figure 32. Horizontal velocity profiles at $x = 5, 9$ and 10 cm on the fine mesh.

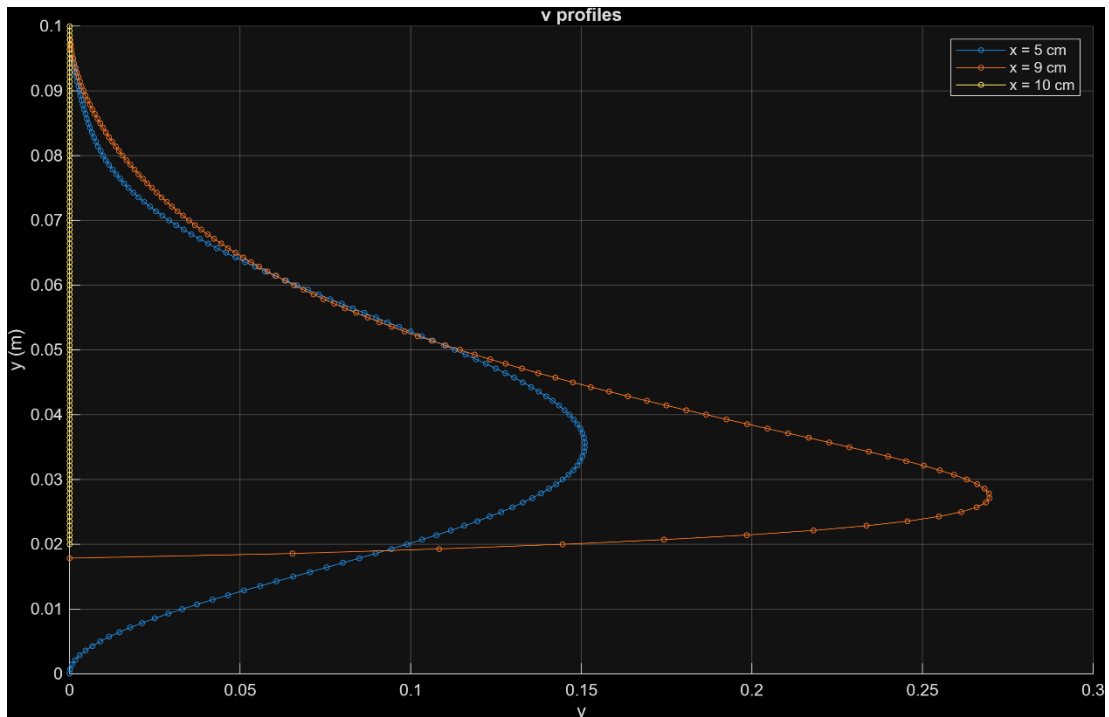


Figure 33. Vertical velocity profiles at $x = 5, 9$ and 10 cm on the fine mesh.

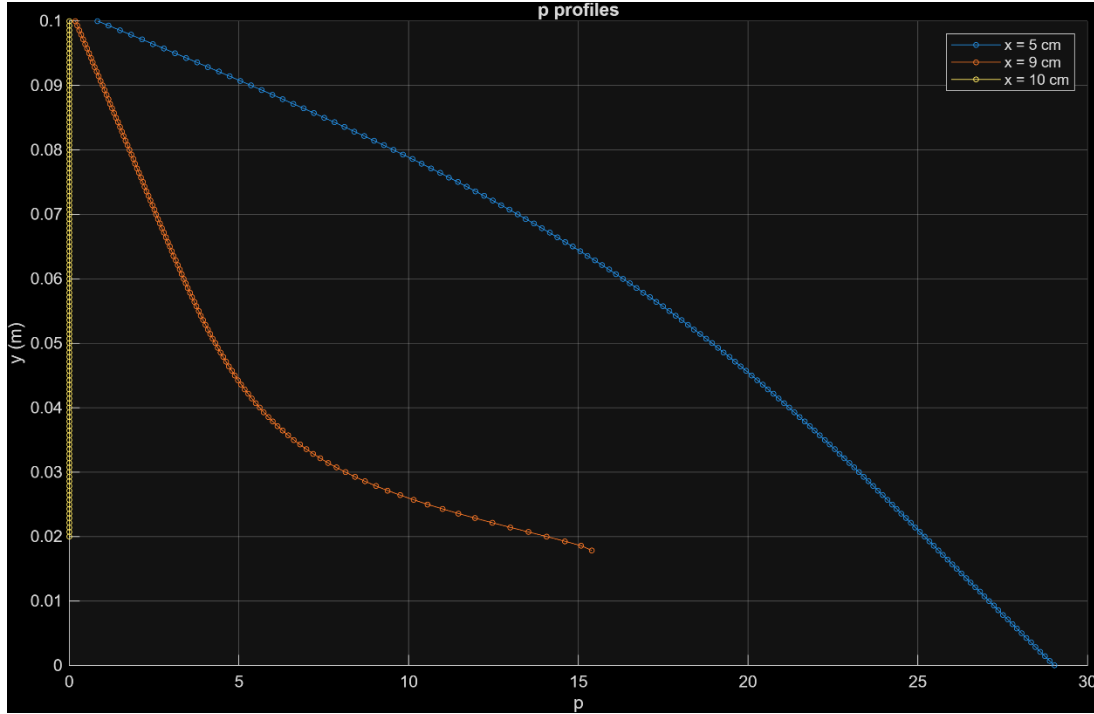


Figure 34. Pressure profiles at $x = 5, 9$ and 10 cm the fine mesh.

6 Successive Over-Relaxation (SOR) Method

The SOR scheme uses the same sweep order and the same sequence as the Gauss-Seidel method, but each new value is obtained as a weighted combination of the previous value and the Gauss-Seidel value:

$$\phi_{i,j}^{k+1} = (1 - \omega)\phi_{i,j}^k + \omega\phi_{i,j}^{GS}$$

$\omega > 1$ over-relaxes the correction and accelerates the transfer of boundary information through the domain.

The relaxation factor for the stream function equation is based on the optimum value for the Laplace operator on a rectangular grid, This is only the starting point value for ω and is changed to find the best amount by trial and error.

$$\sigma = \frac{\cos(\pi/p) + \beta^2 \cos(\pi/q)}{1 + \beta^2}, \quad \omega_{opt} = \frac{2}{1 + \sqrt{1 - \sigma^2}}$$

where p and q are the numbers of intervals in the x and y directions and β is the ratio of the mesh spacings.

the theoretical value proved too aggressive and led to divergence. The value actually used was therefore taken slightly below it, and a separate, smaller relaxation factor was applied to the vorticity equation.

The scheme was run on the coarse, intermediate and fine meshes; the corresponding iteration counts, wall-clock times and relaxation factors are reported below, followed by the contours and line plots obtained on each of the three meshes.

Table 3. Relaxation factors used by the SOR scheme on the three meshes.

Mesh	$\omega_{\text{opt}}(\psi)$	$\omega_{\text{opt}}(\zeta)$
Coarse	1.70	1.40
Intermediate	1.74	1.44
Fine	1.76	1.46

Table 4. Iteration count and wall-clock time of the SOR scheme on the three meshes.

Mesh	Iterations	Wall-clock time (s)
Coarse (61×61)	1429	0.49
Intermediate (101×101)	4,173	2.73
Fine (141×141)	8,041	9.85

6.1 contours (coarse mesh)

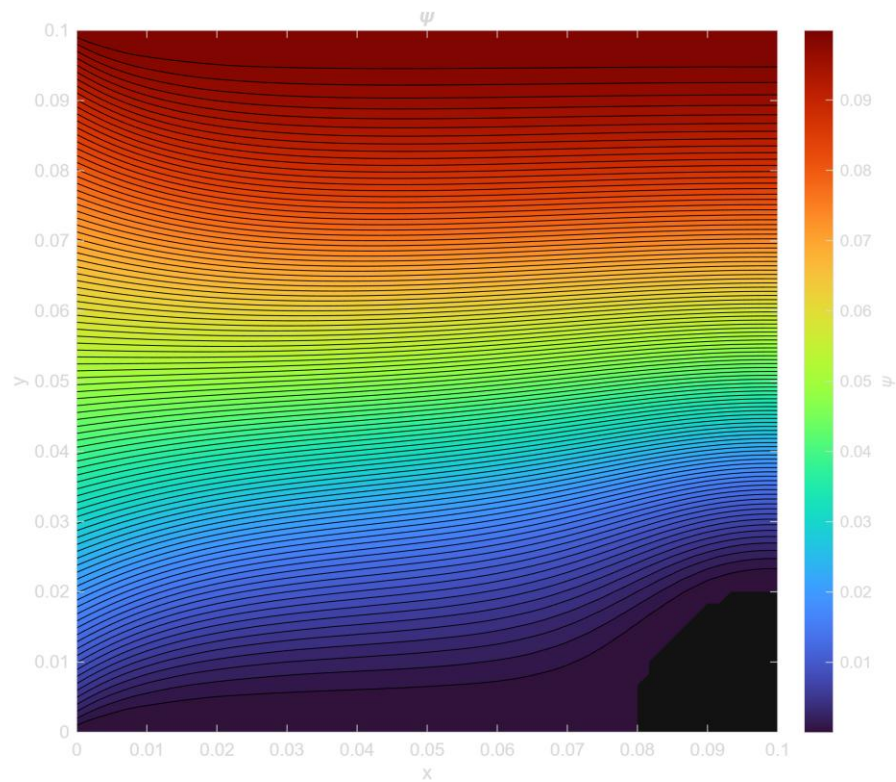


Figure 35. Stream function contour on the coarse mesh.

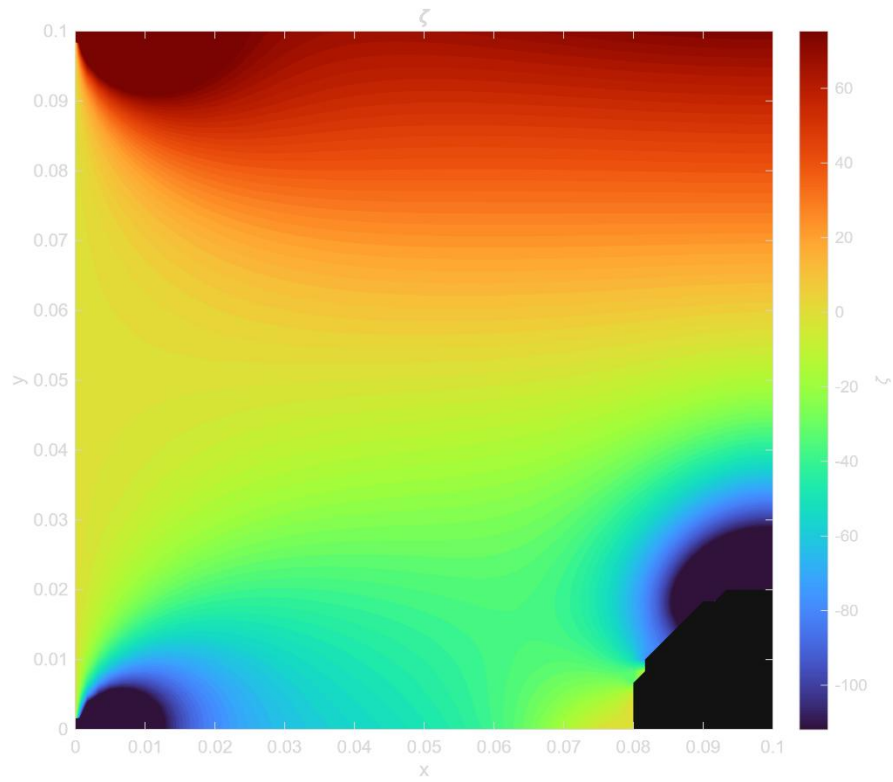


Figure 36. Vorticity contour on the coarse mesh.

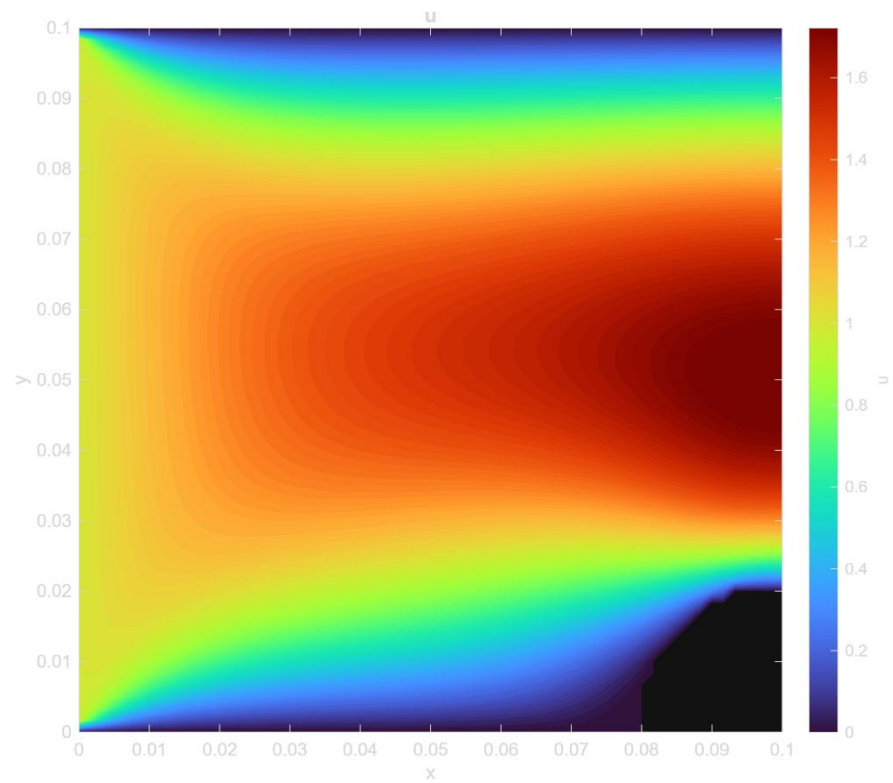


Figure 37. Horizontal velocity contour on the coarse mesh.

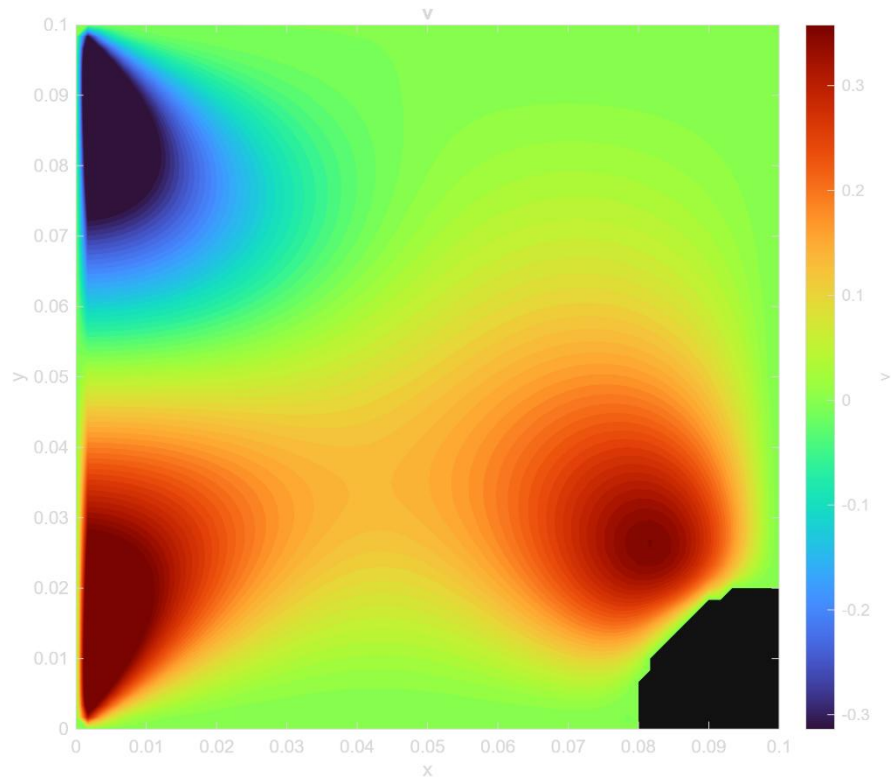


Figure 38. Vertical velocity contour on the coarse mesh.

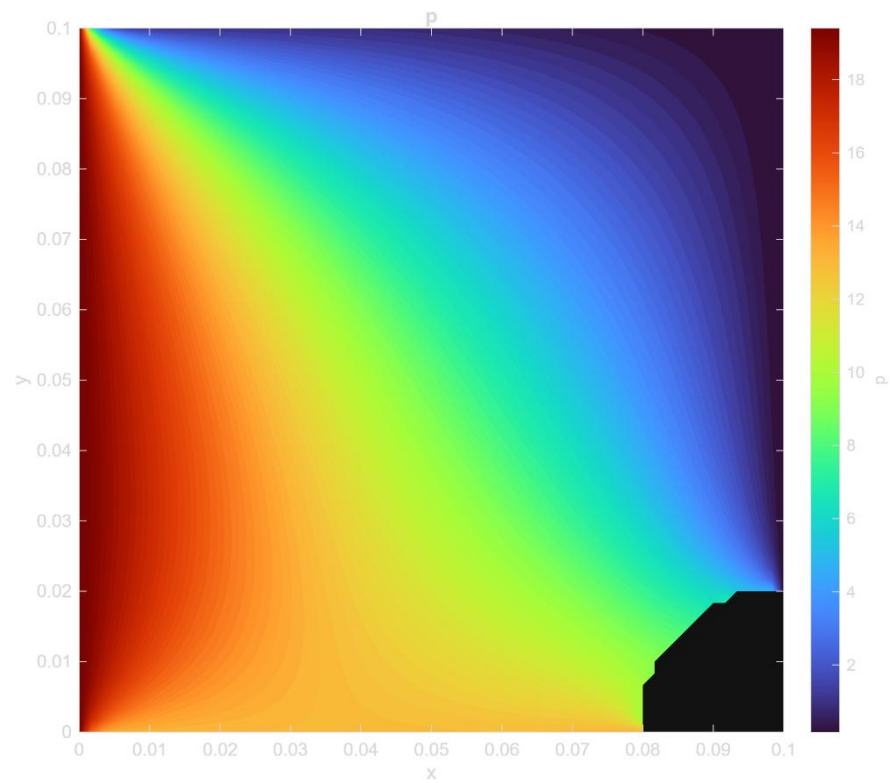


Figure 39. Pressure contour on the coarse mesh.

6.2 line plots (coarse mesh)

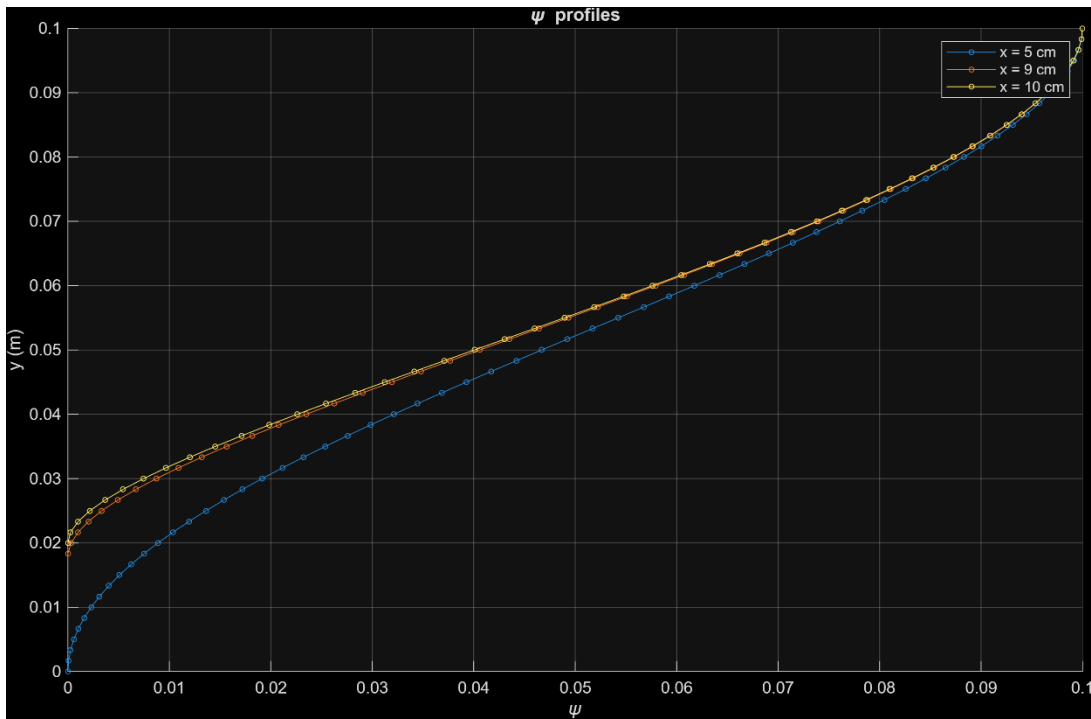


Figure 40. Stream function profiles at $x = 5, 9$ and 10 cm on the coarse mesh.

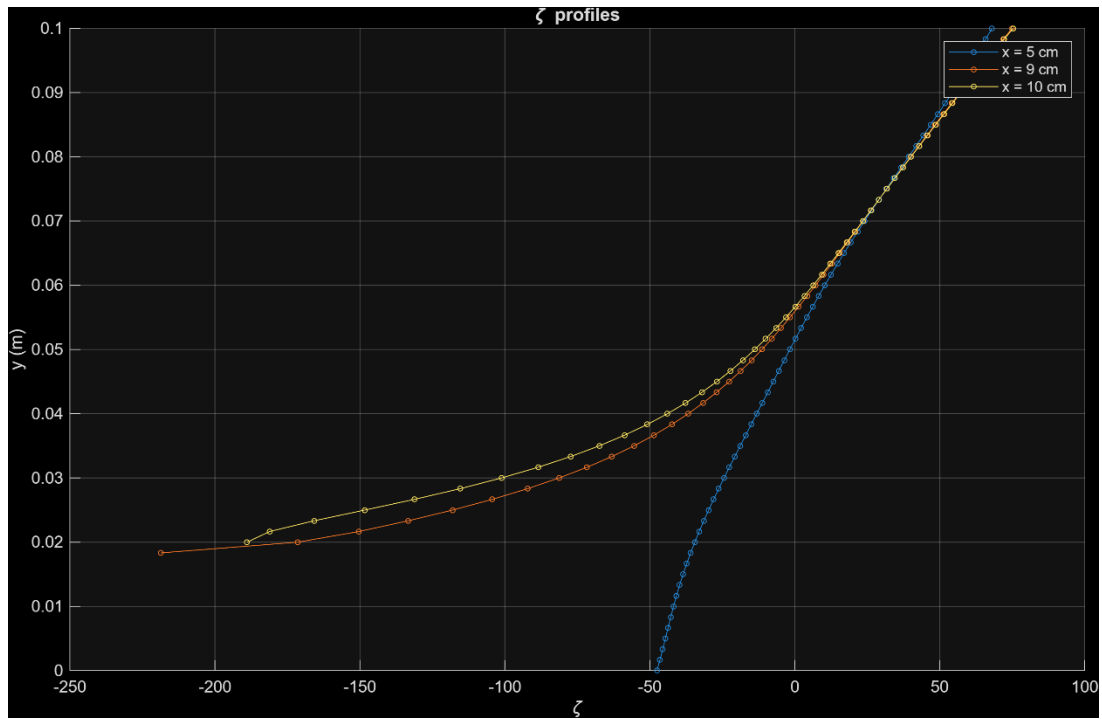


Figure 41. Vorticity profiles at $x = 5, 9$ and 10 cm on the coarse mesh

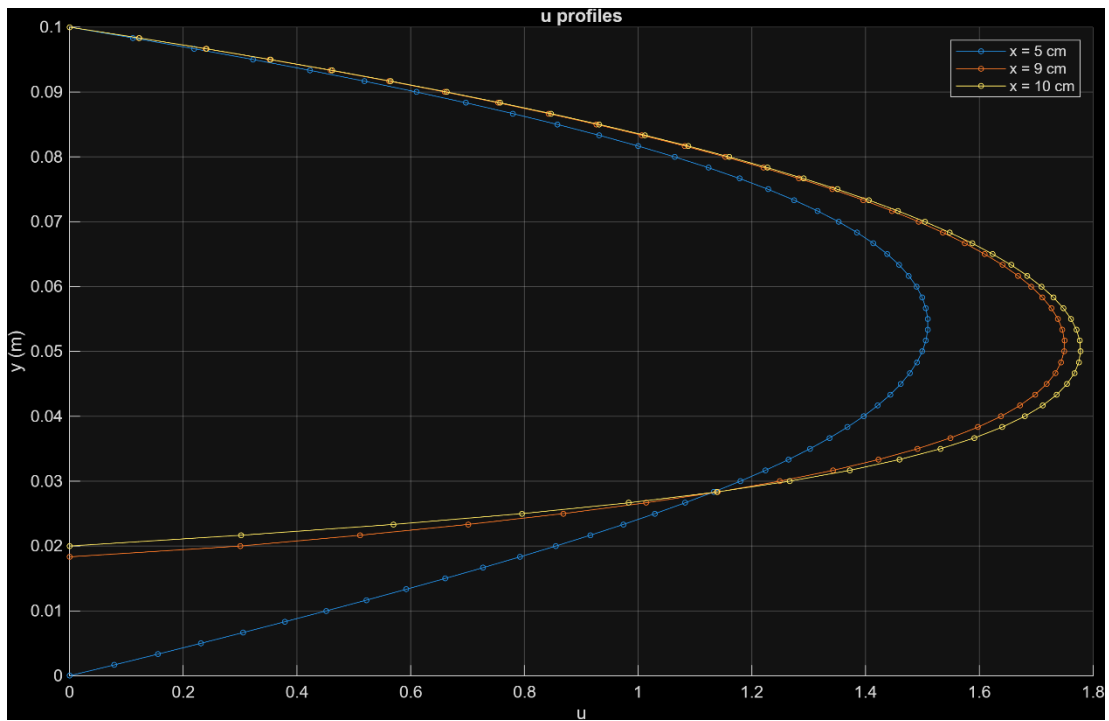


Figure 42. Horizontal velocity profiles at $x = 5, 9$ and 10 cm on the coarse mesh.

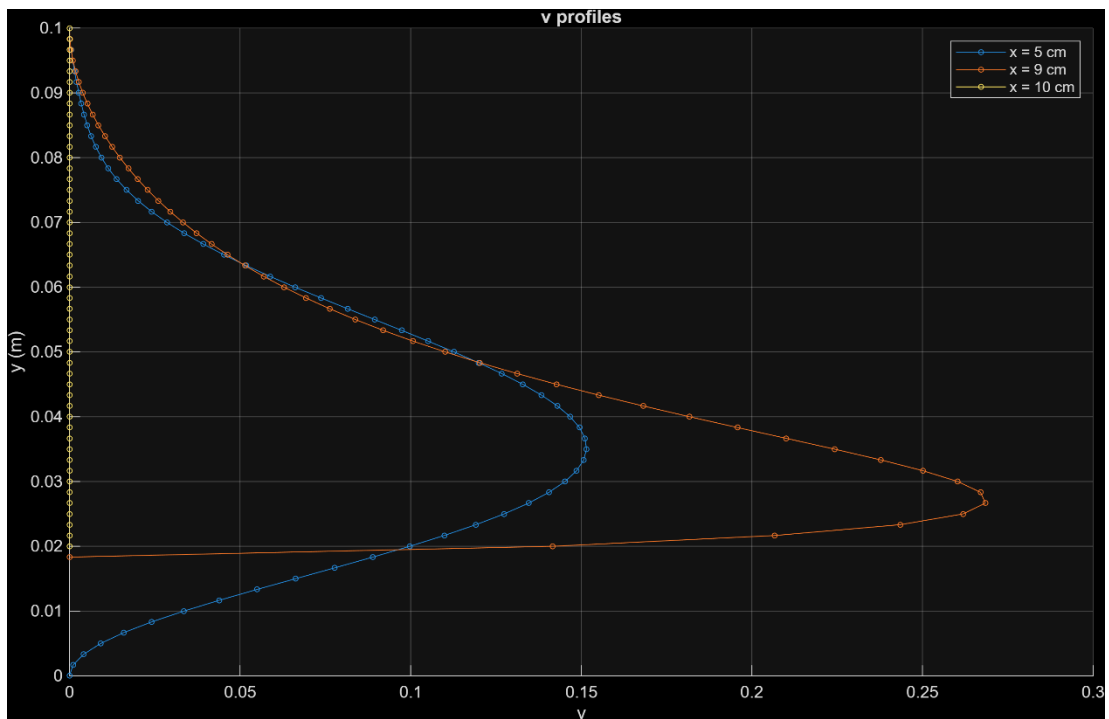


Figure 43. Vertical velocity profiles at $x = 5, 9$ and 10 cm on the coarse mesh.

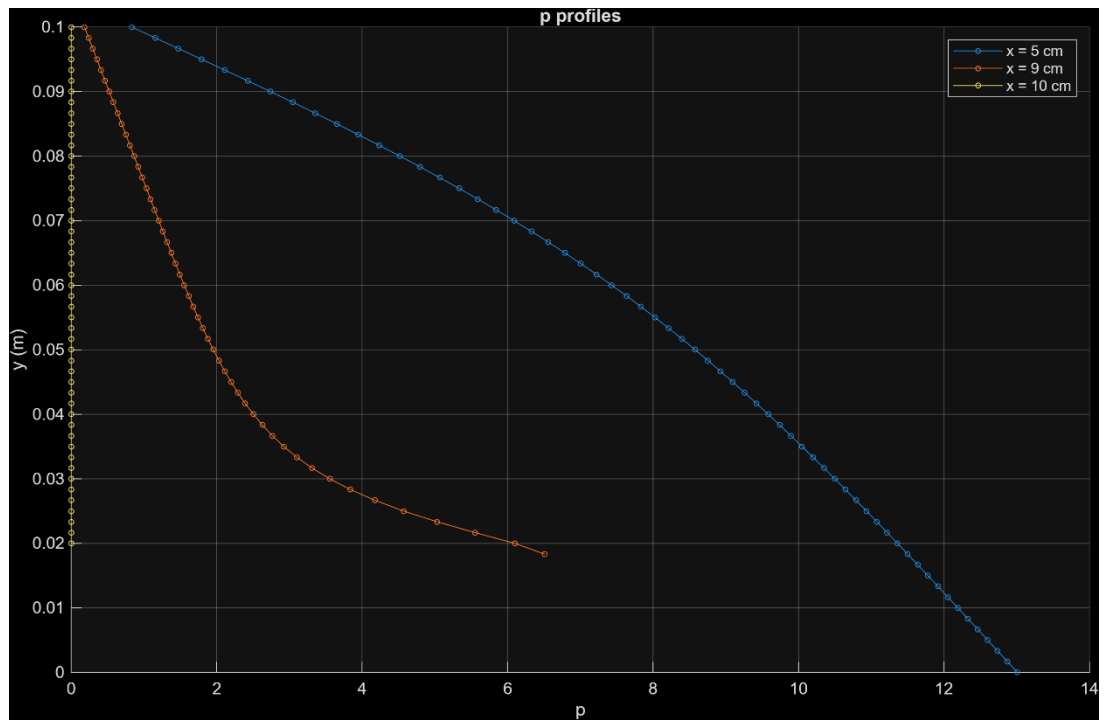


Figure 44. Pressure profiles at $x = 5, 9$ and 10 cm on the coarse mesh.

6.3 contours (intermediate mesh)

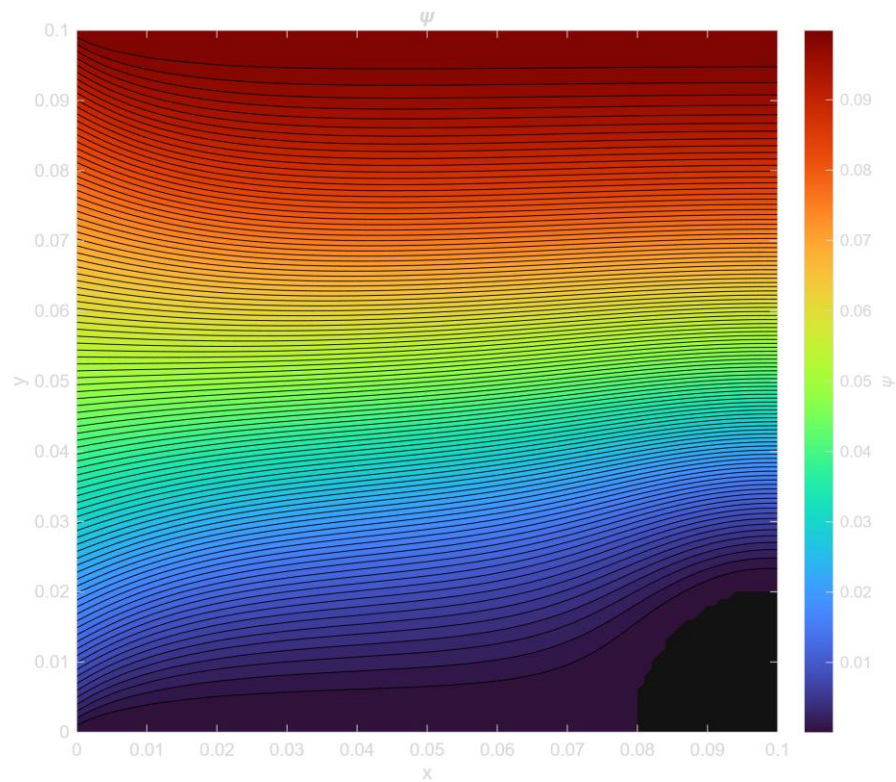


Figure 45. Stream function contour on the intermediate mesh.

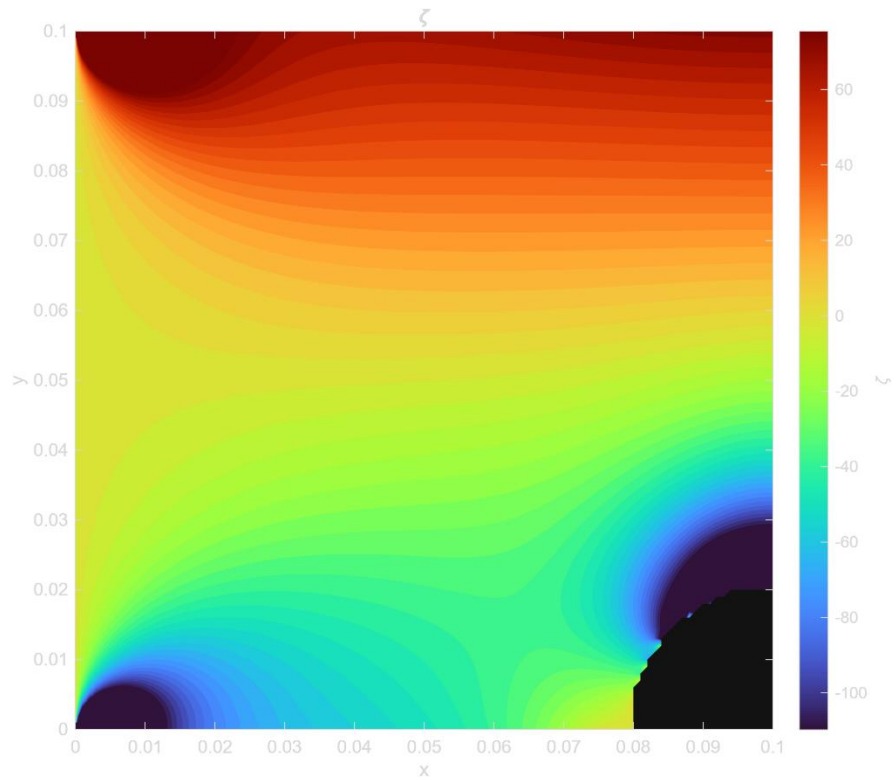


Figure 46. Vorticity contour on the intermediate mesh.

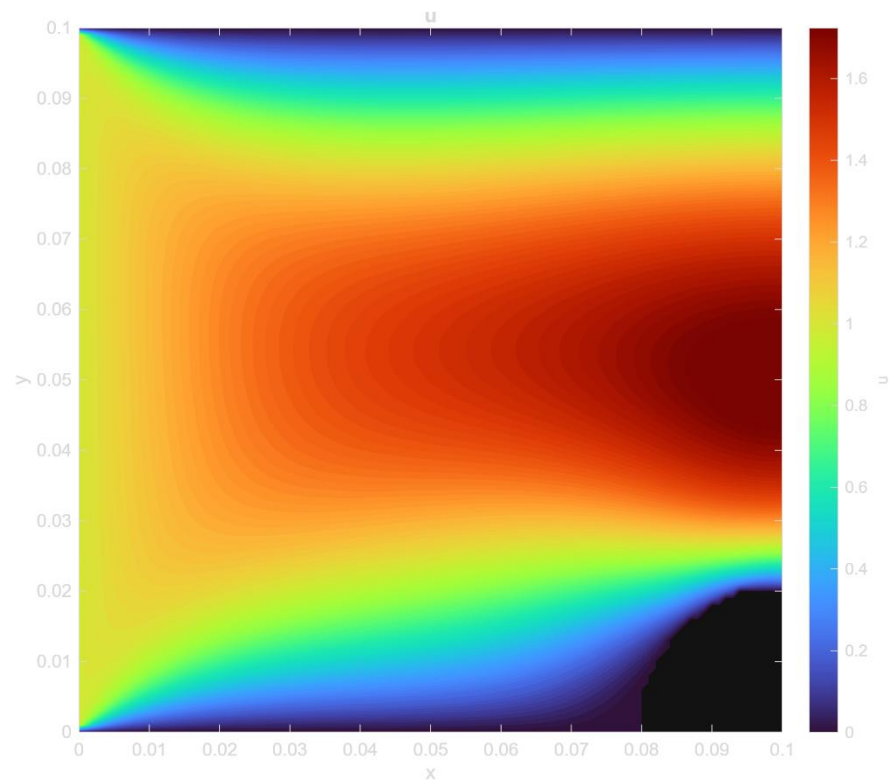


Figure 47. Horizontal velocity contour on the intermediate mesh.

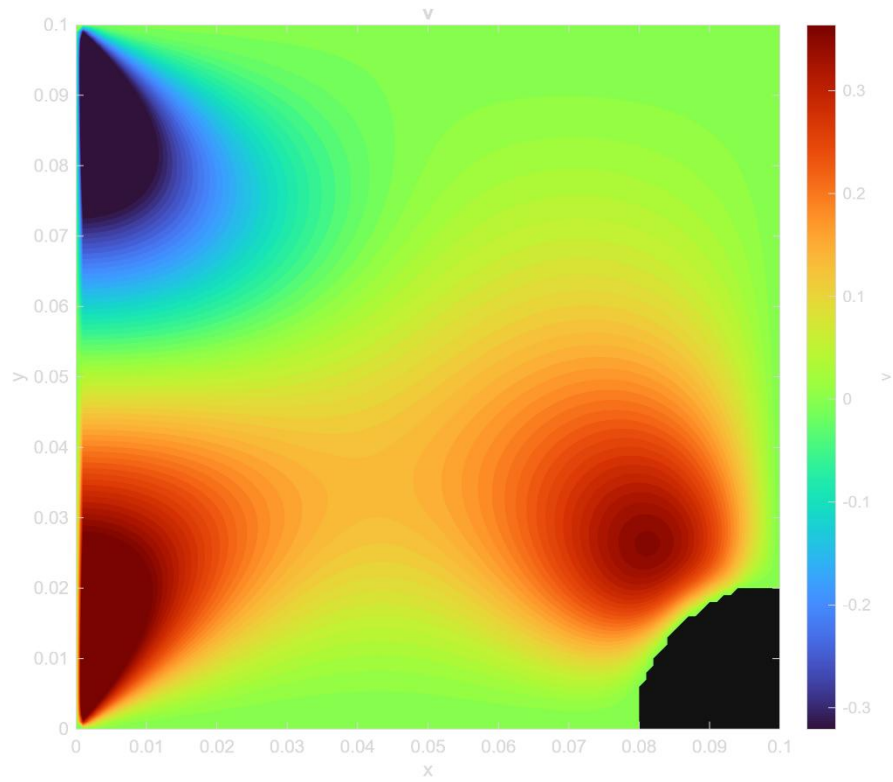


Figure 48. Vertical velocity contour on the intermediate mesh.

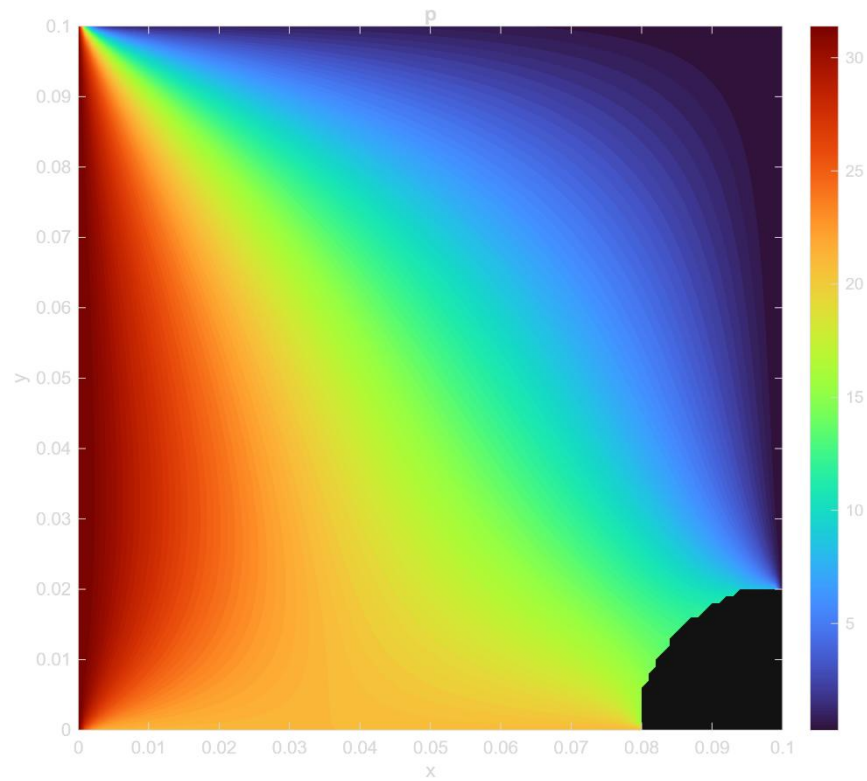


Figure 49. Pressure contour on the intermediate mesh.

6.4 line plots (intermediate mesh)

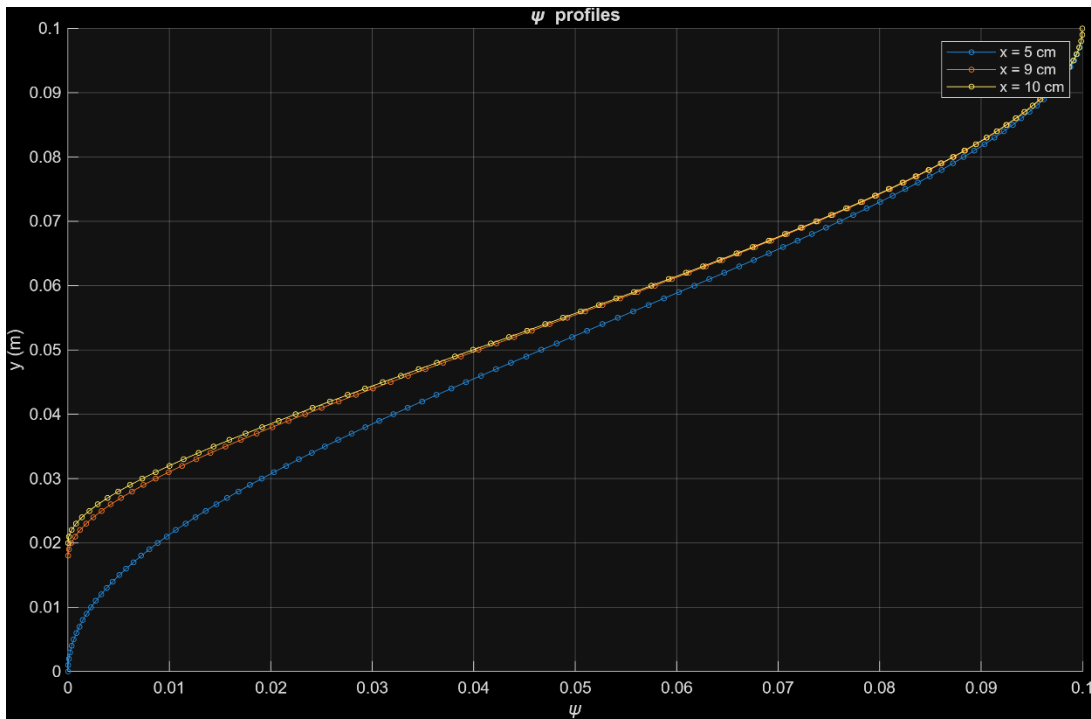


Figure 50. Stream function profiles at $x = 5, 9$ and 10 cm on the intermediate mesh.

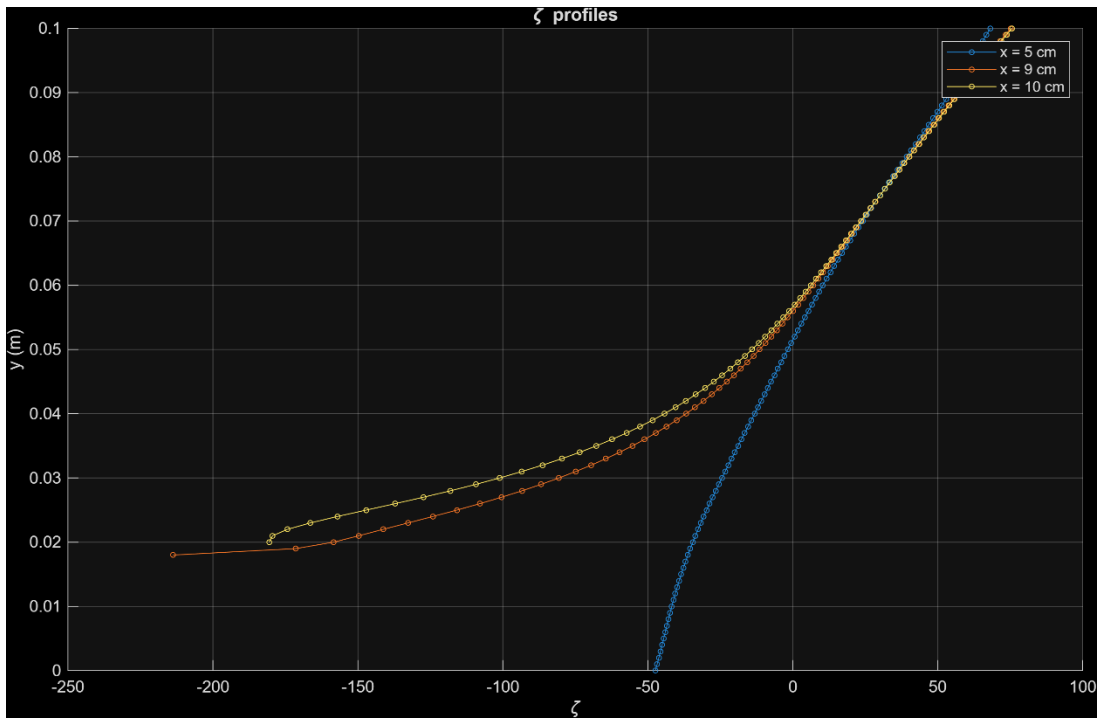


Figure 51. Vorticity profiles at $x = 5, 9$ and 10 cm on the intermediate mesh.

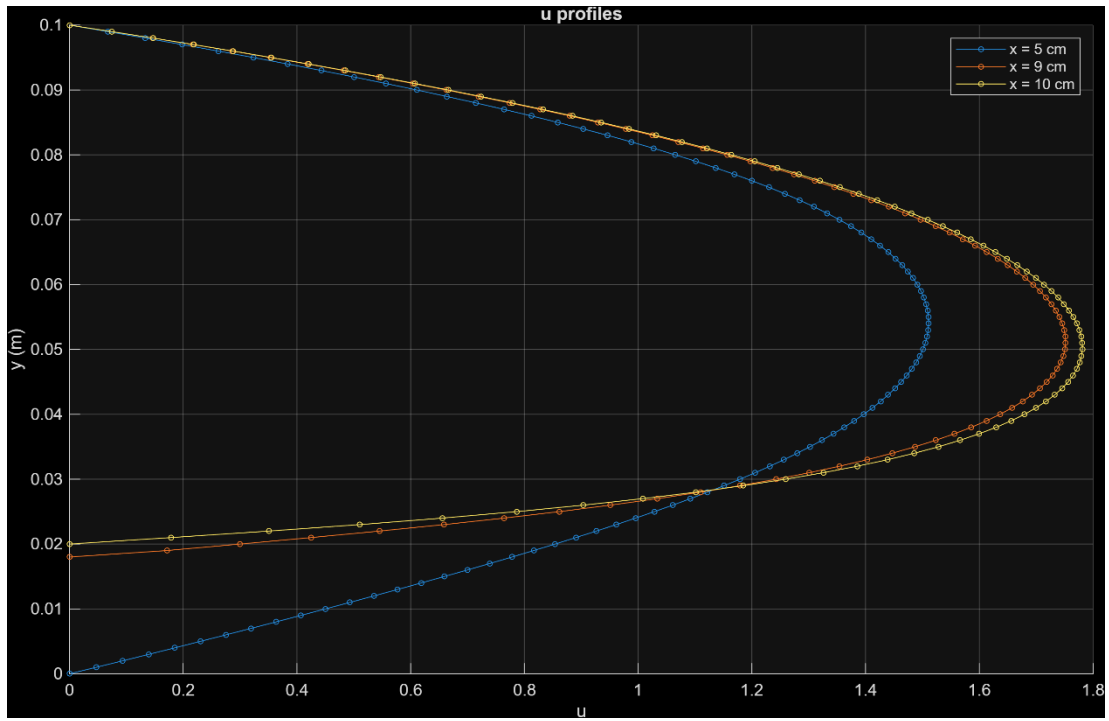


Figure 52. Horizontal velocity profiles at $x = 5, 9$ and 10 cm on the intermediate mesh.

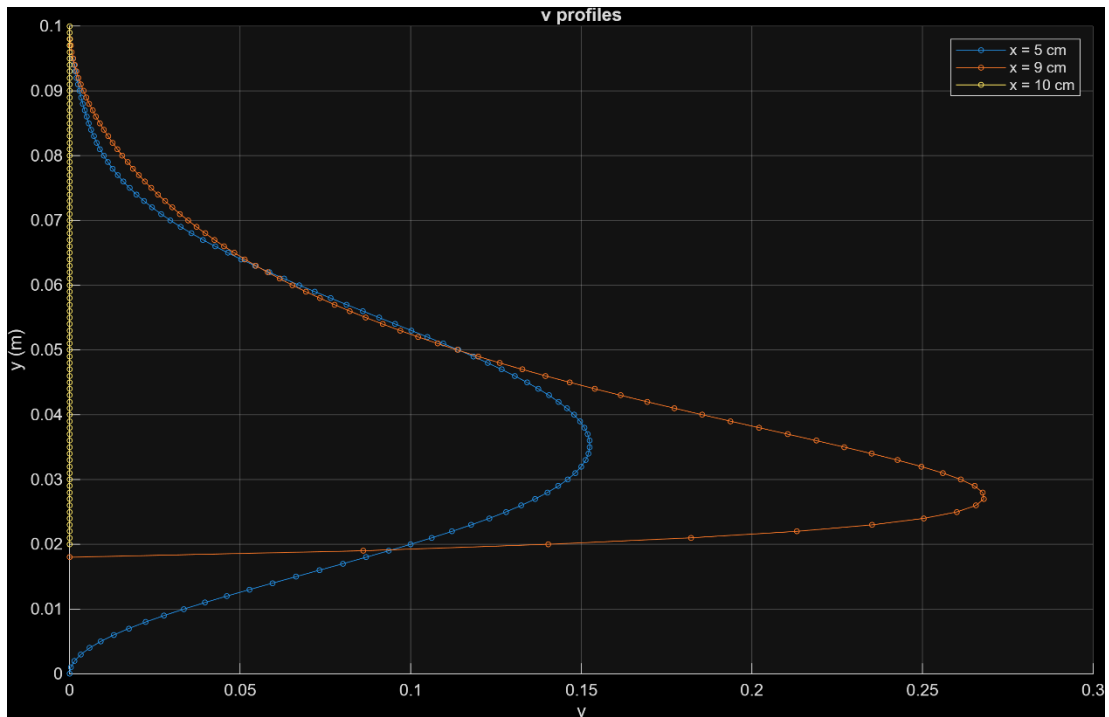


Figure 53. Vertical velocity profiles at $x = 5, 9$ and 10 cm on the intermediate mesh.

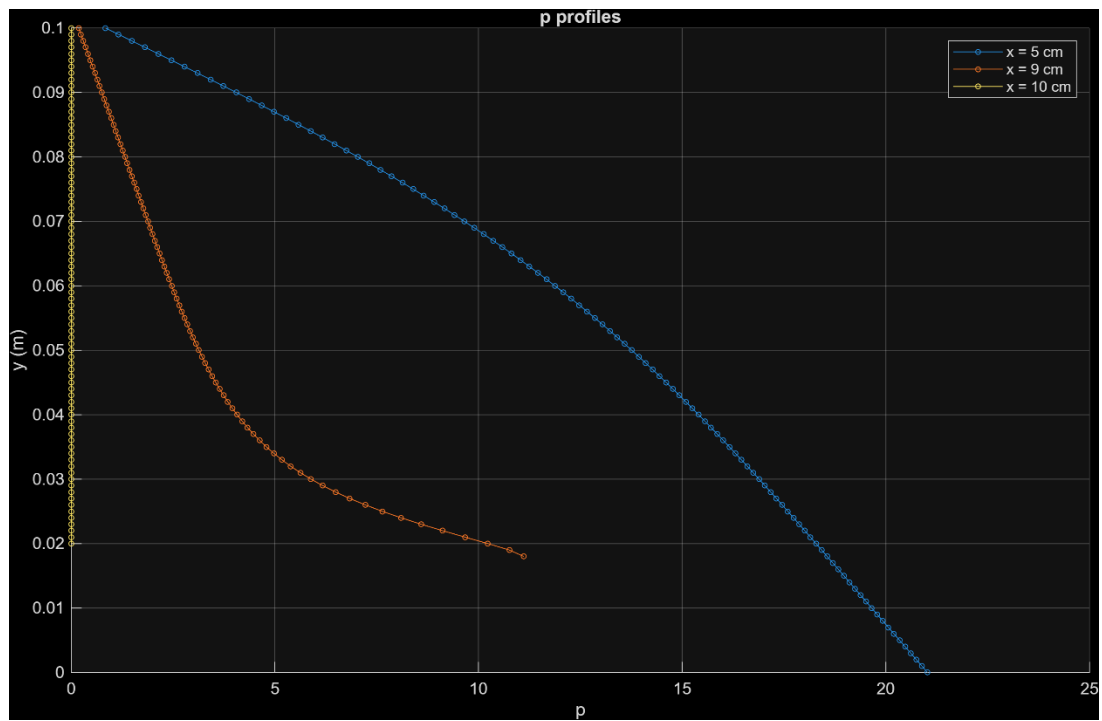


Figure 54. Pressure profiles at $x = 5, 9$ and 10 cm obtained with the Gauss–Seidel scheme on the intermediate mesh.

6.5 contours (fine mesh)

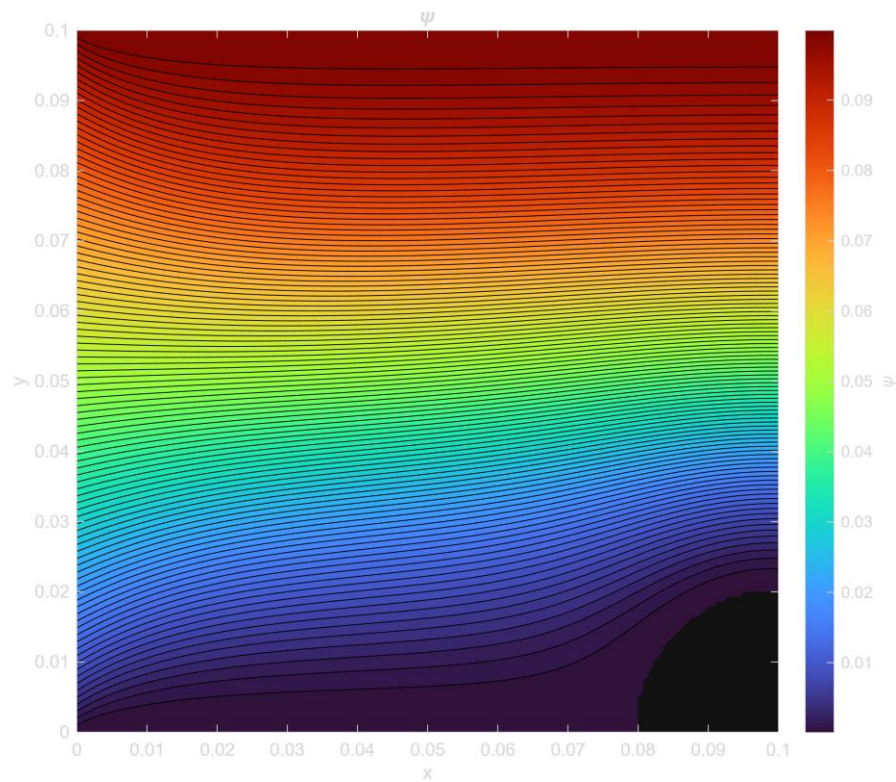


Figure 55. Stream function contour on the fine mesh.

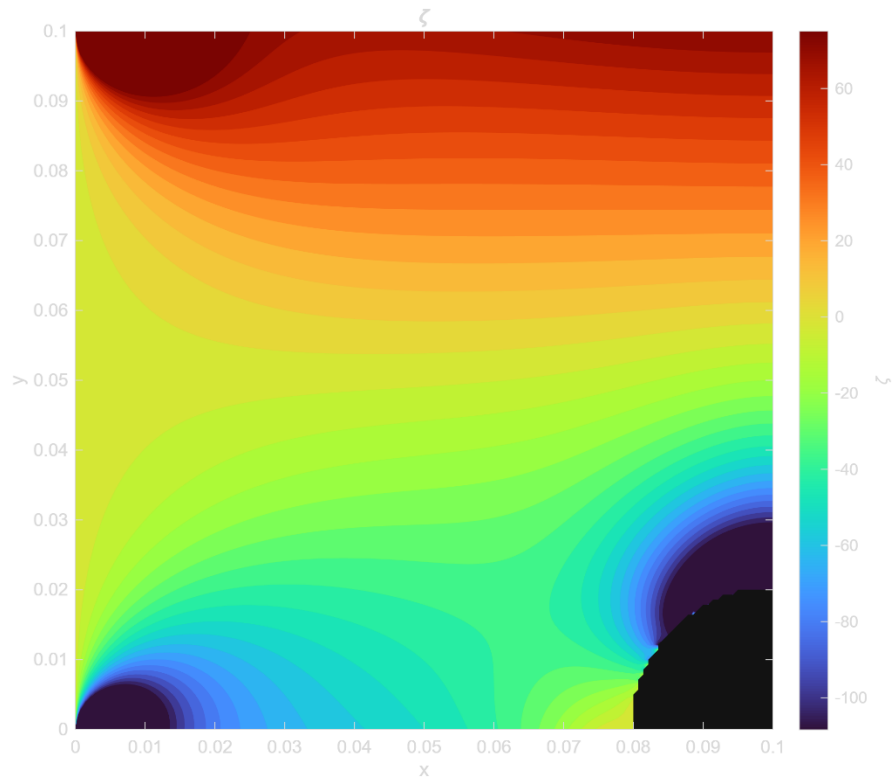


Figure 56. Vorticity contour on the fine mesh.

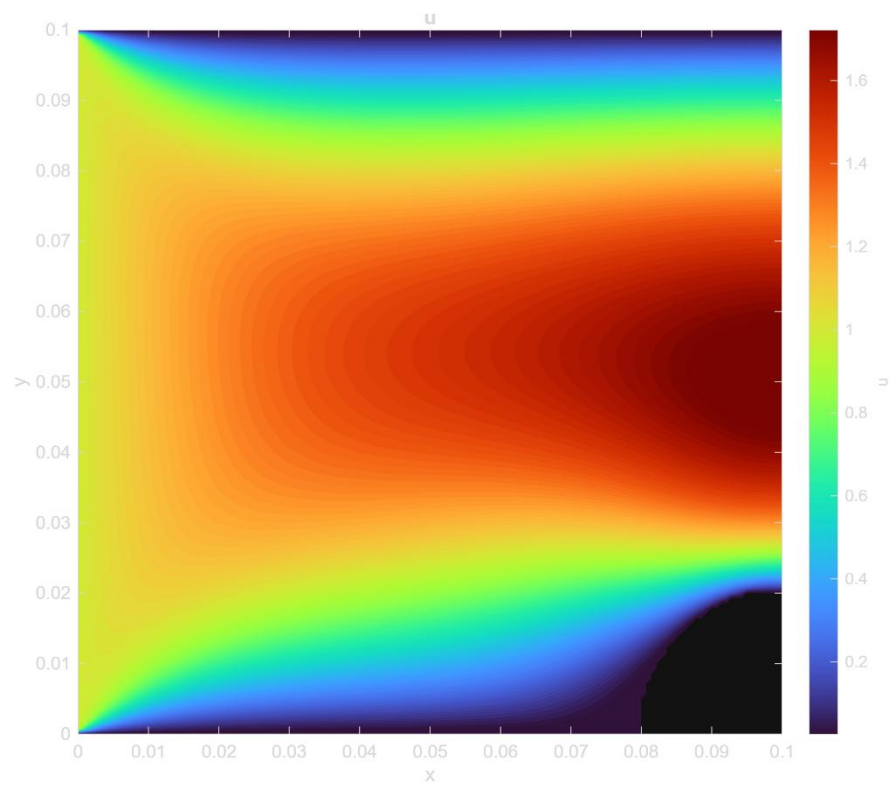


Figure 57. Horizontal velocity contour on the fine mesh.

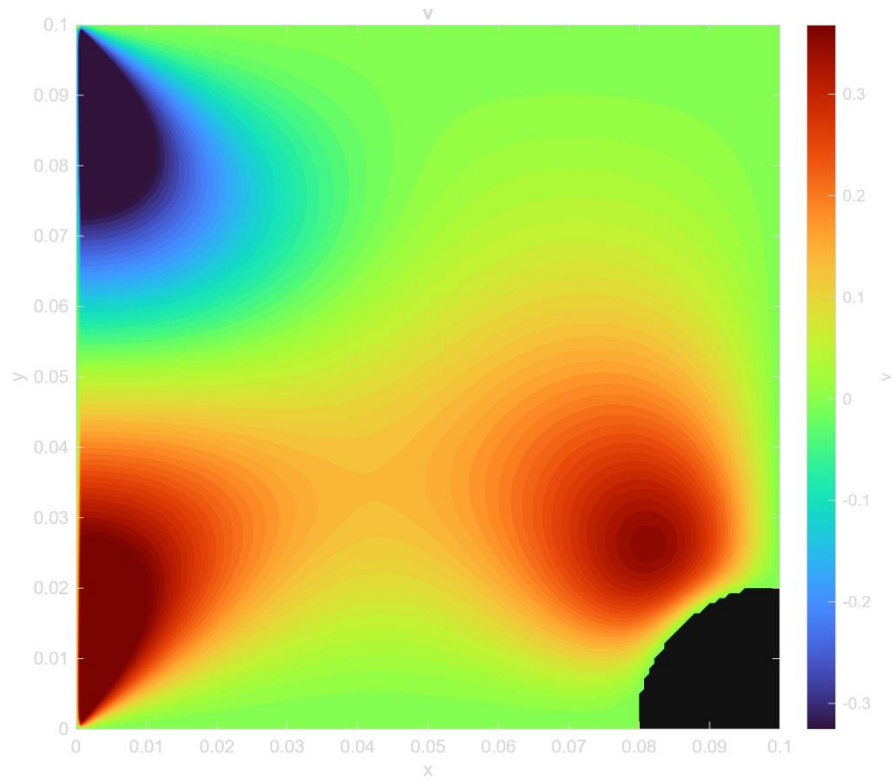


Figure 58. Vertical velocity contour on the fine mesh.

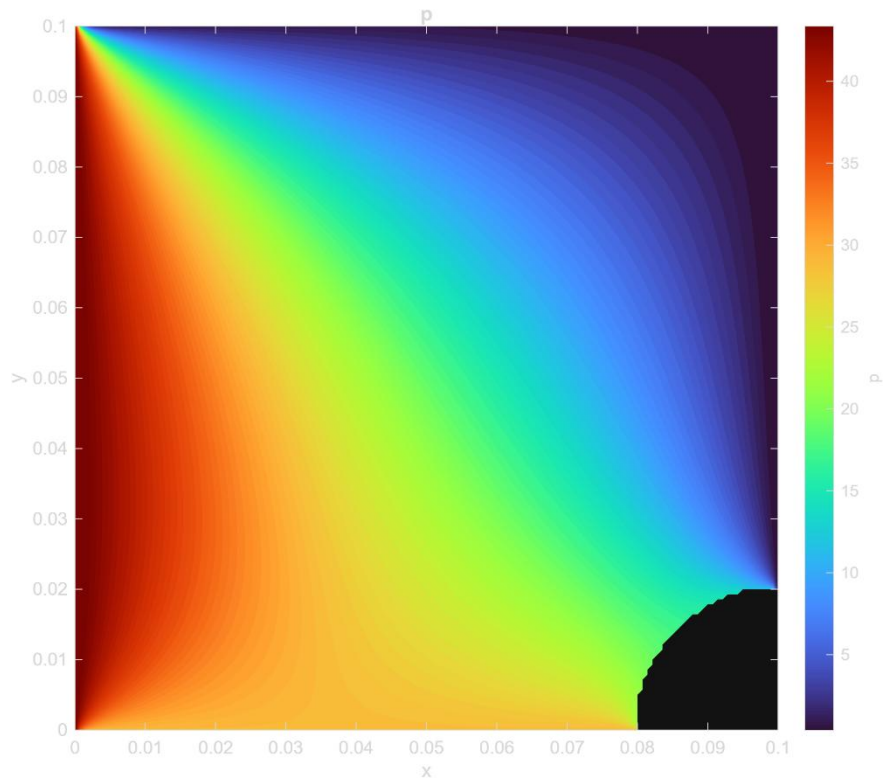


Figure 59. Pressure contour on the fine mesh.

6.6 line plots (fine mesh)

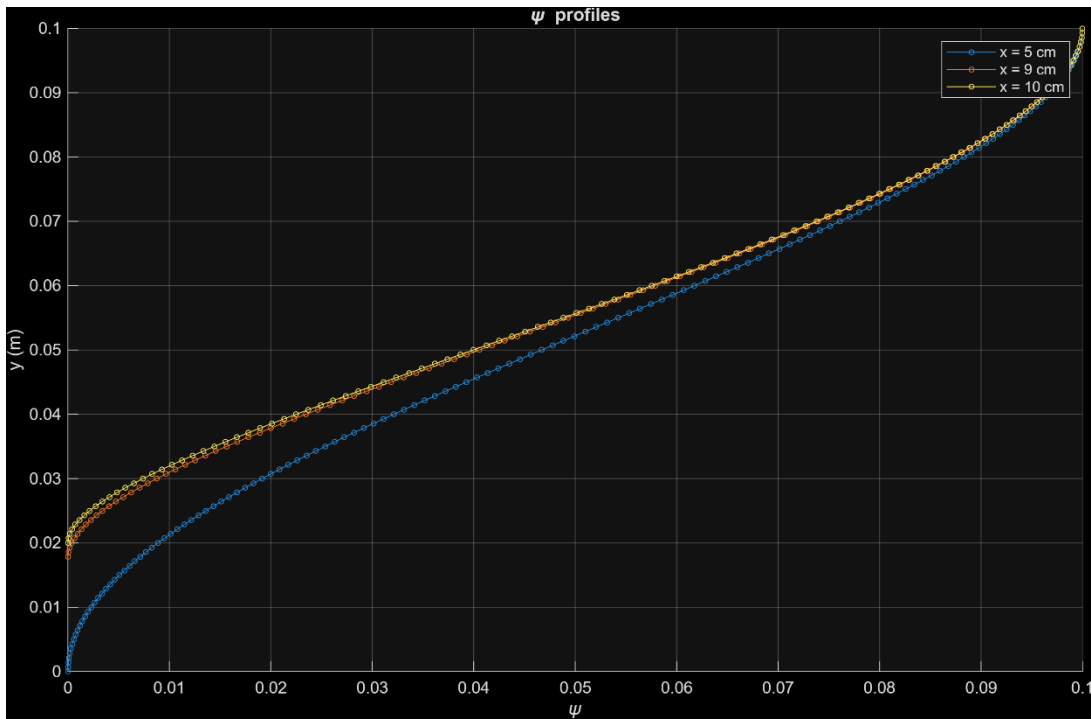


Figure 60. Stream function profiles at $x = 5, 9$ and 10 cm on the fine mesh.

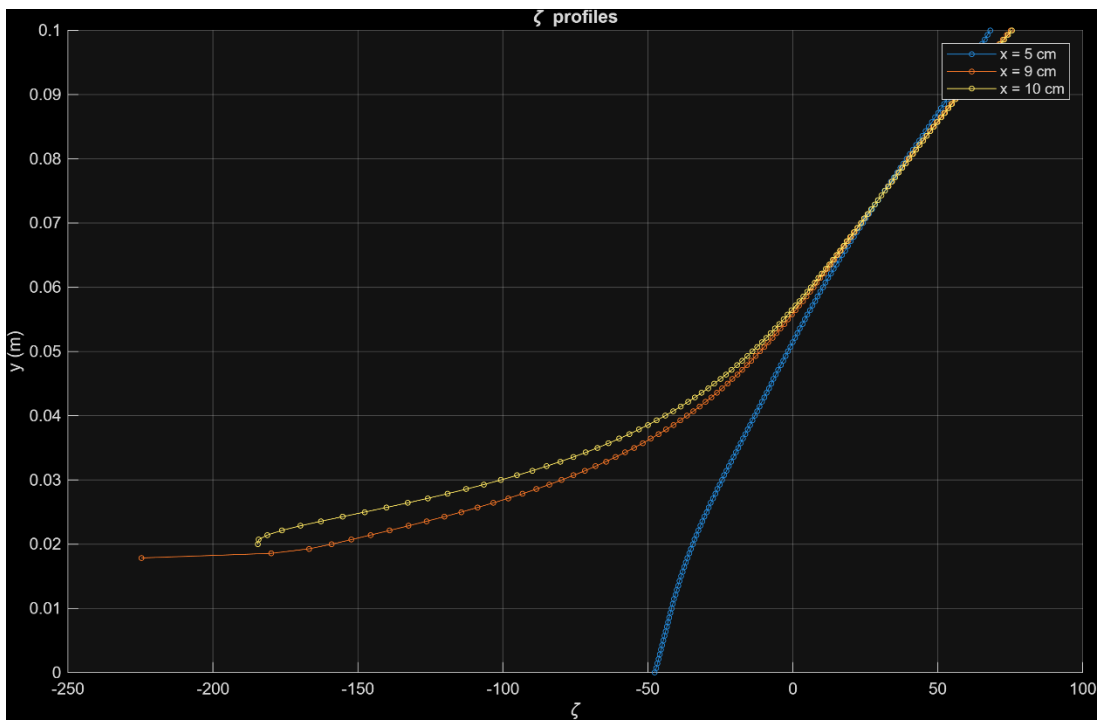


Figure 61. Vorticity profiles at $x = 5, 9$ and 10 cm on the fine mesh.

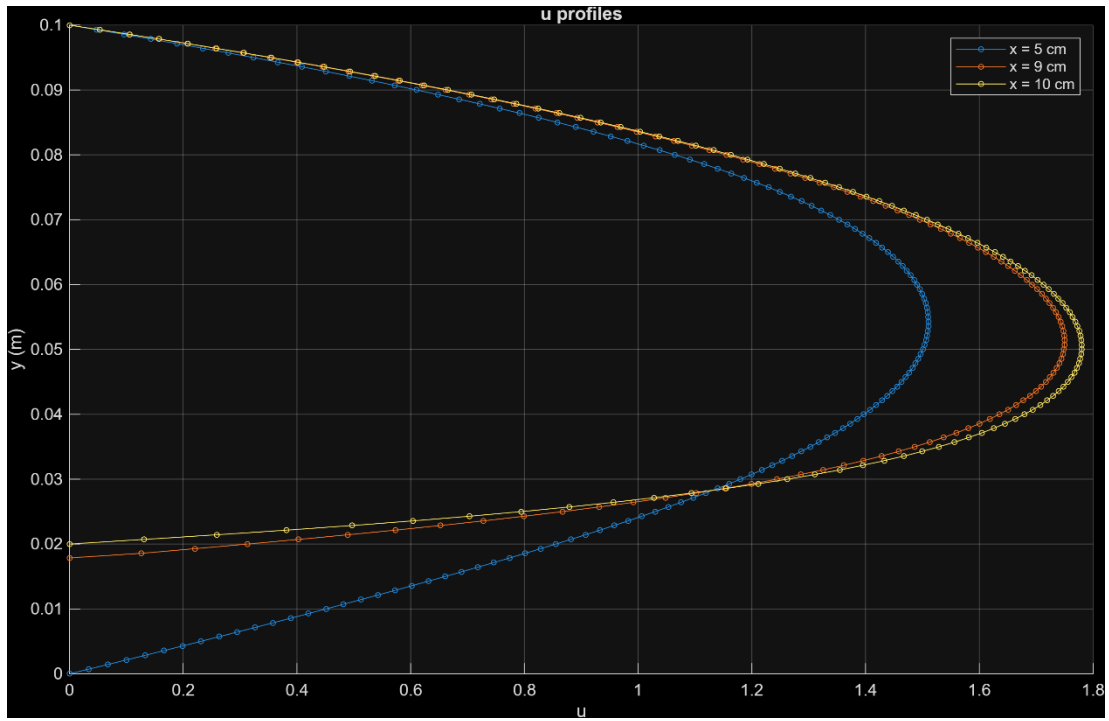


Figure 62. Horizontal velocity profiles at $x = 5, 9$ and 10 cm on the fine mesh.

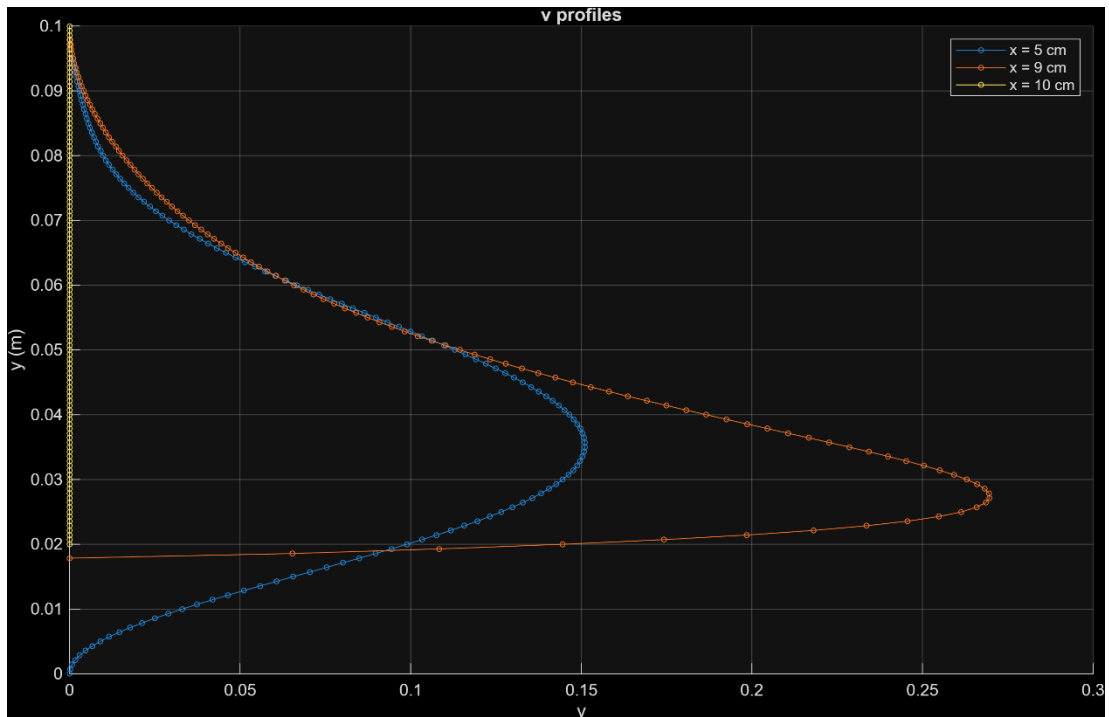


Figure 63. Vertical velocity profiles at $x = 5, 9$ and 10 cm on the fine mesh.

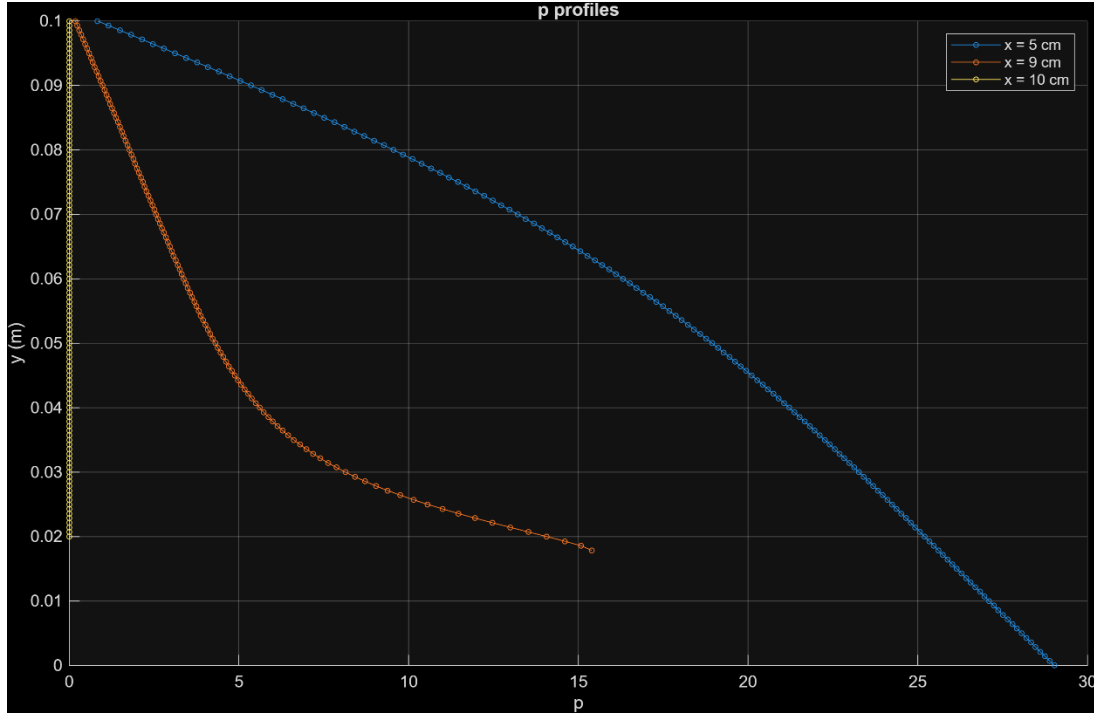


Figure 64. Pressure profiles at $x = 5, 9$ and 10 cm the fine mesh.

7 Successive Line Over-Relaxation (SLOR) Method

The SLOR scheme replaces the point-by-point update of SOR with a line update. The domain is swept row by row in the y direction, and along each row all node values are solved simultaneously: the terms coupling the node to its left and right neighbors are kept implicit on the left-hand side, while the terms coupling it to the rows below and above are treated explicitly and moved to the right-hand side. Each row therefore yields a tridiagonal system.

$$A_i \phi_{i-1,j} + B_i \phi_{i,j} + C_i \phi_{i+1,j} = \text{RHS}$$

which is solved directly with the Thomas algorithm. The first and last nodes of every row are boundary nodes and are enforced by setting the corresponding diagonal entry to unity and the right-hand side to the known boundary value. Over-relaxation is then applied to the line solution in the same form as in SOR:

$$\phi_{i,j}^{k+1} = (1 - \omega) \phi_{i,j}^k + \omega \phi_{i,j}^{\text{line}}$$

The length of each tridiagonal system is not constant. For every row the number of fluid nodes is counted first, stopping at the first node adjacent to the cylinder, and the coefficient matrix is built at that size.

Solving a whole line at once propagates information across the full width of the domain in a single sweep instead of one node at a time. However the cost per iteration is higher.

The relaxation factors were obtained in the same way as for SOR, with the stream function factor set 0.2 below the theoretical optimum and the vorticity factor a further 0.7 below that.

Table 5. Relaxation factors used by the SLOR scheme on the three meshes.

Mesh	$\omega_{\text{opt}}(\psi)$	$\omega_{\text{opt}}(\zeta)$
Coarse	1.7	1
Intermediate	1.74	1.04
Fine	1.76	1.06

Mesh	Iterations	Wall-clock time (s)
Coarse (61×61)	2,131	1.18
Intermediate (101×101)	6,682	11.78
Fine (141×141)	17,048	68.79

The scheme was run on the coarse, intermediate and fine meshes; the contours and line plots obtained on each are shown below.

7.1 contours (coarse mesh)

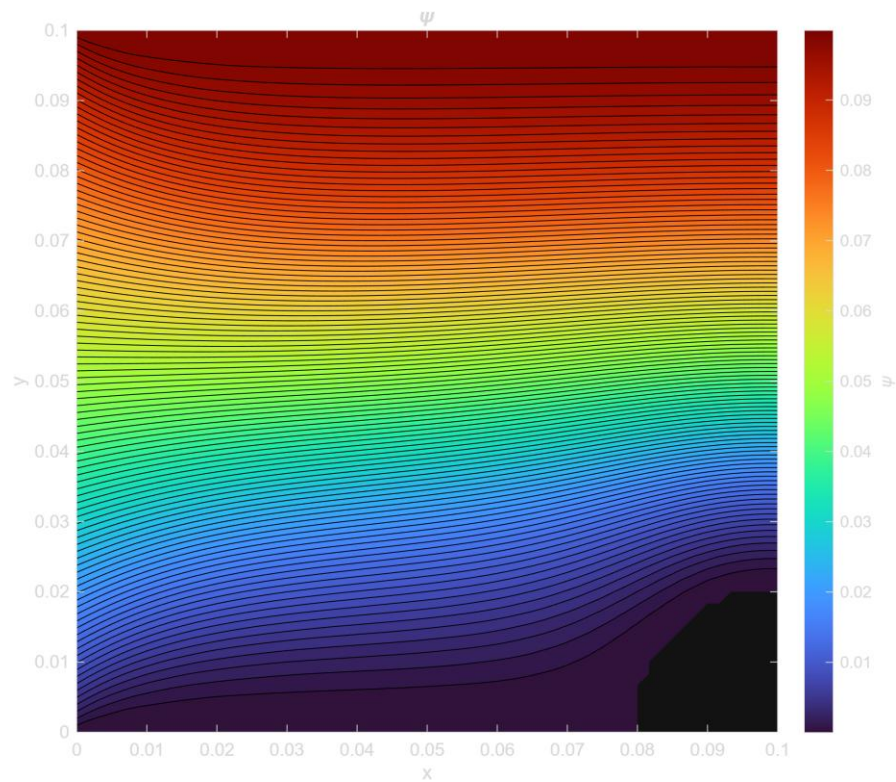


Figure 65. Stream function contour on the coarse mesh.

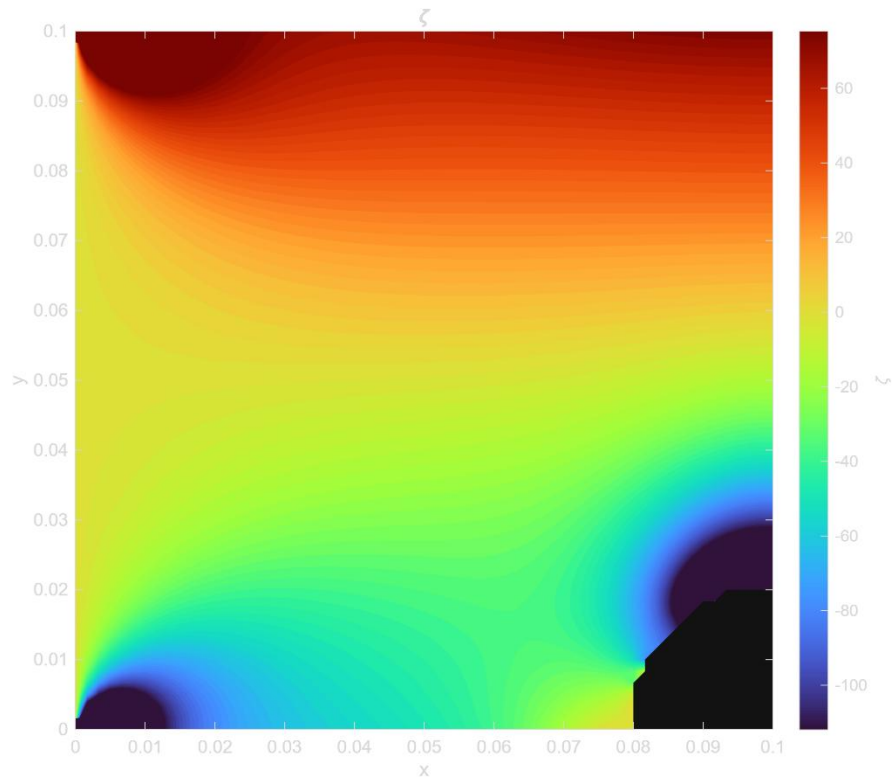


Figure 66. Vorticity contour on the coarse mesh.

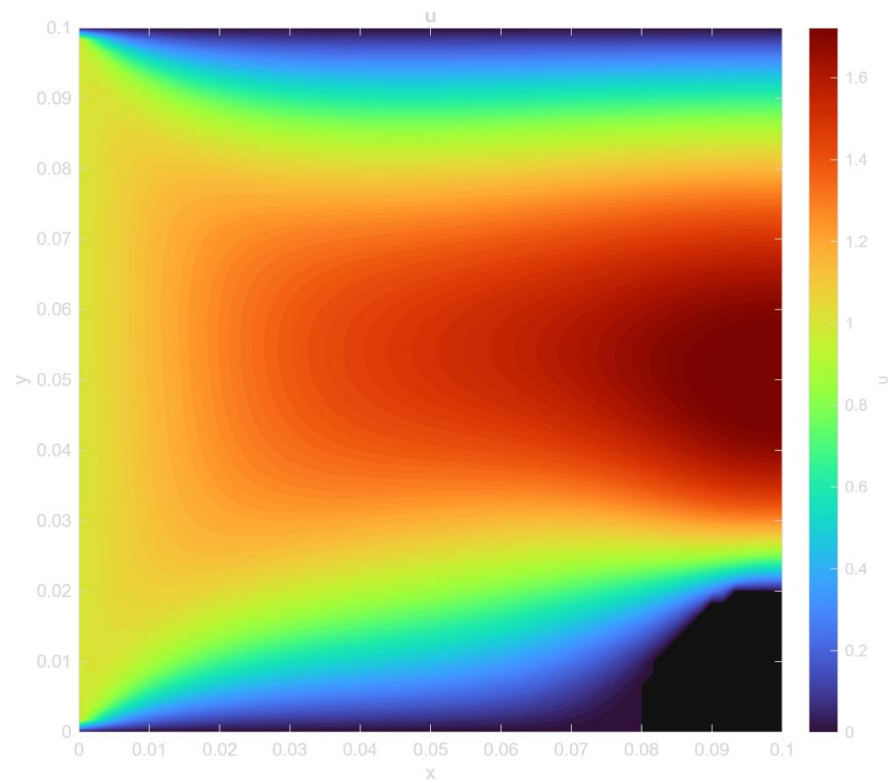


Figure 67. Horizontal velocity contour on the coarse mesh.

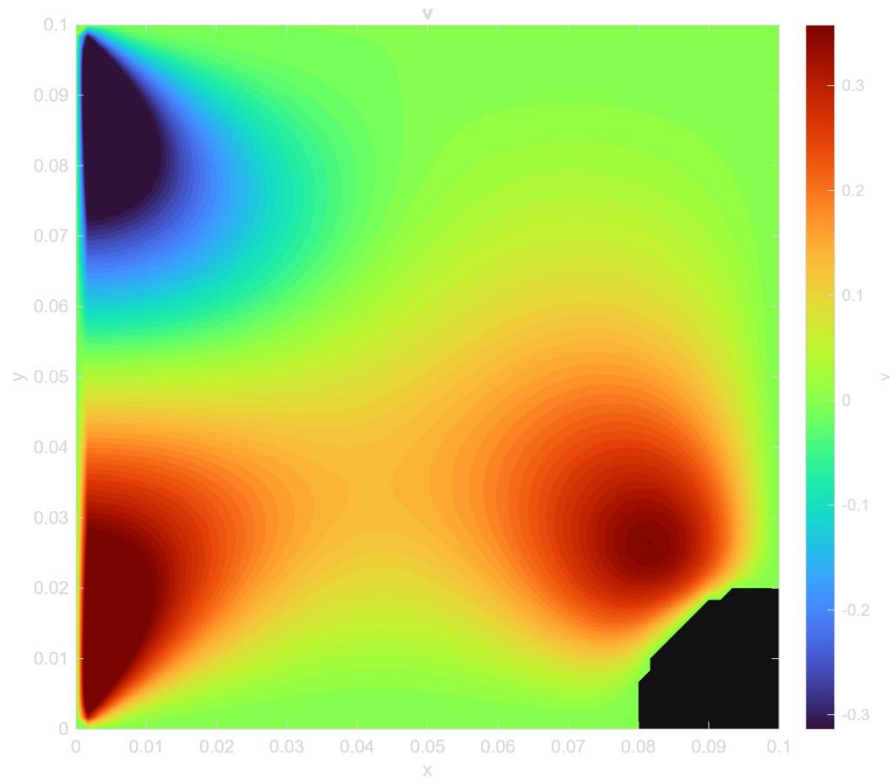


Figure 68. Vertical velocity contour on the coarse mesh.

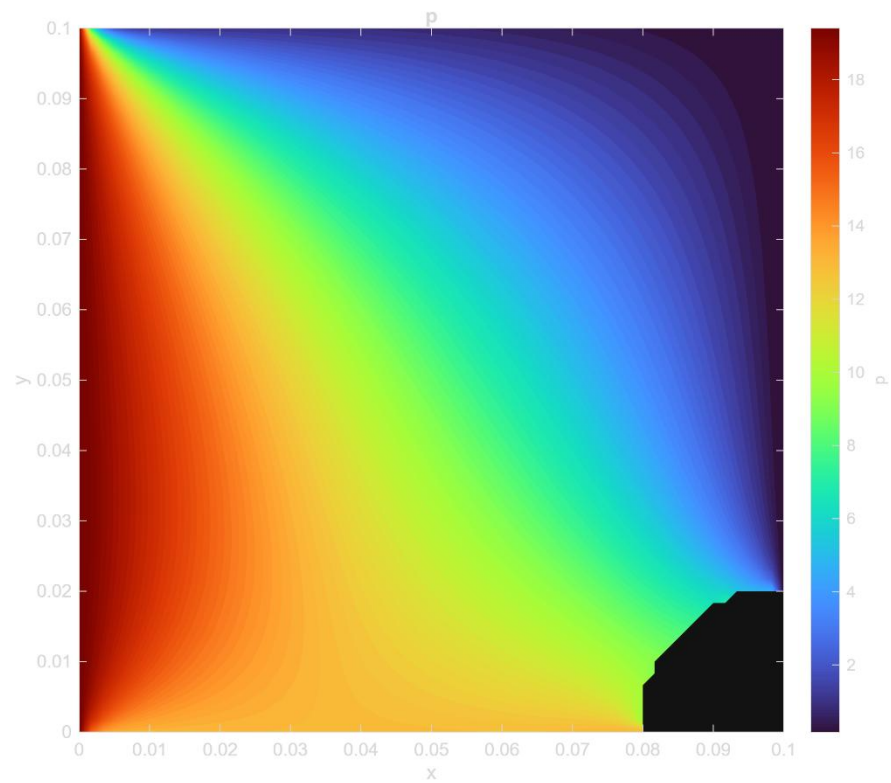


Figure 69. Pressure contour on the coarse mesh.

7.2 line plots (coarse mesh)

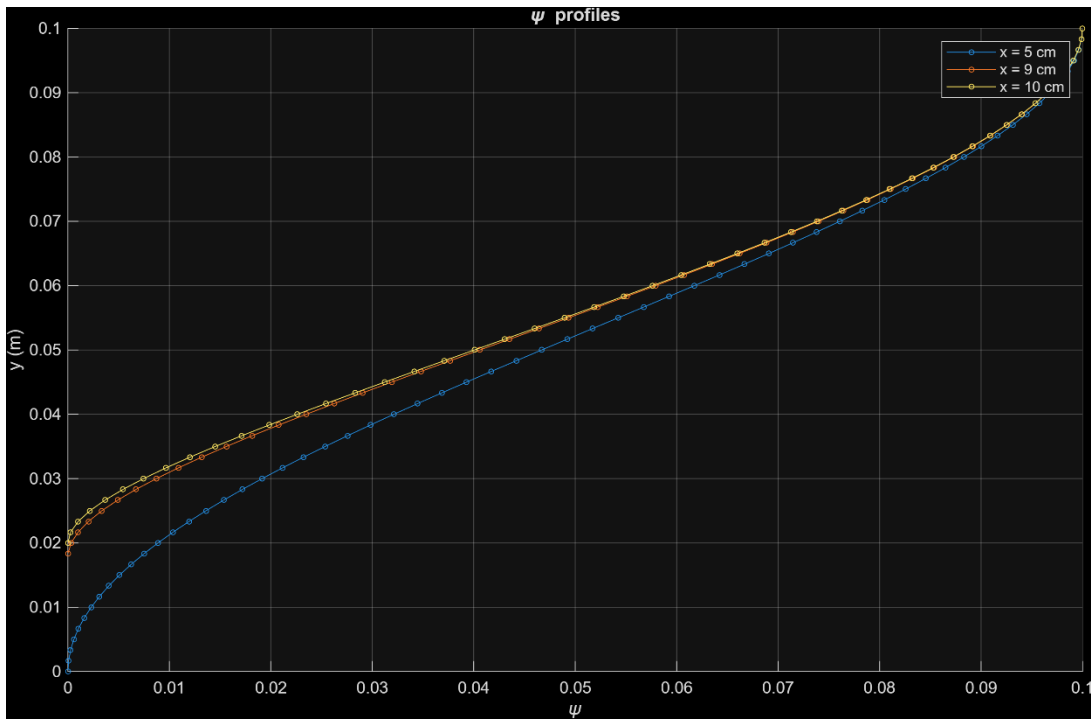


Figure 70. Stream function profiles at $x = 5, 9$ and 10 cm on the coarse mesh.

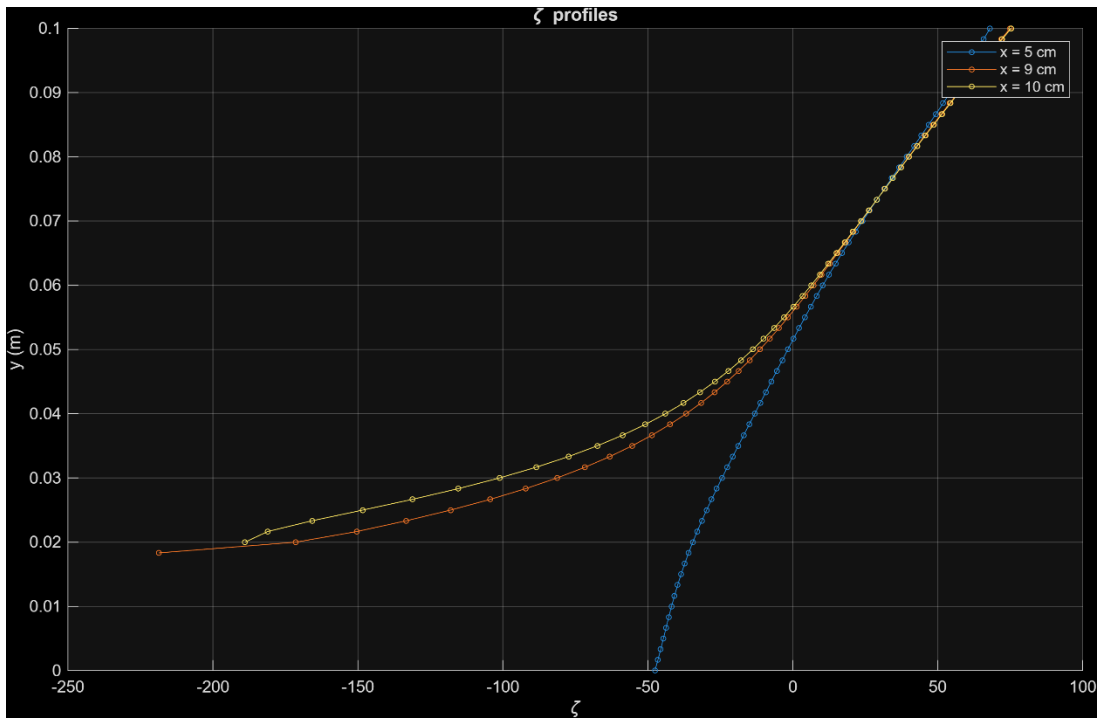


Figure 71. Vorticity profiles at $x = 5, 9$ and 10 cm on the coarse mesh

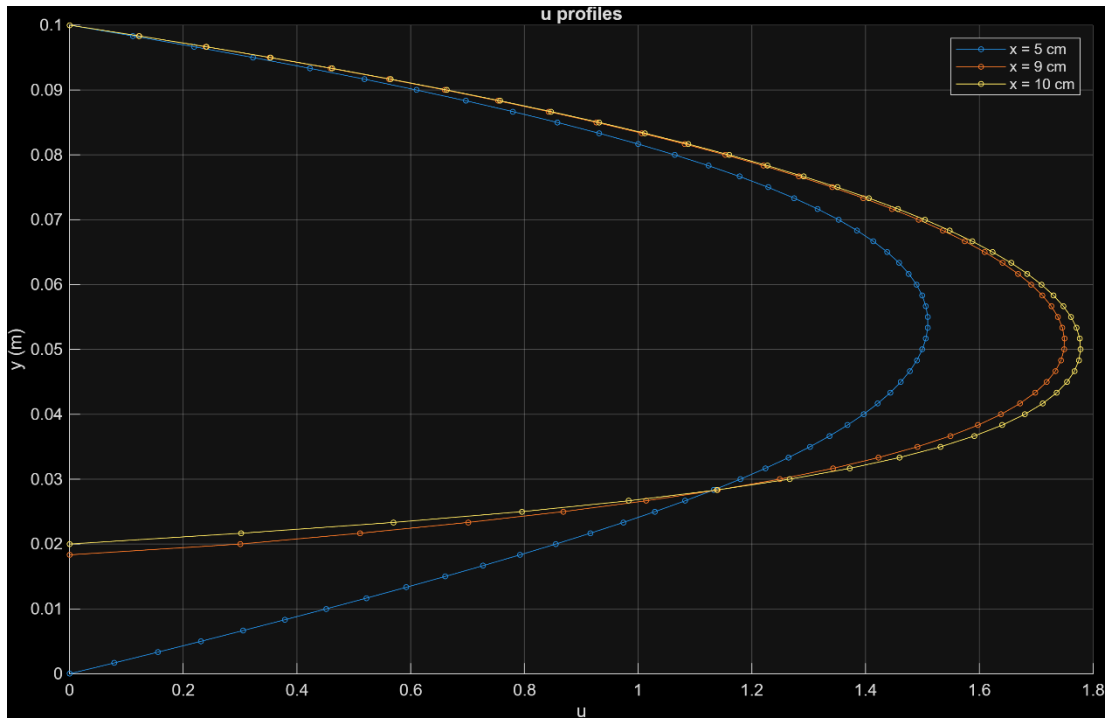


Figure 72. Horizontal velocity profiles at $x = 5, 9$ and 10 cm on the coarse mesh.

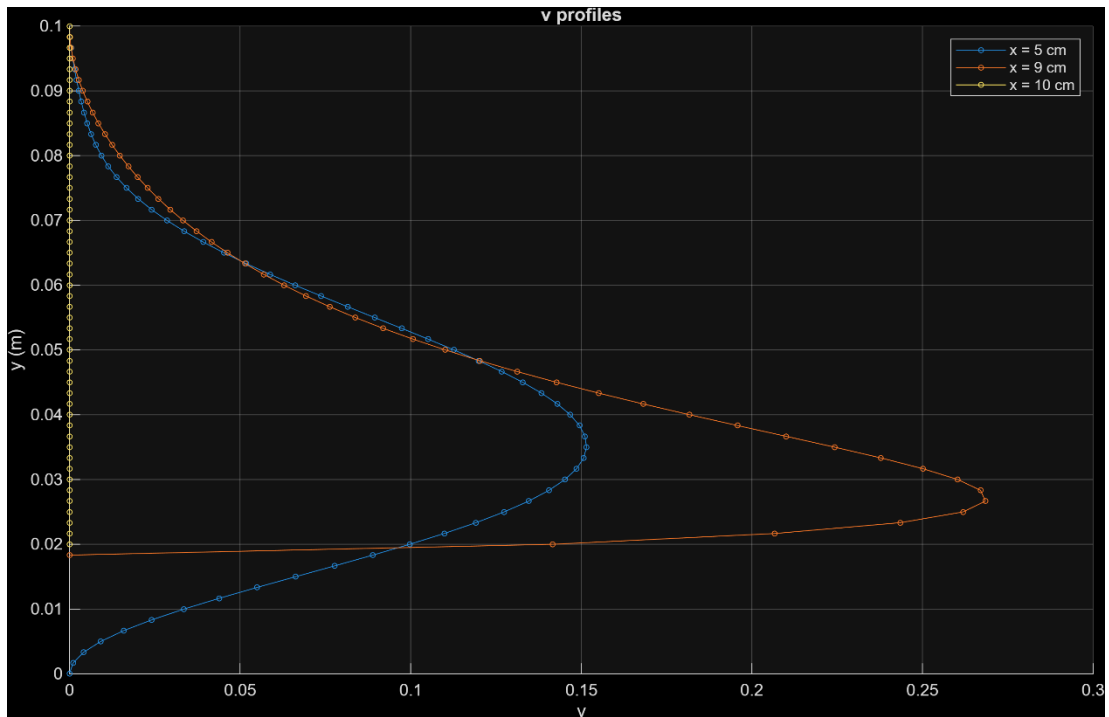


Figure 73. Vertical velocity profiles at $x = 5, 9$ and 10 cm on the coarse mesh.

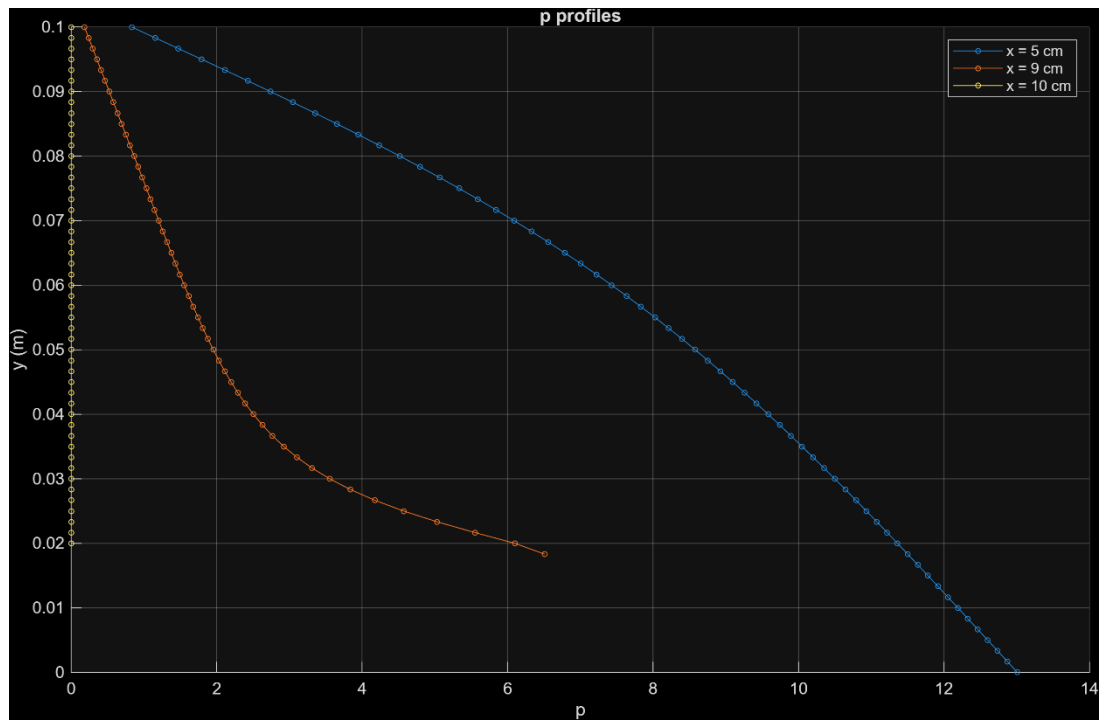


Figure 74. Pressure profiles at $x = 5, 9$ and 10 cm on the coarse mesh.

7.3 contours (intermediate mesh)

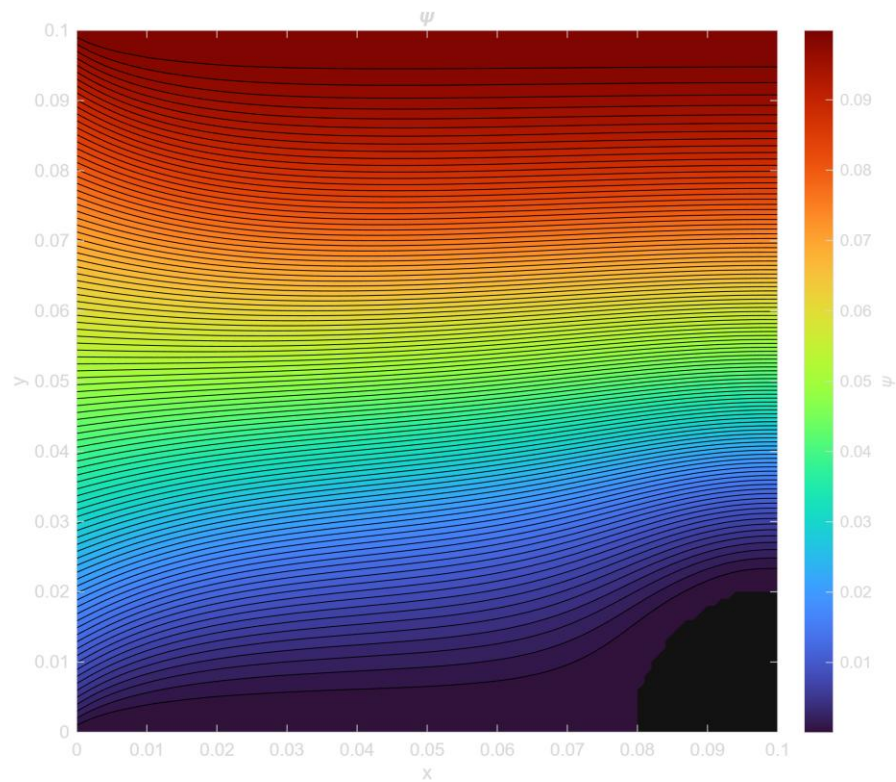


Figure 75. Stream function contour on the intermediate mesh.

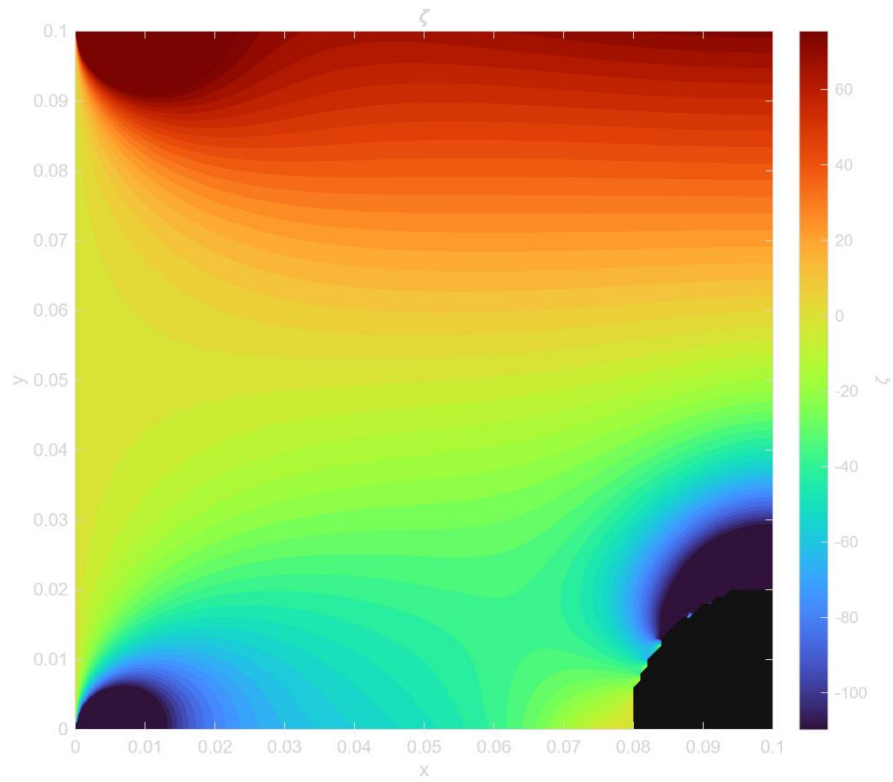


Figure 76. Vorticity contour on the intermediate mesh.

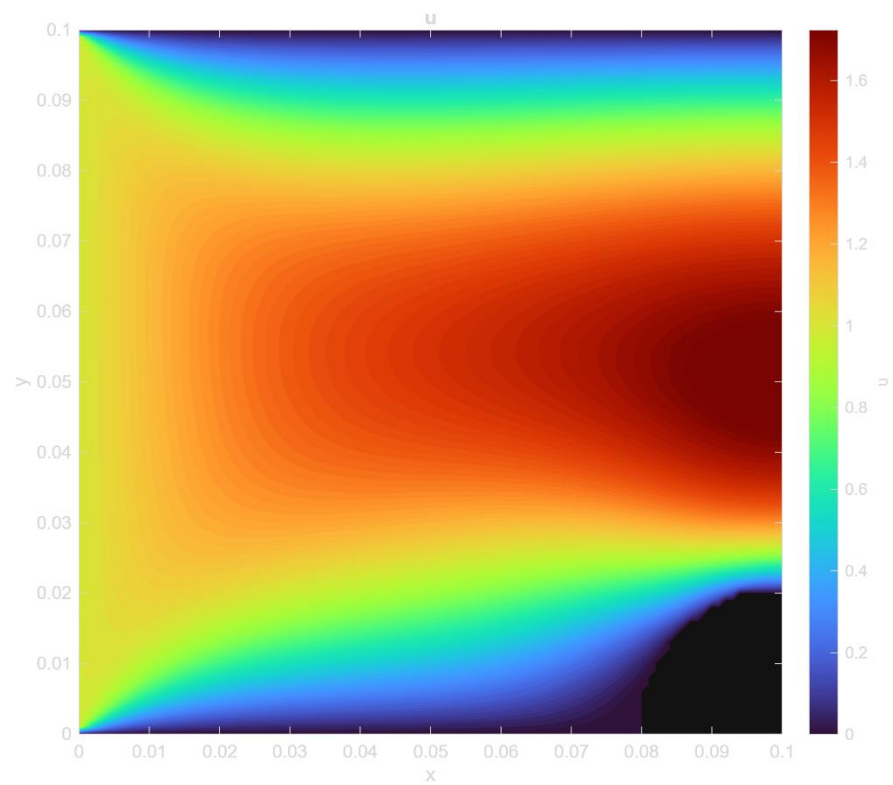


Figure 77. Horizontal velocity contour on the intermediate mesh.

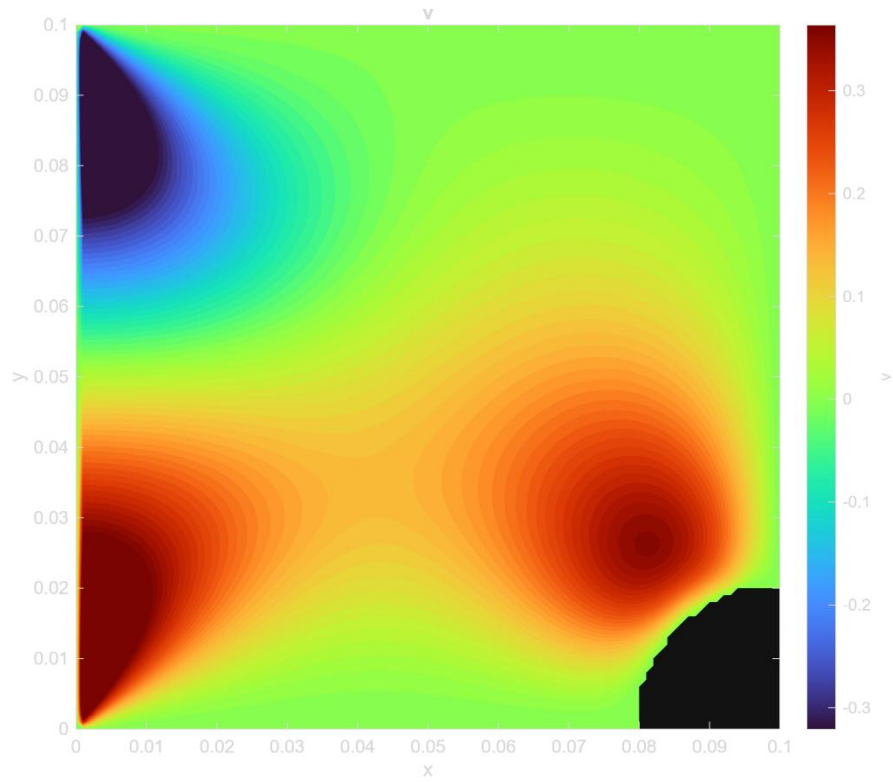


Figure 78. Vertical velocity contour on the intermediate mesh.

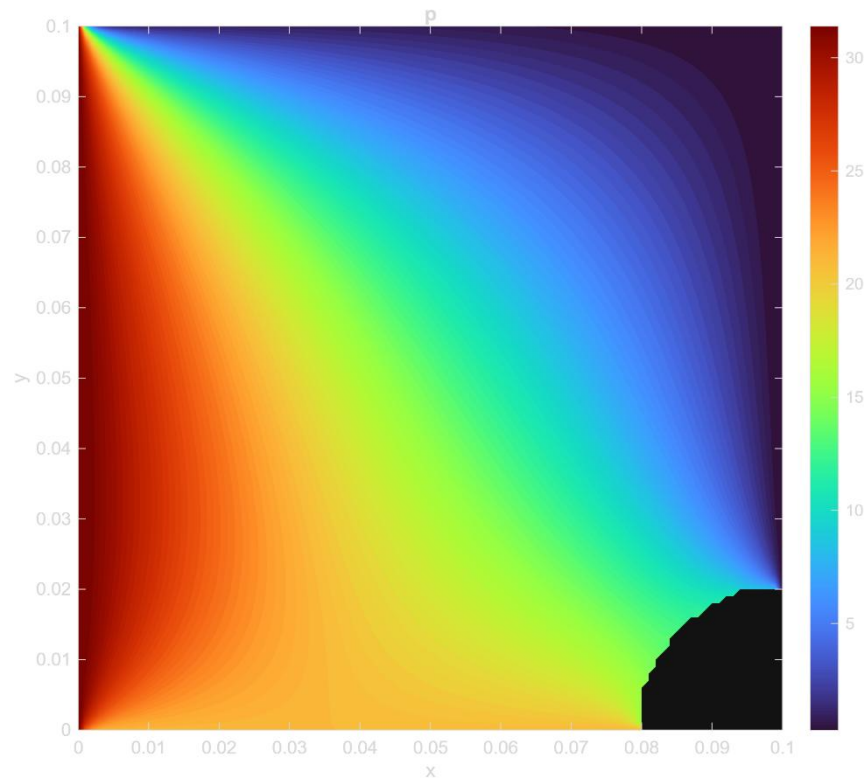


Figure 79. Pressure contour on the intermediate mesh.

7.4 line plots (intermediate mesh)

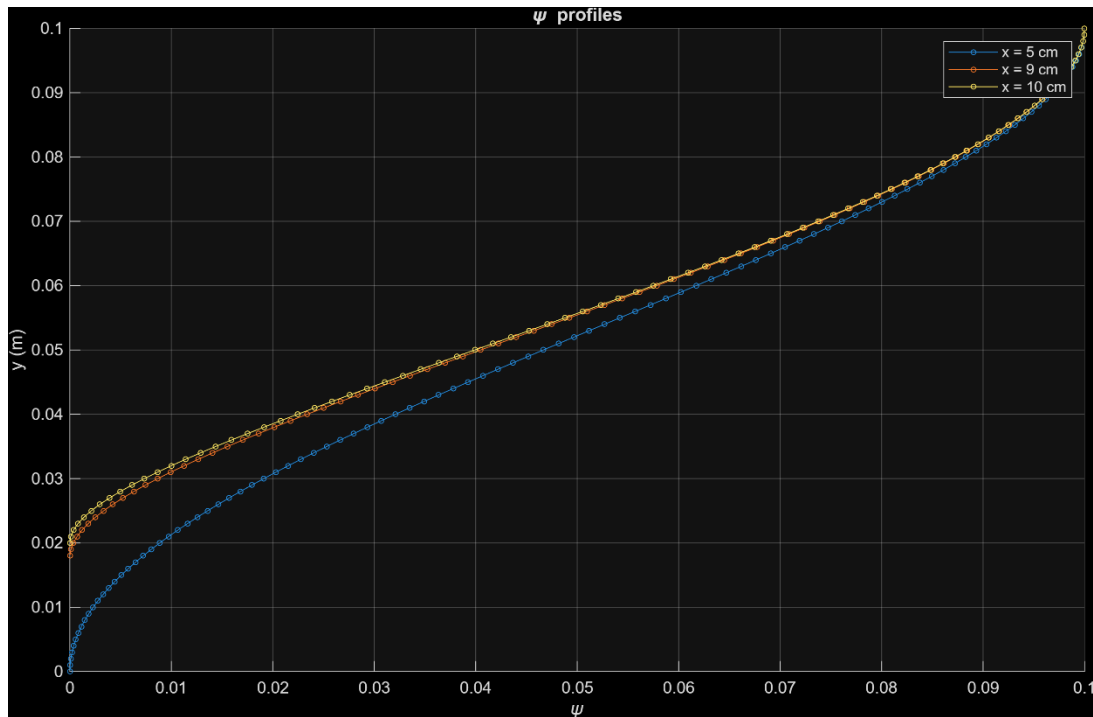


Figure 80. Stream function profiles at $x = 5, 9$ and 10 cm on the intermediate mesh.

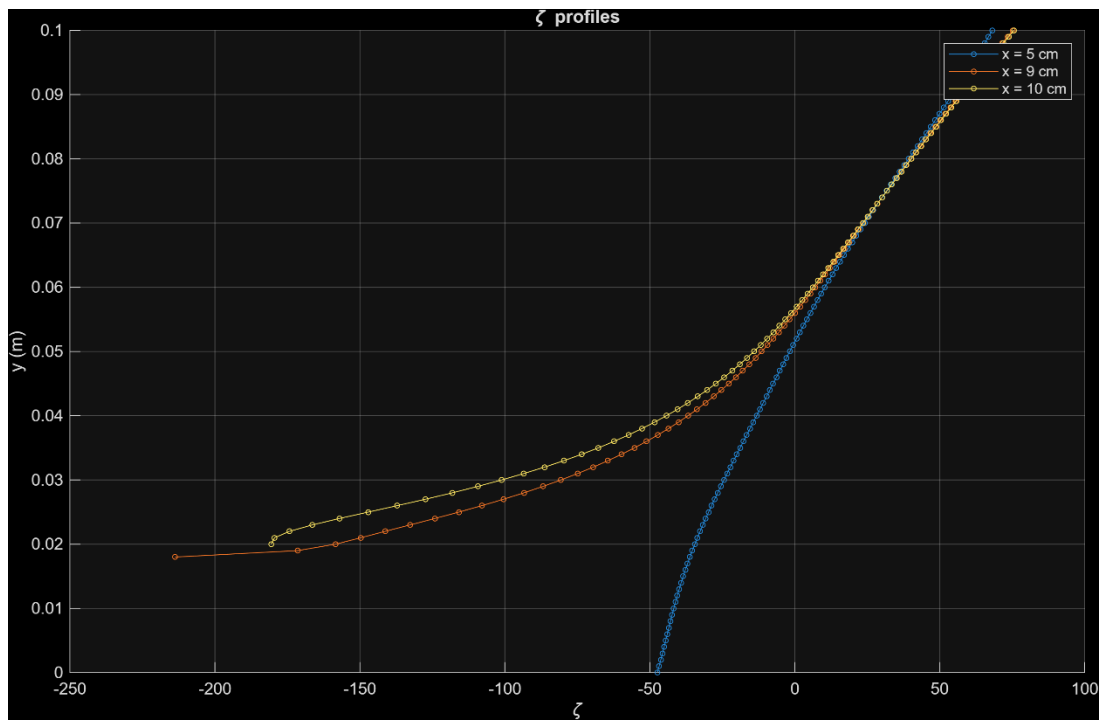


Figure 81. Vorticity profiles at $x = 5, 9$ and 10 cm on the intermediate mesh.

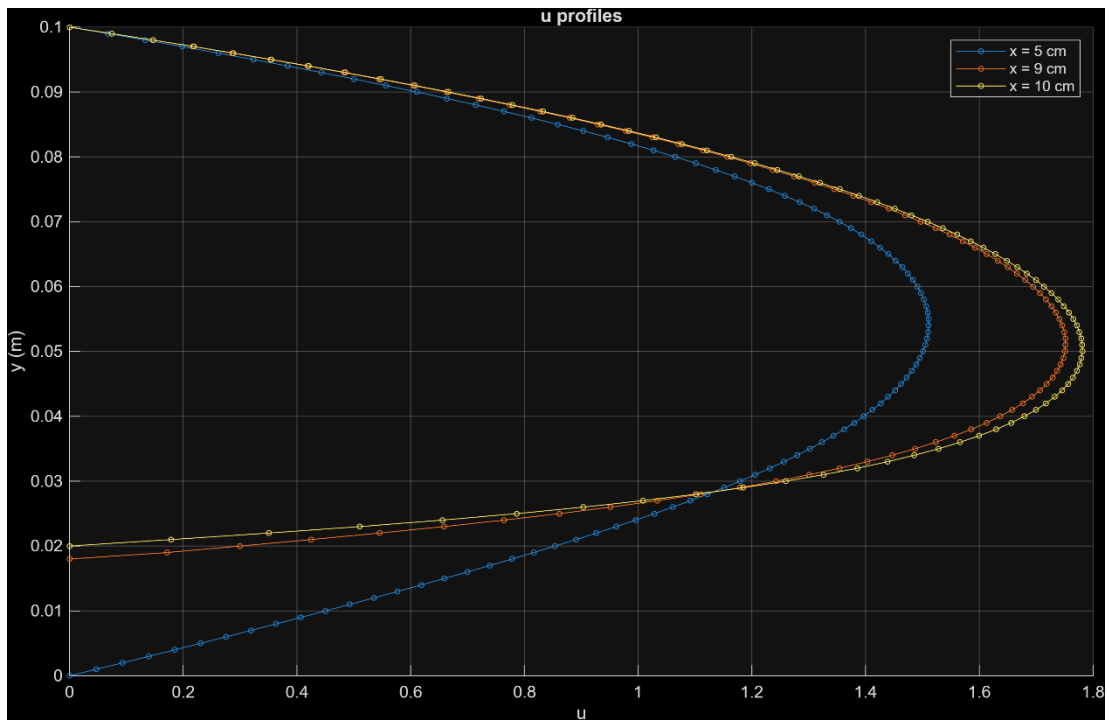


Figure 82. Horizontal velocity profiles at $x = 5, 9$ and 10 cm on the intermediate mesh.

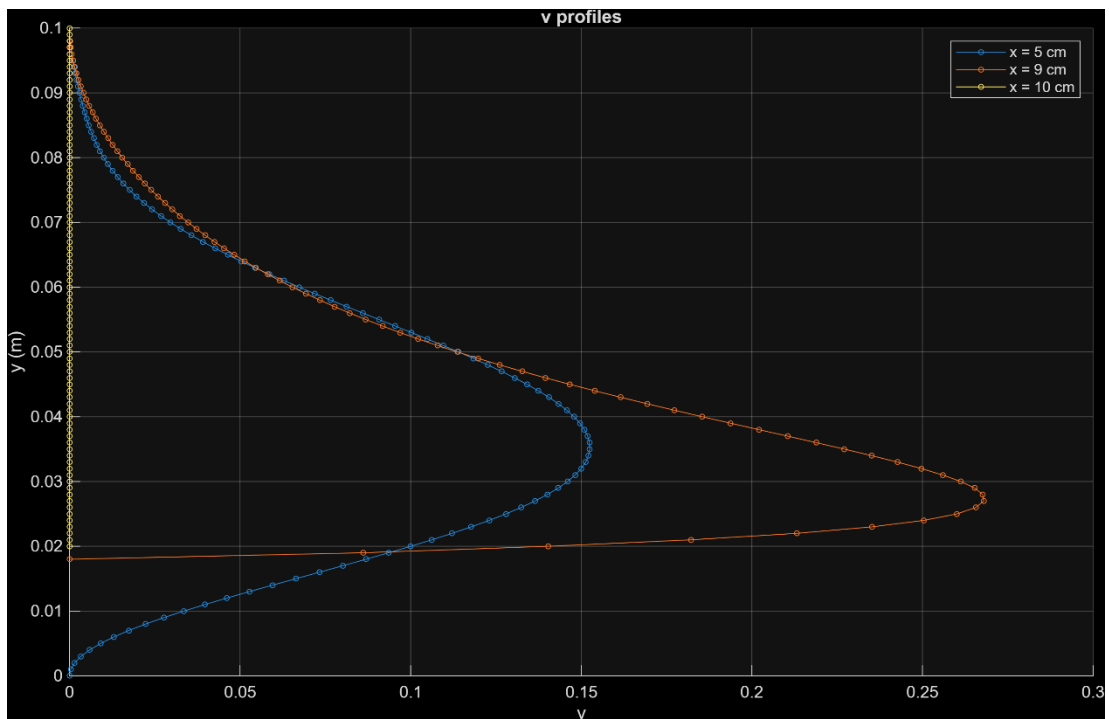


Figure 83. Vertical velocity profiles at $x = 5, 9$ and 10 cm on the intermediate mesh.

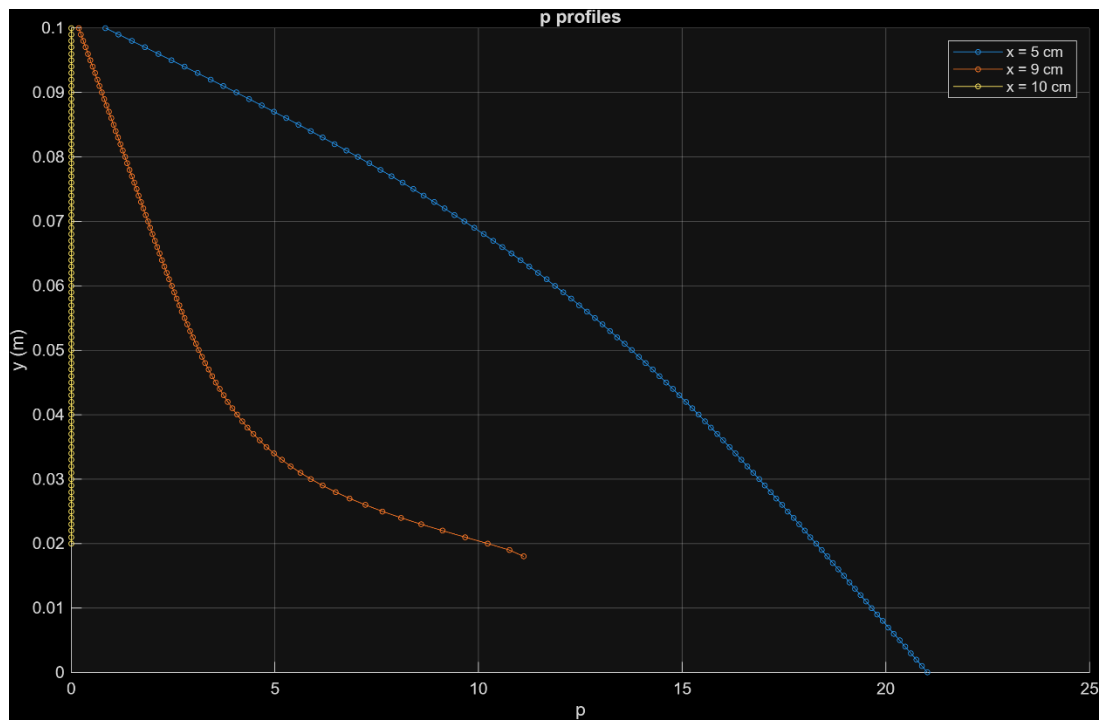


Figure 84. Pressure profiles at $x = 5, 9$ and 10 cm obtained with the Gauss–Seidel scheme on the intermediate mesh.

7.5 contours (fine mesh)

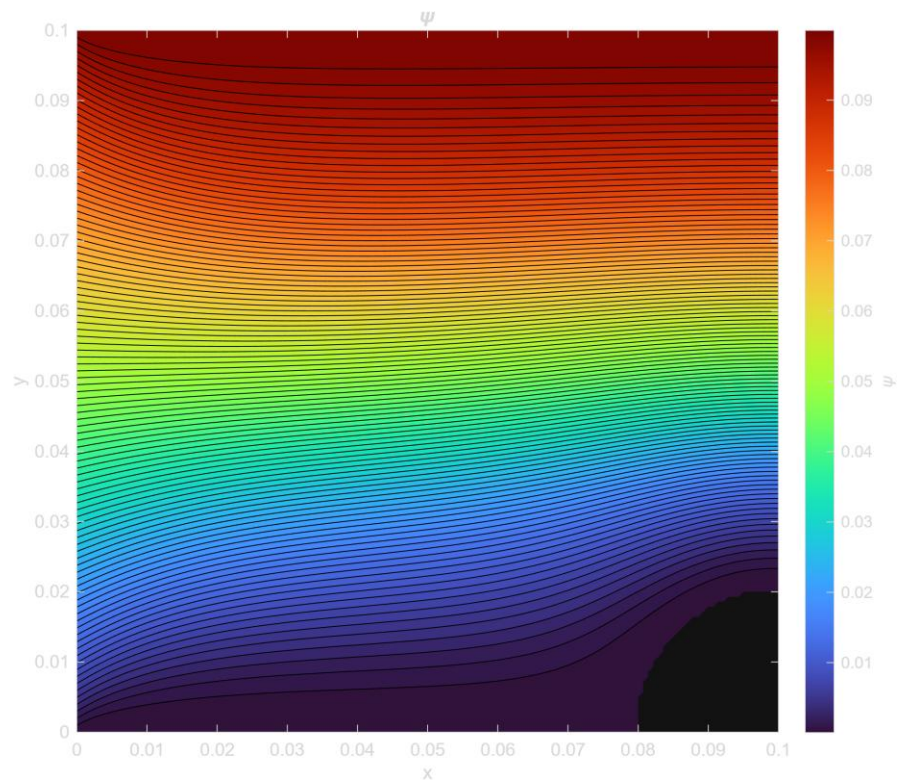


Figure 85. Stream function contour on the fine mesh.

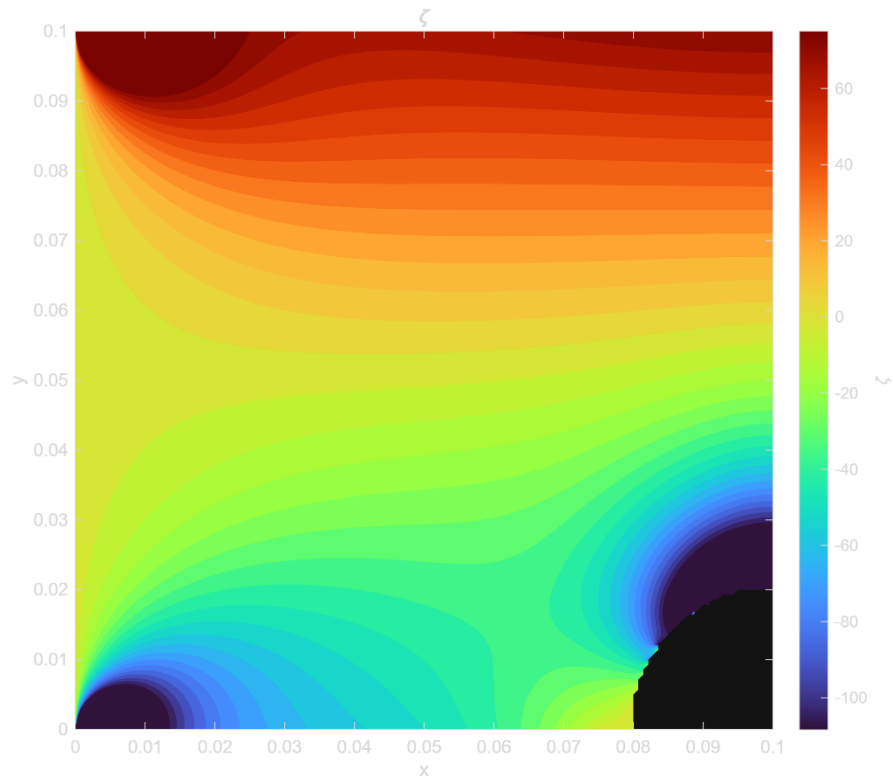


Figure 86. Vorticity contour on the fine mesh.

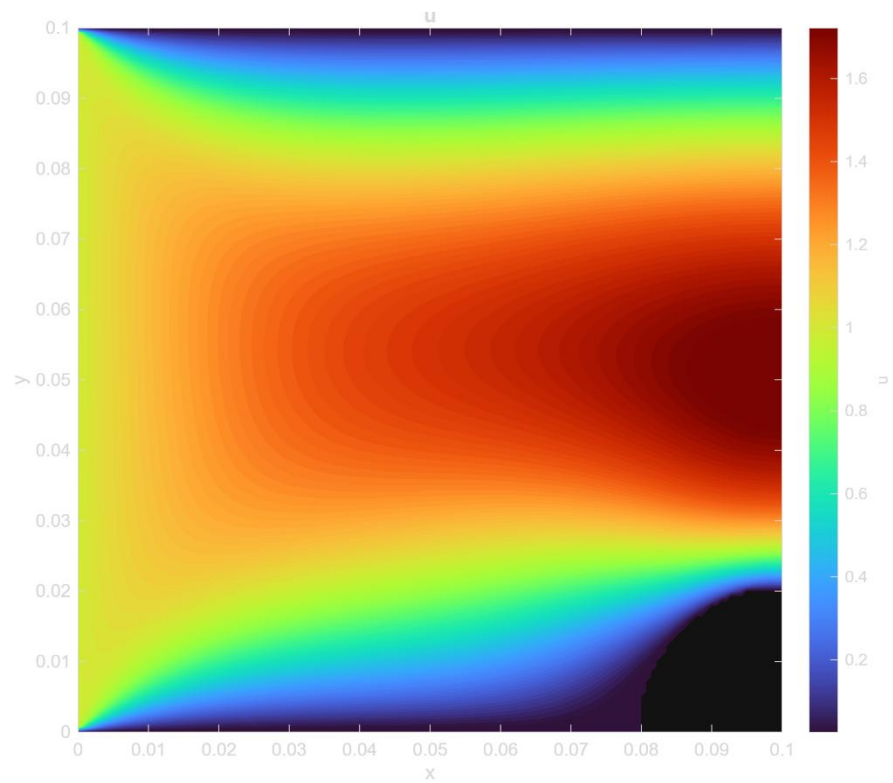


Figure 87. Horizontal velocity contour on the fine mesh.

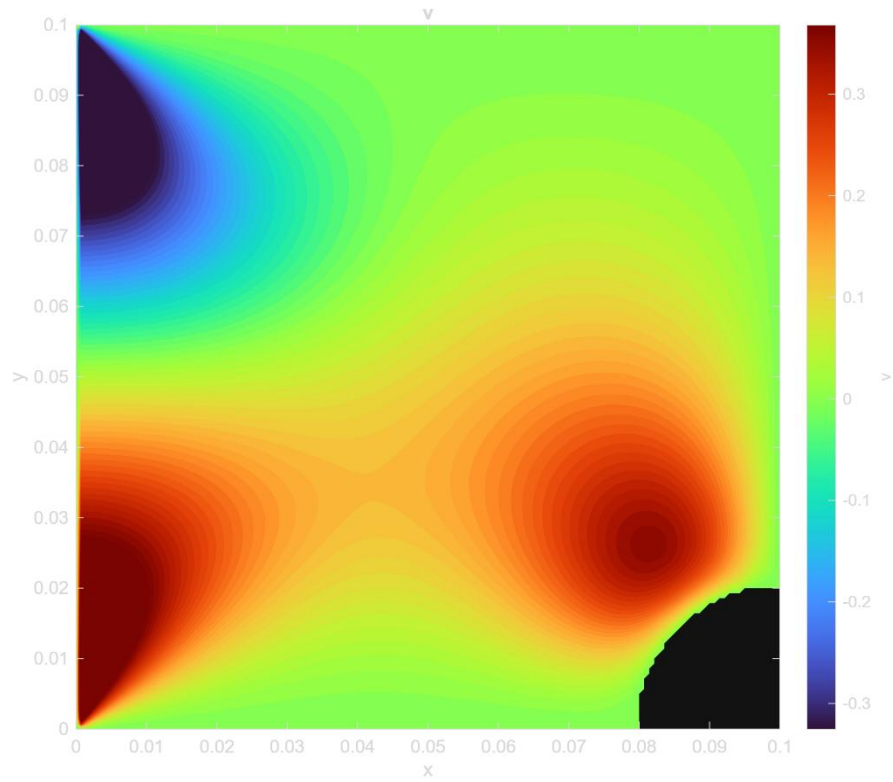


Figure 88. Vertical velocity contour on the fine mesh.

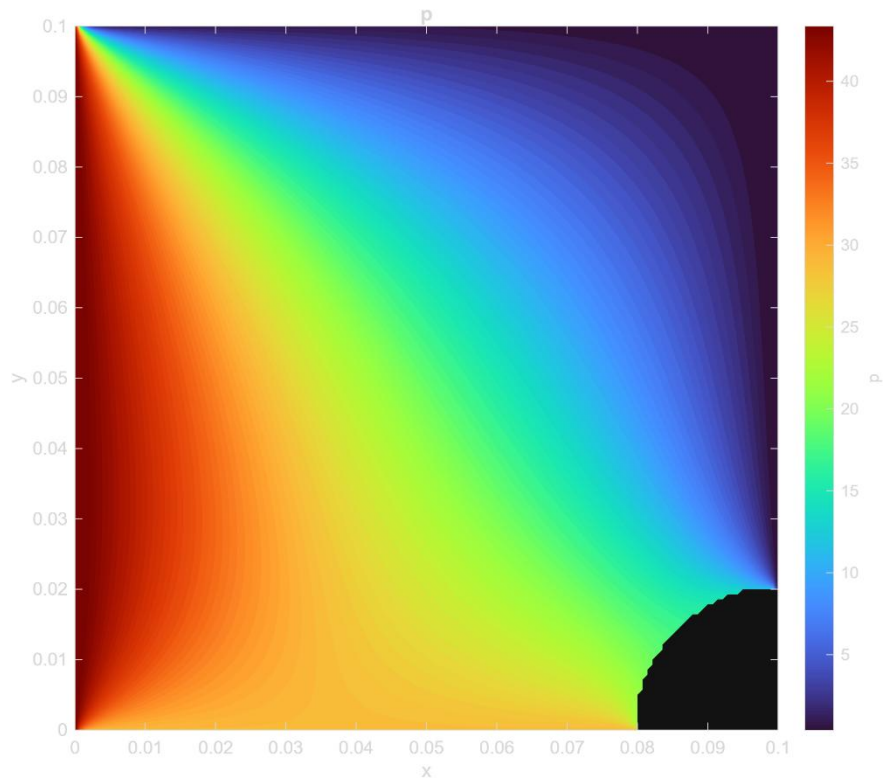


Figure 89. Pressure contour on the fine mesh.

7.6 line plots (fine mesh)

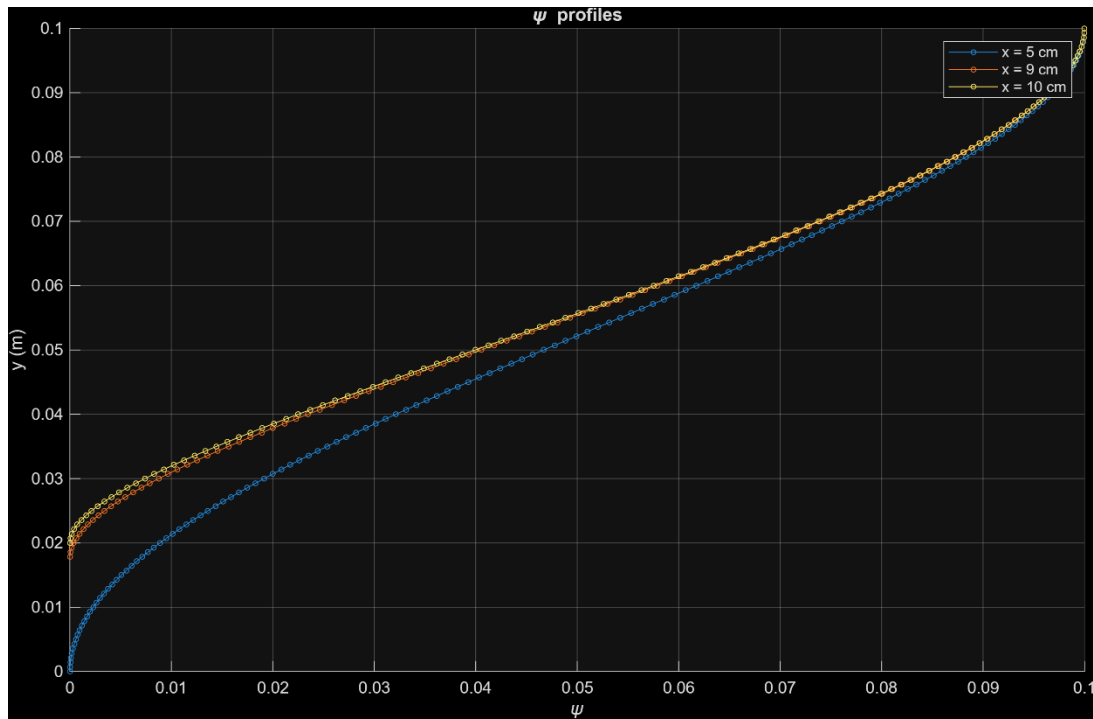


Figure 90. Stream function profiles at $x = 5, 9$ and 10 cm on the fine mesh.

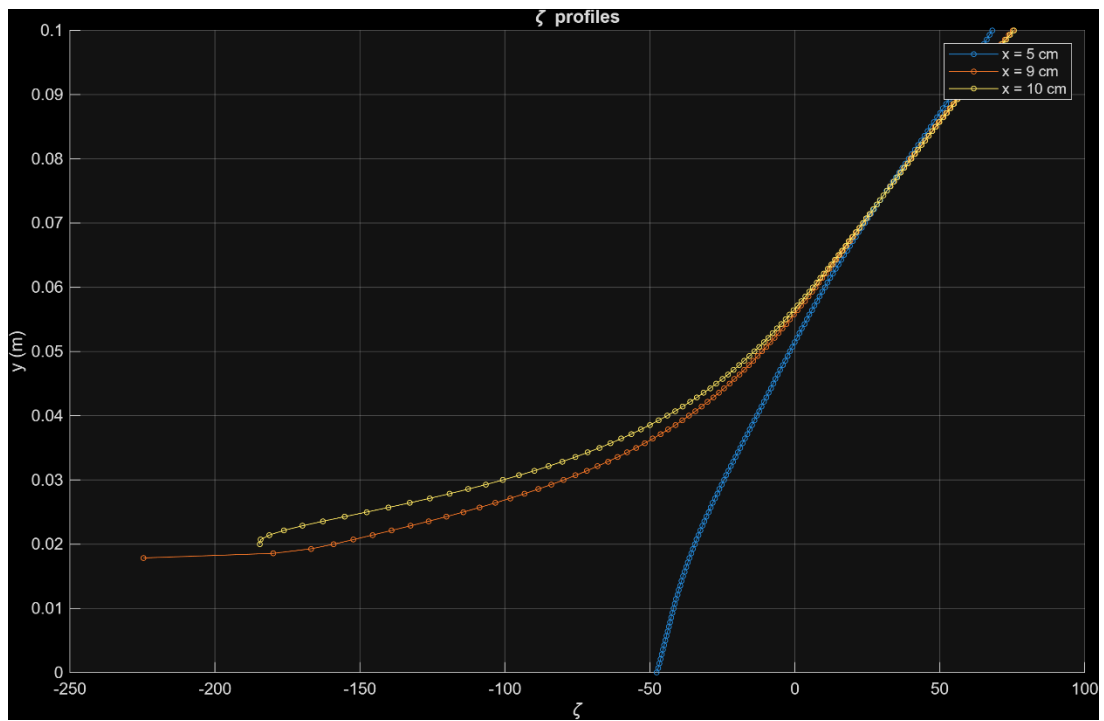


Figure 91. Vorticity profiles at $x = 5, 9$ and 10 cm on the fine mesh.

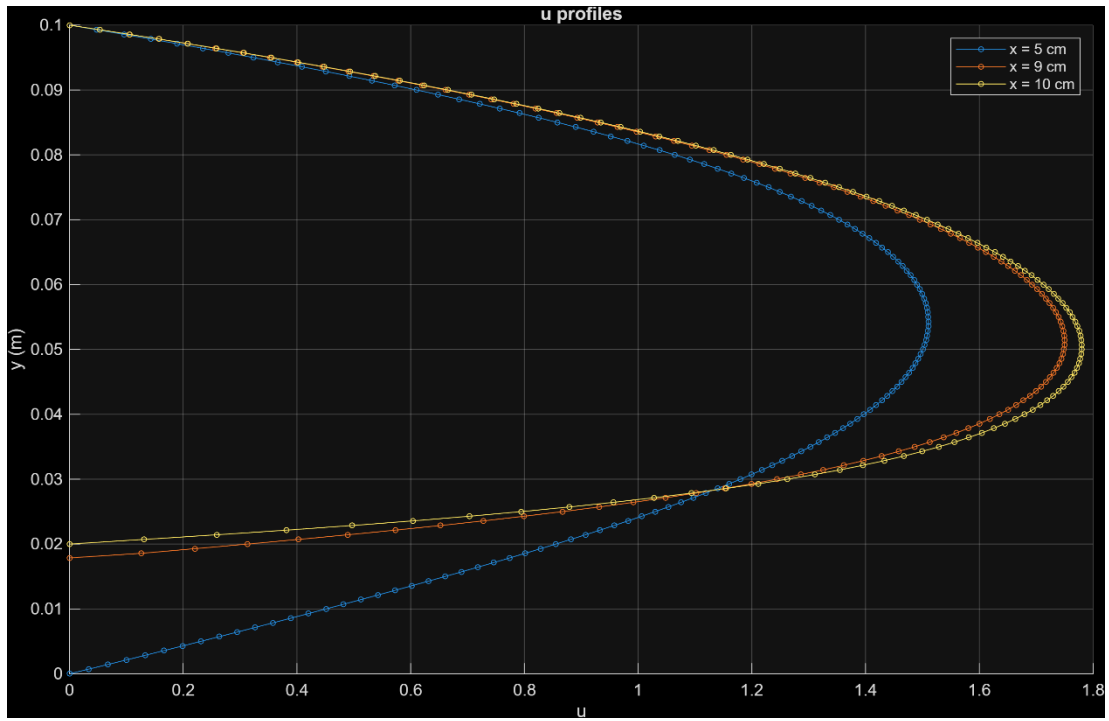


Figure 92. Horizontal velocity profiles at $x = 5, 9$ and 10 cm on the fine mesh.

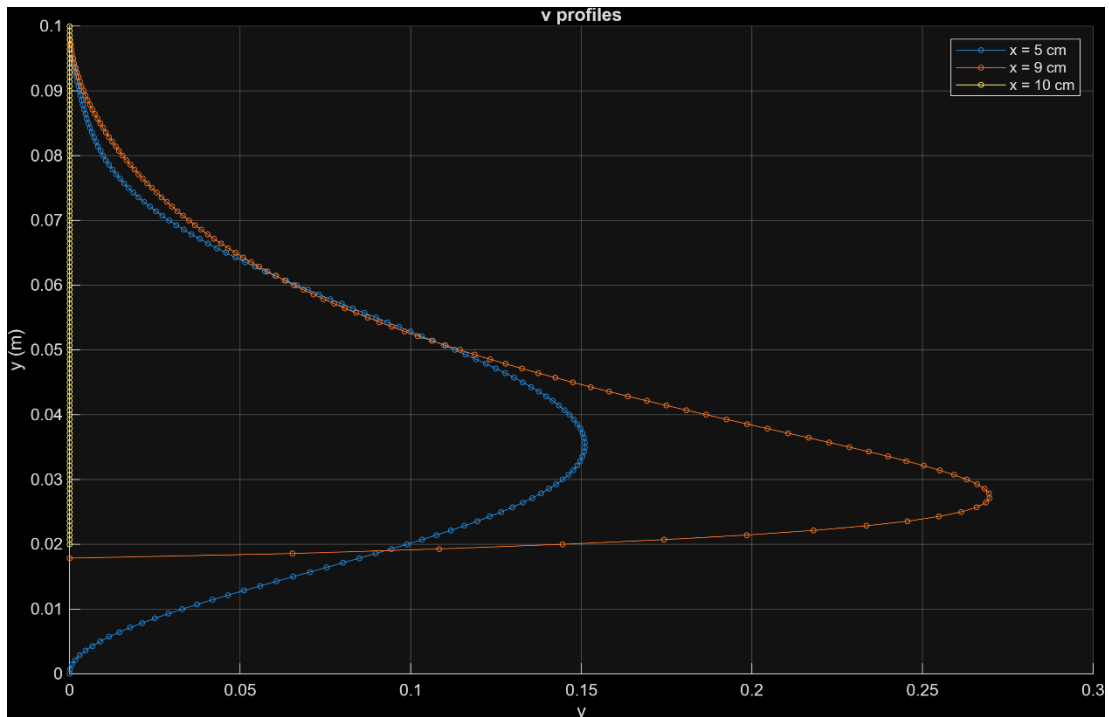


Figure 93. Vertical velocity profiles at $x = 5, 9$ and 10 cm on the fine mesh.

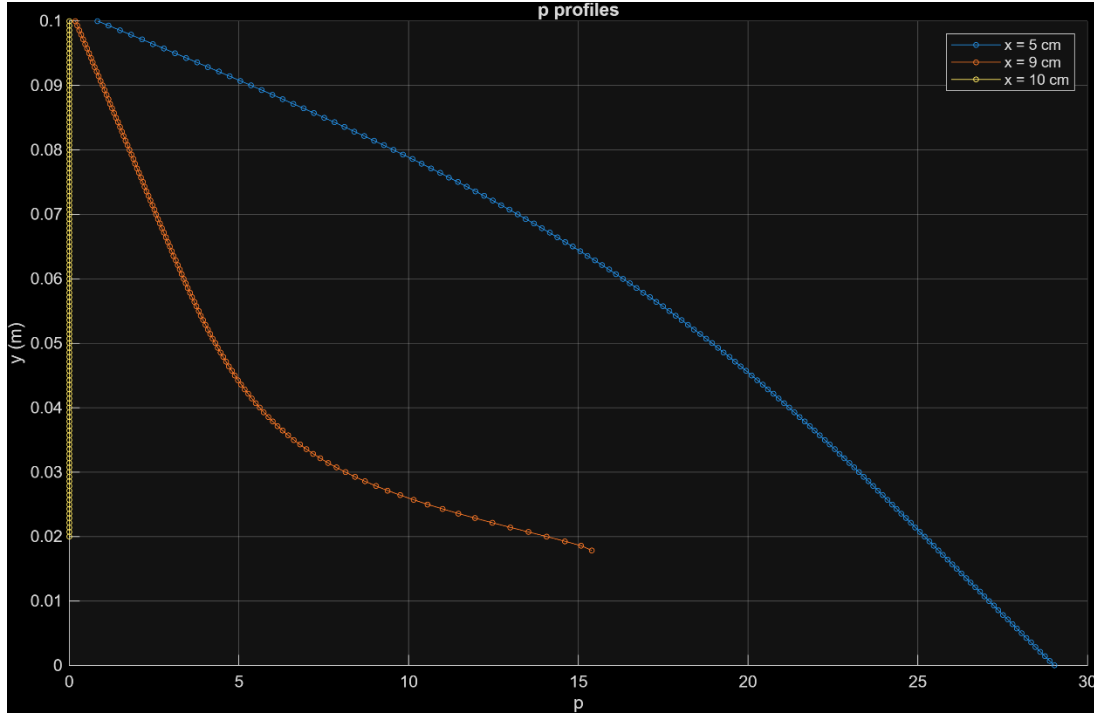


Figure 94. Pressure profiles at $x = 5, 9$ and 10 cm the fine mesh.

8 Two-Level Multigrid Method

By using a fine mesh and a coarse mesh whose nodes coincide with every second fine-mesh node in each direction, so that the fine mesh of 141×141 nodes is paired with a coarse mesh of 71×71 nodes. Point Gauss-Seidel is used to solve the equations in each level.

Each V-cycle proceeds as follows. Six Gauss-Seidel sweeps are first performed on the fine mesh, which is enough to damp the high-frequency components of the error but leaves the smooth components almost untouched. The residual of the discretized Poisson equation is then calculated at every interior fluid node.

Residual equation:

$$r_{i,j} = [\zeta_{i,j} + \beta A_2 \psi_{i-1,j} + \alpha B_2 \psi_{i,j-1} + A_2 \psi_{i+1,j} + B_2 \psi_{i,j+1}] - [(1 + \beta)A_2 + (1 + \alpha)B_2] \psi_{i,j}$$

$$A_2 = \frac{2}{\beta(\beta + 1)(\Delta x^-)^2}, B_2 = \frac{2}{\alpha(\alpha + 1)(\Delta y^-)^2}$$

and transferred to the coarse mesh by injection, which requires no interpolation because the coarse nodes are a subset of the fine nodes. On the coarse mesh the delta equation is solved by Gauss-Seidel to convergence. This equation has the same form as the stream function

equation, with the correction replacing the stream function and the residual replacing the vorticity term.

Delta equation ($L(\Delta\psi) = -r$):

$$\Delta\psi_{i,j} = \frac{r_{i,j} + \beta A_2 \delta\psi_{i-1,j} + \alpha B_2 \delta\psi_{i,j-1} + A_2 \delta\psi_{i+1,j} + B_2 \delta\psi_{i,j+1}}{(1 + \beta)A_2 + (1 + \alpha)B_2}$$

The converged correction is returned to the fine mesh by injection at the coinciding nodes and by linear interpolation everywhere else. The V-cycle is repeated until the maximum correction falls below the tolerance, at which point the smooth error components have been removed on the coarse mesh.

multigrid was run only with the fine mesh as the base level, as required by the project statement.

Table 6. Iteration count and wall-clock time of the two-level multigrid scheme.

Mesh pair	Iterations	Wall-clock time (s)
Fine (141×141) with coarse level (71×71)	25,472	56.60

The contours and line plots obtained with this scheme are shown below.

8.1 contours (fine mesh)

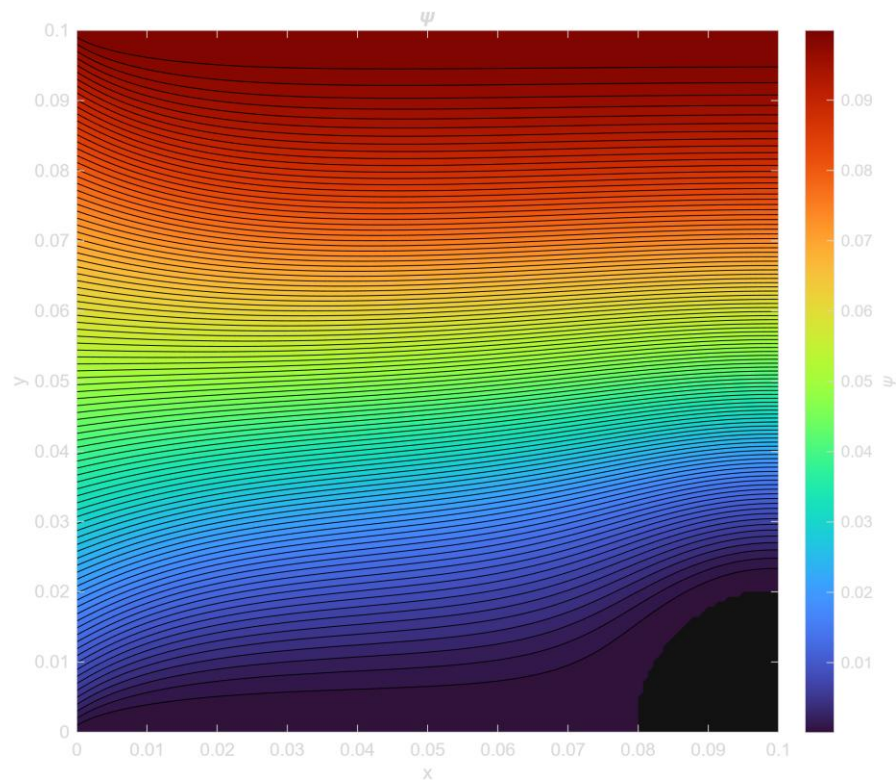


Figure 95. Stream function contour on the fine mesh.

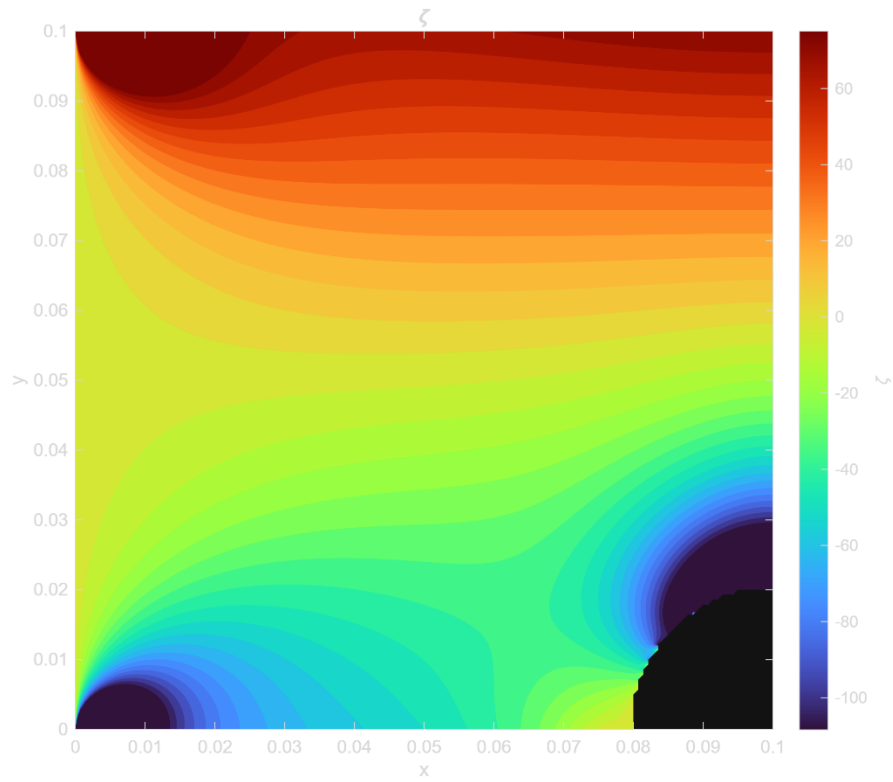


Figure 96. Vorticity contour on the fine mesh.

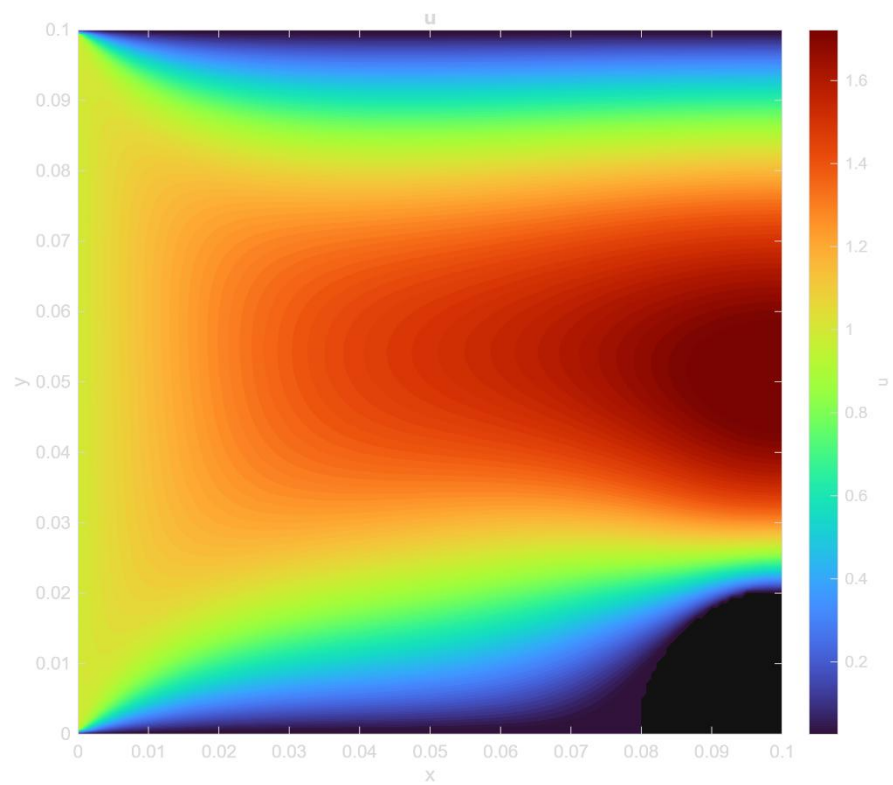


Figure 97. Horizontal velocity contour on the fine mesh.

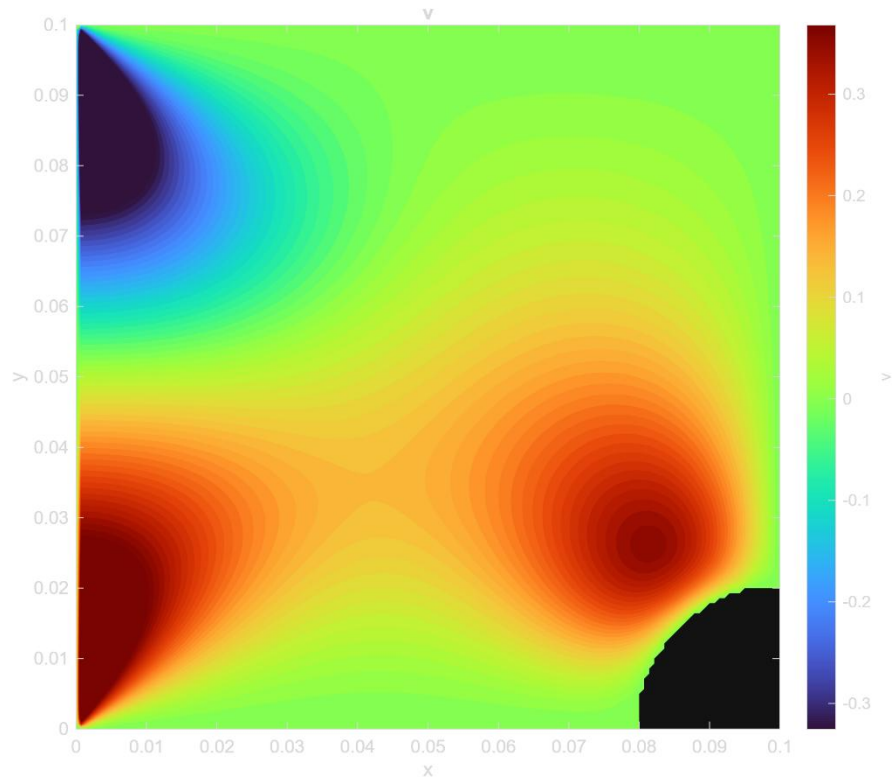


Figure 98. Vertical velocity contour on the fine mesh.

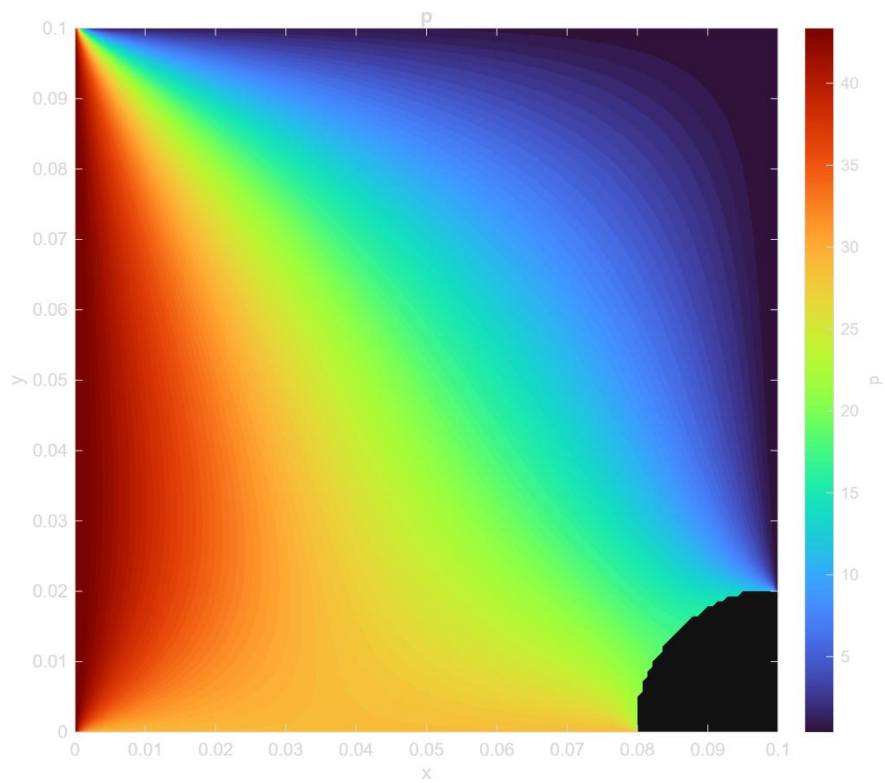


Figure 99. Pressure contour on the fine mesh.

8.2 line plots (fine mesh)

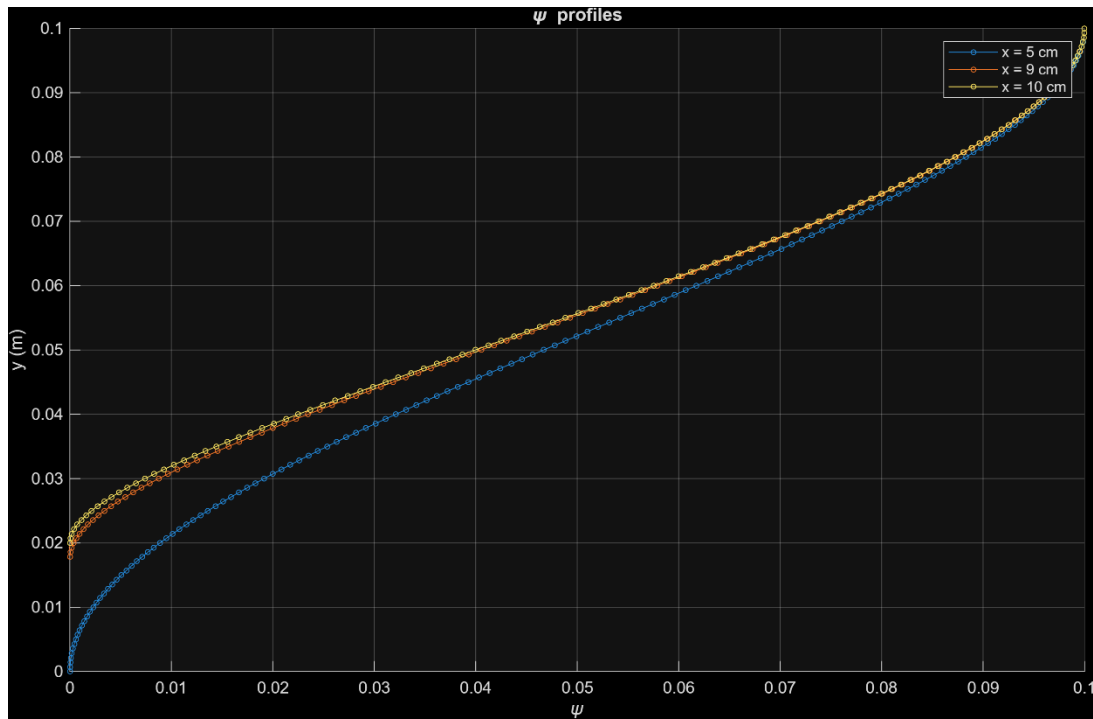


Figure 100. Stream function profiles at $x = 5, 9$ and 10 cm on the fine mesh.

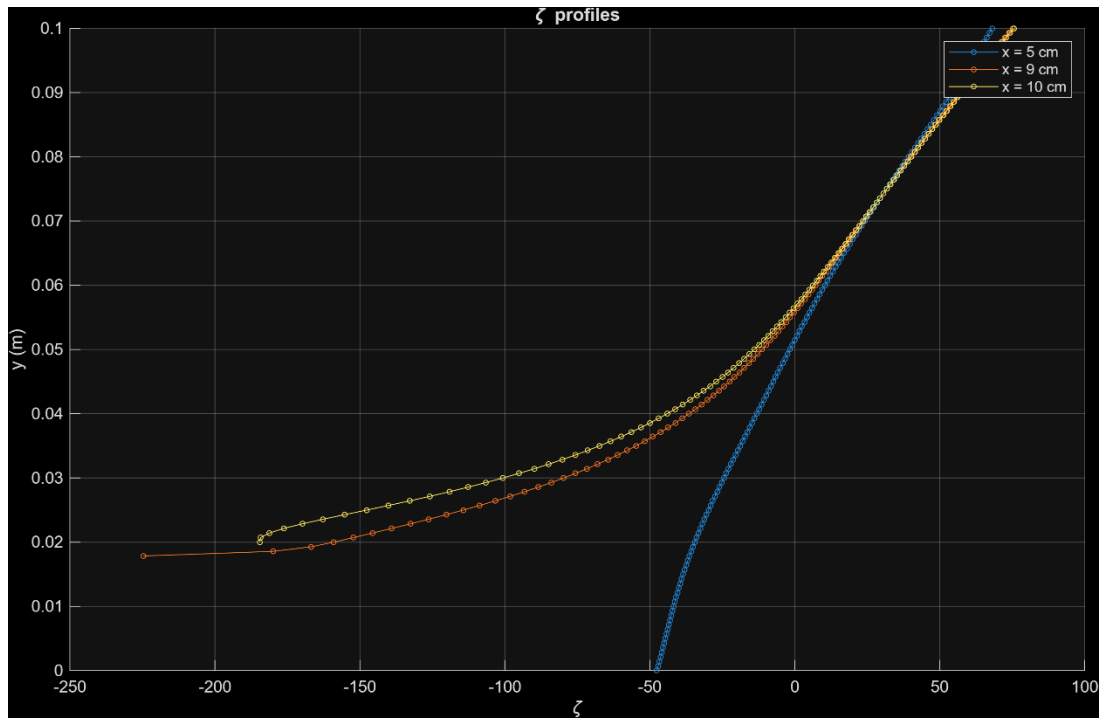


Figure 101. Vorticity profiles at $x = 5, 9$ and 10 cm on the fine mesh.

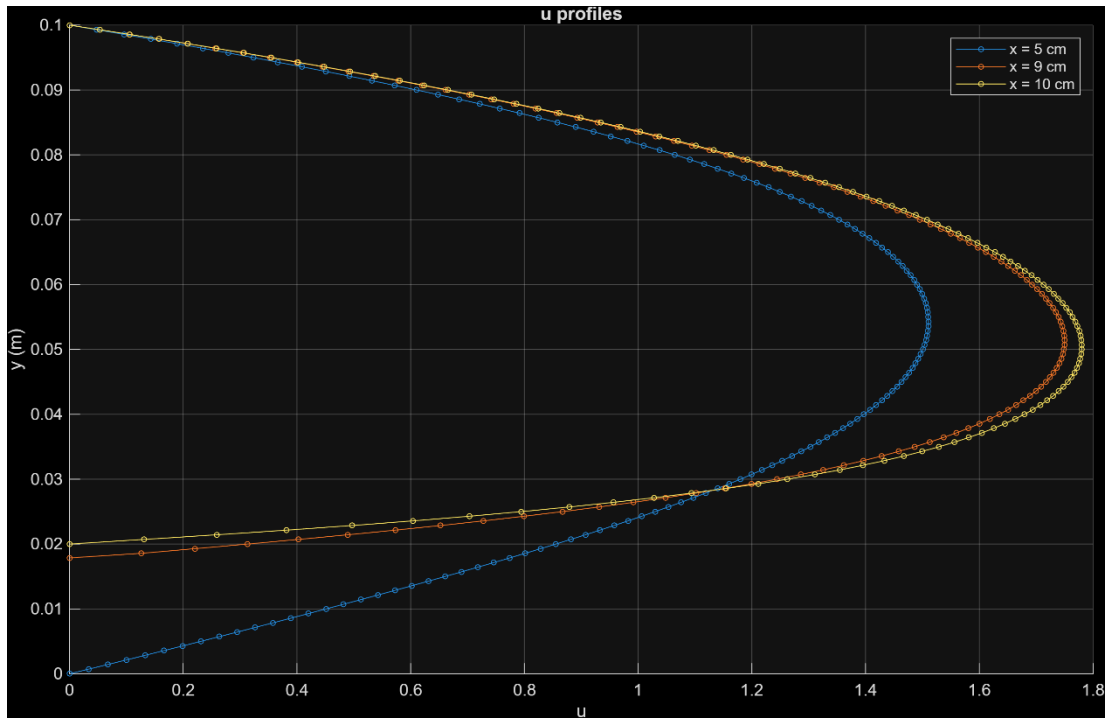


Figure 102. Horizontal velocity profiles at $x = 5, 9$ and 10 cm on the fine mesh.

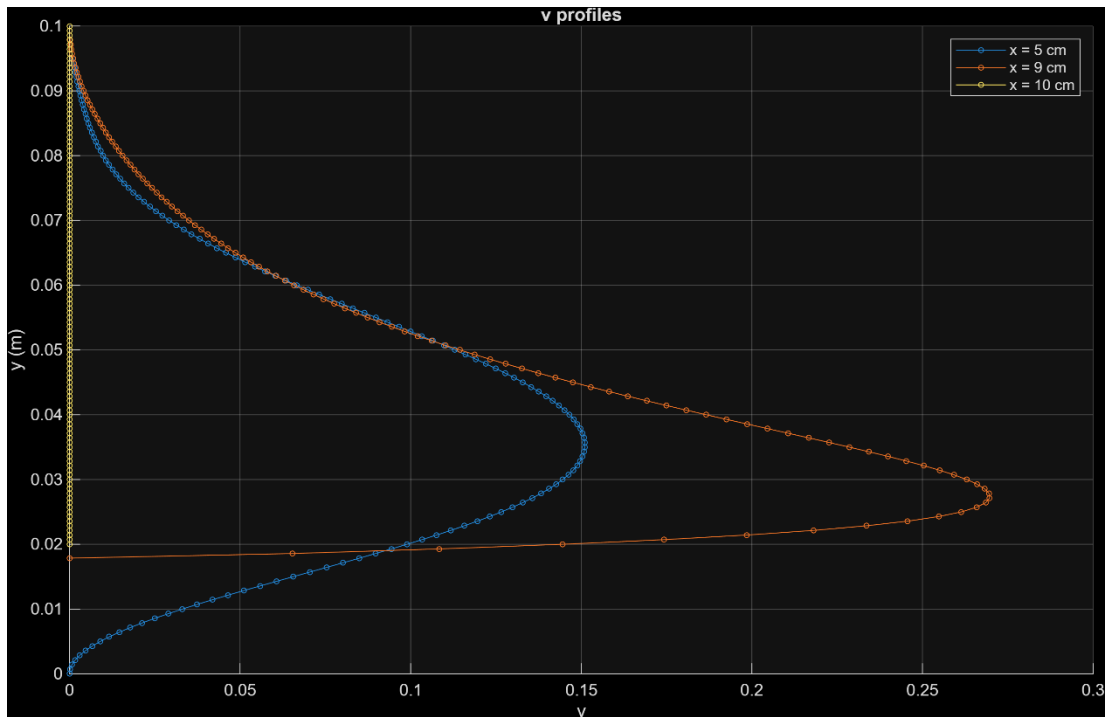


Figure 103. Vertical velocity profiles at $x = 5, 9$ and 10 cm on the fine mesh.

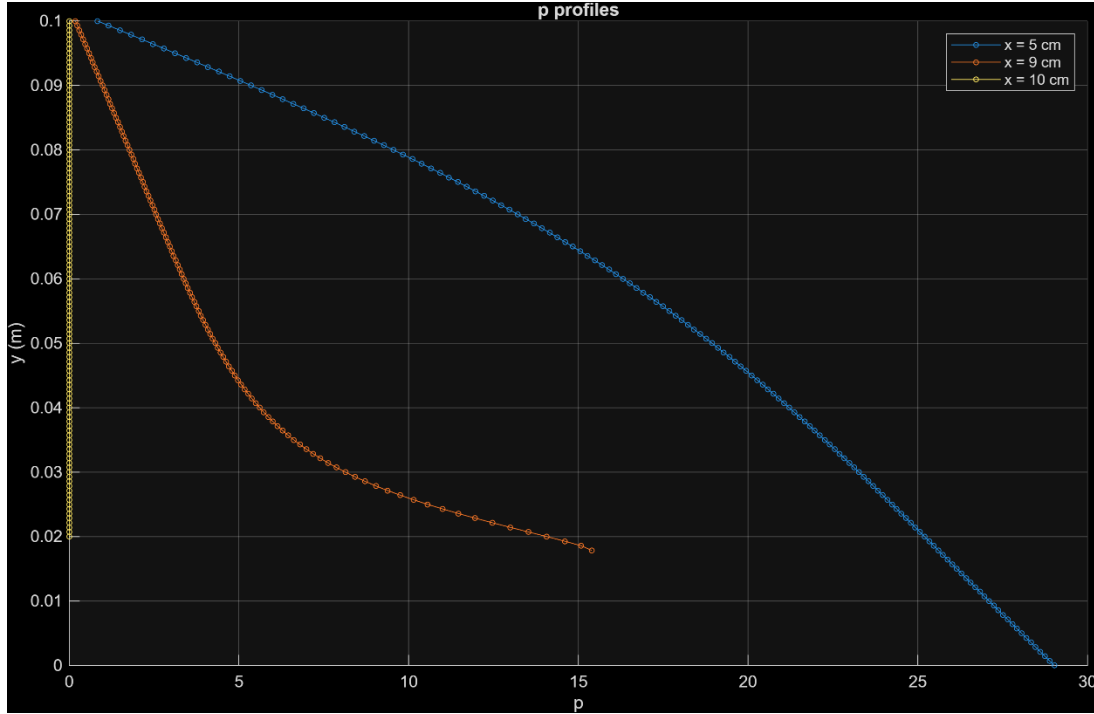


Figure 104. Pressure profiles at $x = 5, 9$ and 10 cm the fine mesh.

9 SLOR Combined with ADI Method

The ADI scheme extends SLOR by alternating the implicit direction within each iteration. The first half-step is identical to SLOR: the domain is swept row by row and each row is solved implicitly in x , with the terms coupling the node to the rows below and above moved to the right-hand side. The second half-step repeats the procedure column by column, solving implicitly in y with the terms coupling the node to its left and right neighbors moved to the right-hand side. Each half-step therefore yields a tridiagonal system.

$$A_i \phi_{i-1,j} + B_i \phi_{i,j} + C_i \phi_{i+1,j} = \text{RHS} \quad (\text{x-implicit})$$

$$A_j \phi_{i,j-1} + B_j \phi_{i,j} + C_j \phi_{i,j+1} = \text{RHS} \quad (\text{y-implicit})$$

which is solved with the Thomas algorithm, and over-relaxation is applied to the line solution in the same form as before:

$$\phi_{i,j}^{k+1} = (1 - \omega) \phi_{i,j}^k + \omega \phi_{i,j}^{\text{line}}$$

As in SLOR, the quarter-cylinder makes the length of the systems variable. For every row the fluid nodes are counted from the inlet up to the first node adjacent to the cylinder; for every column they are counted downward from the upper wall to the first such node, and the index of the lowest included node is recorded so that the system can be assembled at the correct size. Nodes lying on or inside the obstruction are excluded and retain their boundary values.

The relaxation factors are listed in the Table. The theoretical optimum of the Laplace operator could not be used here, the stream function factor was therefore selected by trial, and the vorticity equation was left unrelaxed.

Table 7. Relaxation factors used by the SLOR + ADI scheme on the three meshes.

Mesh	$\omega_{\text{opt}}(\psi)$	$\omega_{\text{opt}}(\zeta)$
Coarse	1.3	1
Intermediate	1.3	1
Fine	1.3	1

Table 8. Iteration count and wall-clock time of the SLOR + ADI scheme on the three meshes.

Mesh	Iterations	Wall-clock time (s)
Coarse (61×61)	2,270	2.49
Intermediate (101×101)	8,411	27.30
Fine (141×141)	20,382	135.40

The contours and line plots obtained on each of the three meshes are shown below.

9.1 contours (coarse mesh)

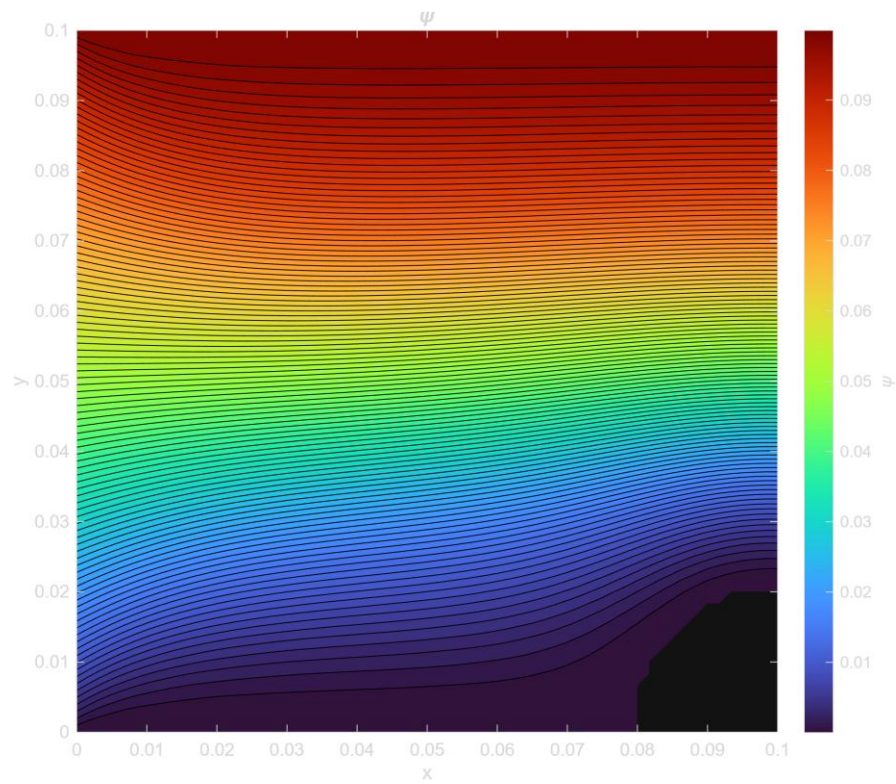


Figure 105. Stream function contour on the coarse mesh.

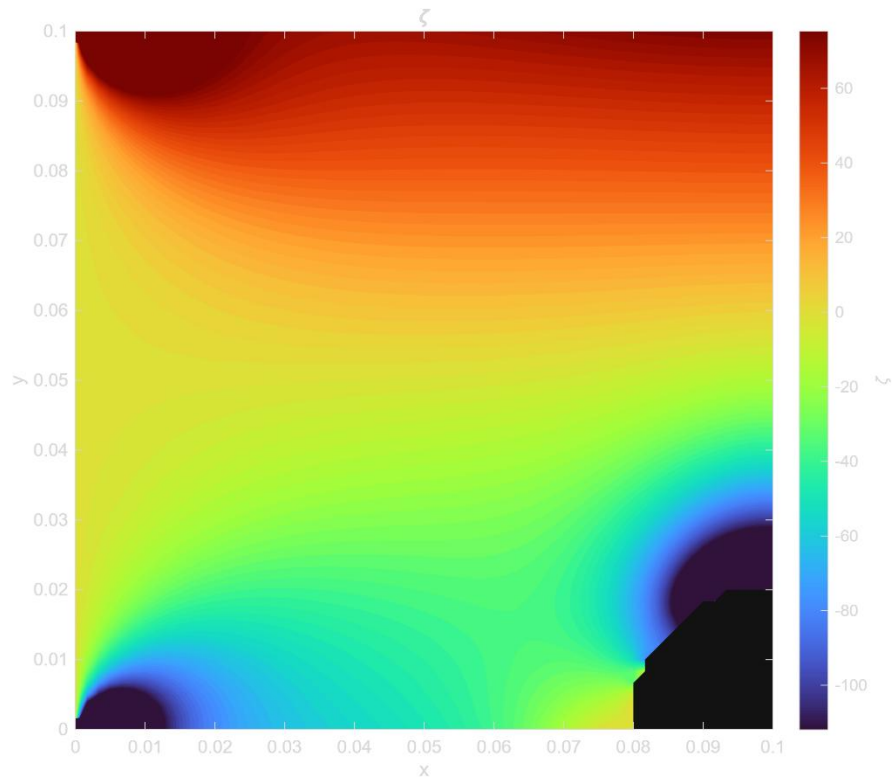


Figure 106. Vorticity contour on the coarse mesh.

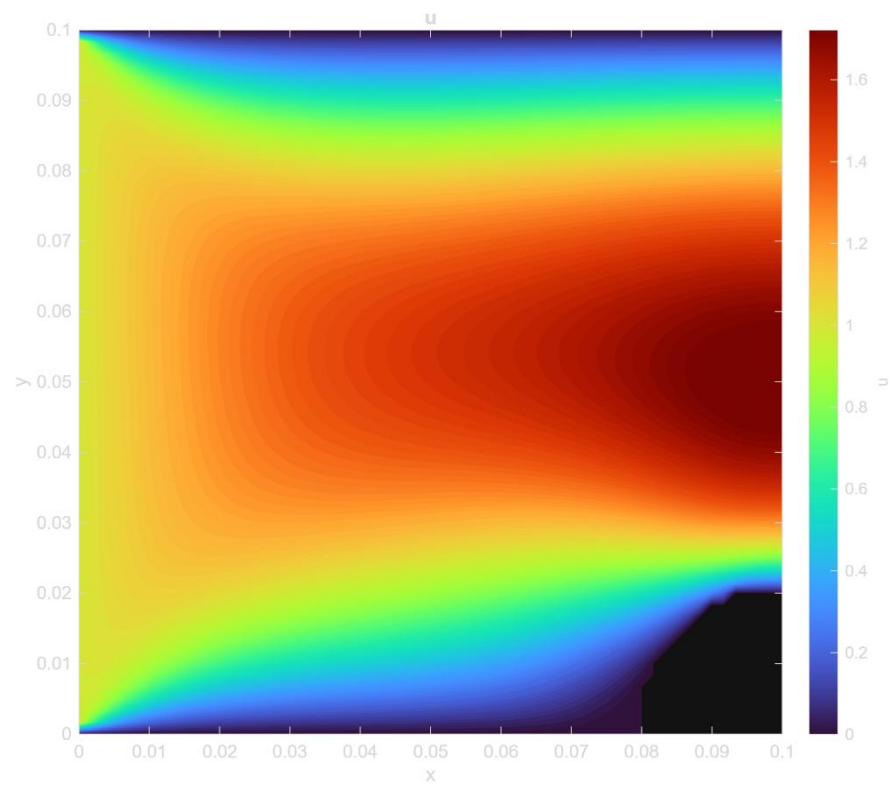


Figure 107. Horizontal velocity contour on the coarse mesh.

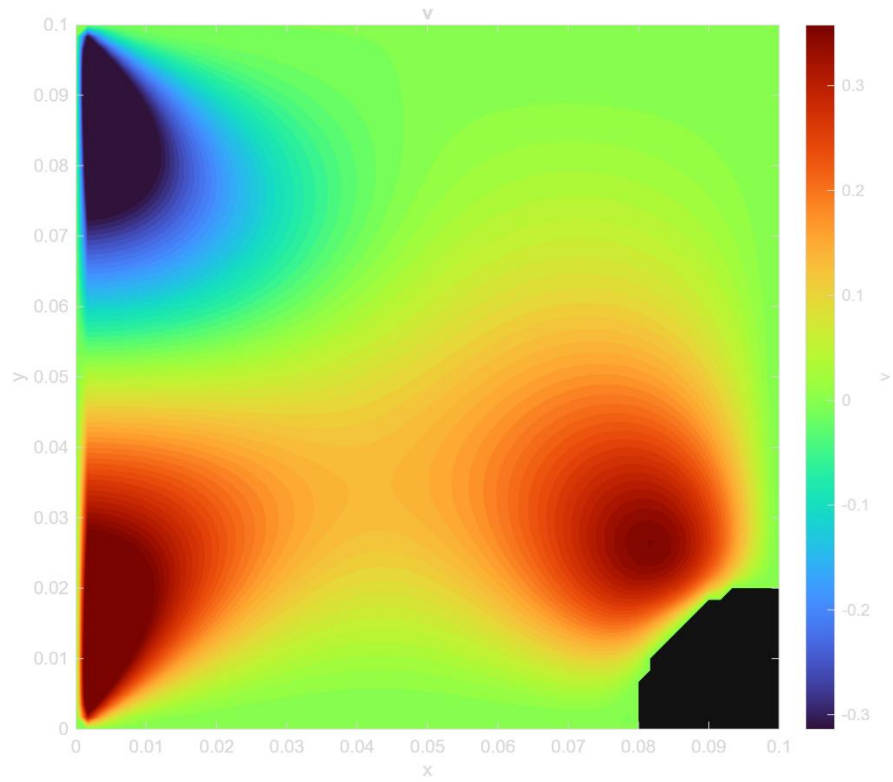


Figure 108. Vertical velocity contour on the coarse mesh.

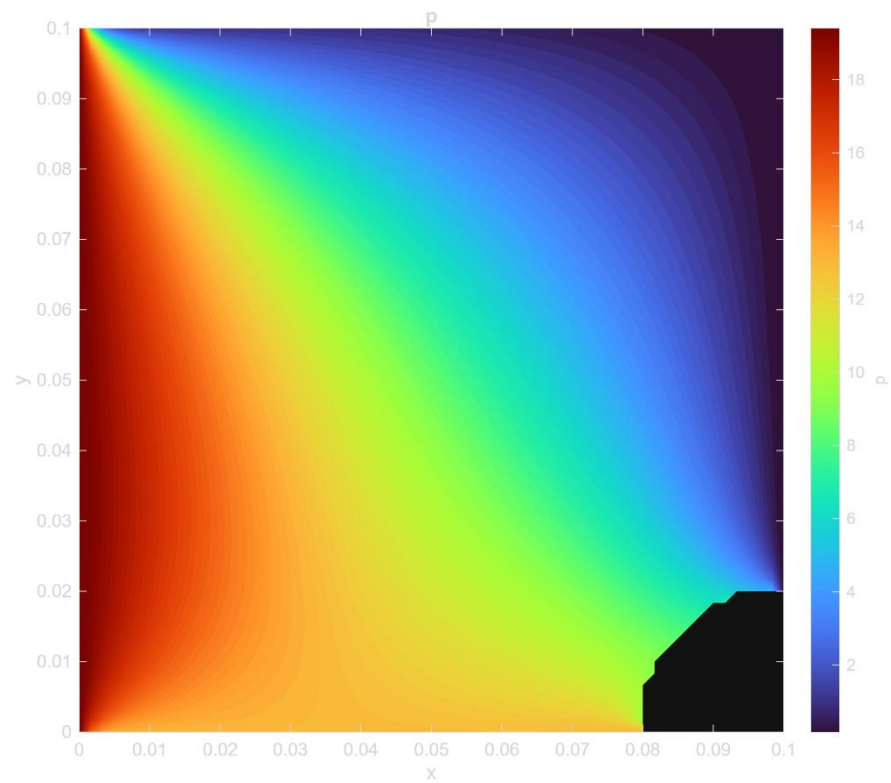


Figure 109. Pressure contour on the coarse mesh.

9.2 line plots (coarse mesh)

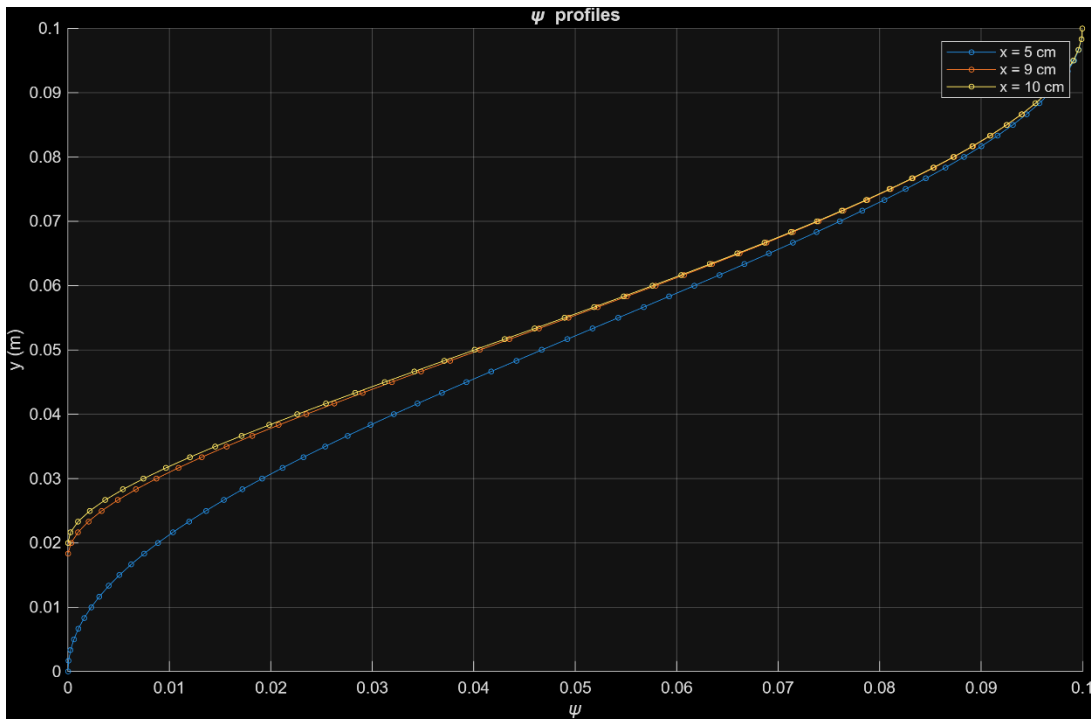


Figure 110. Stream function profiles at $x = 5, 9$ and 10 cm on the coarse mesh.

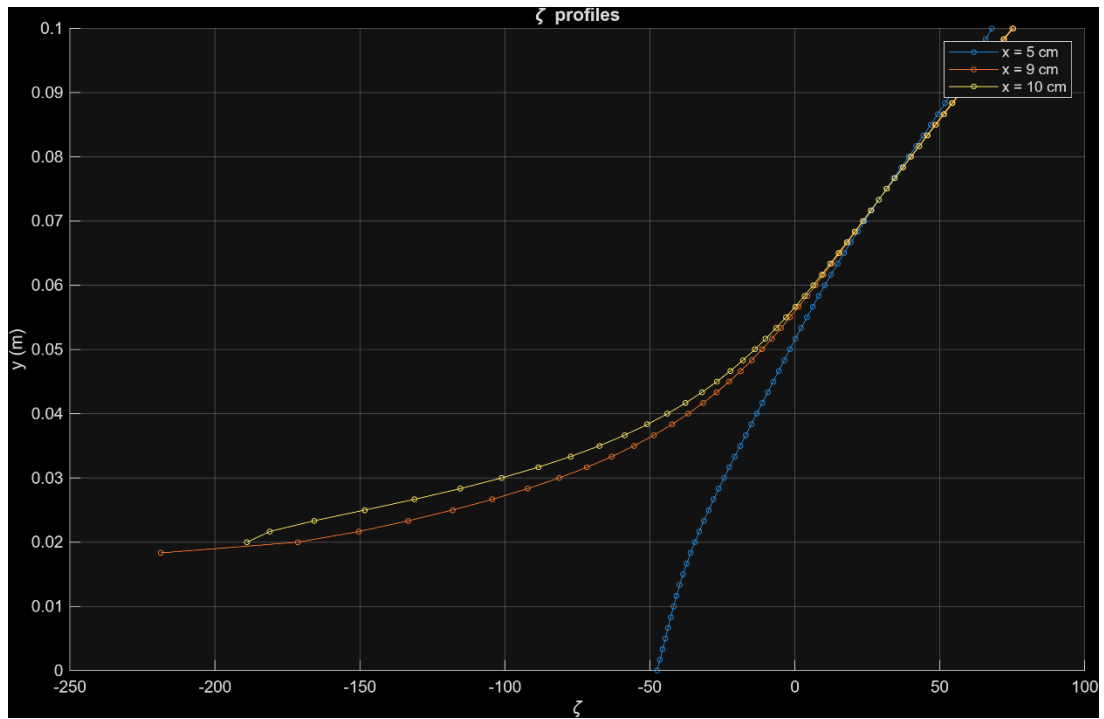


Figure 111. Vorticity profiles at $x = 5, 9$ and 10 cm on the coarse mesh

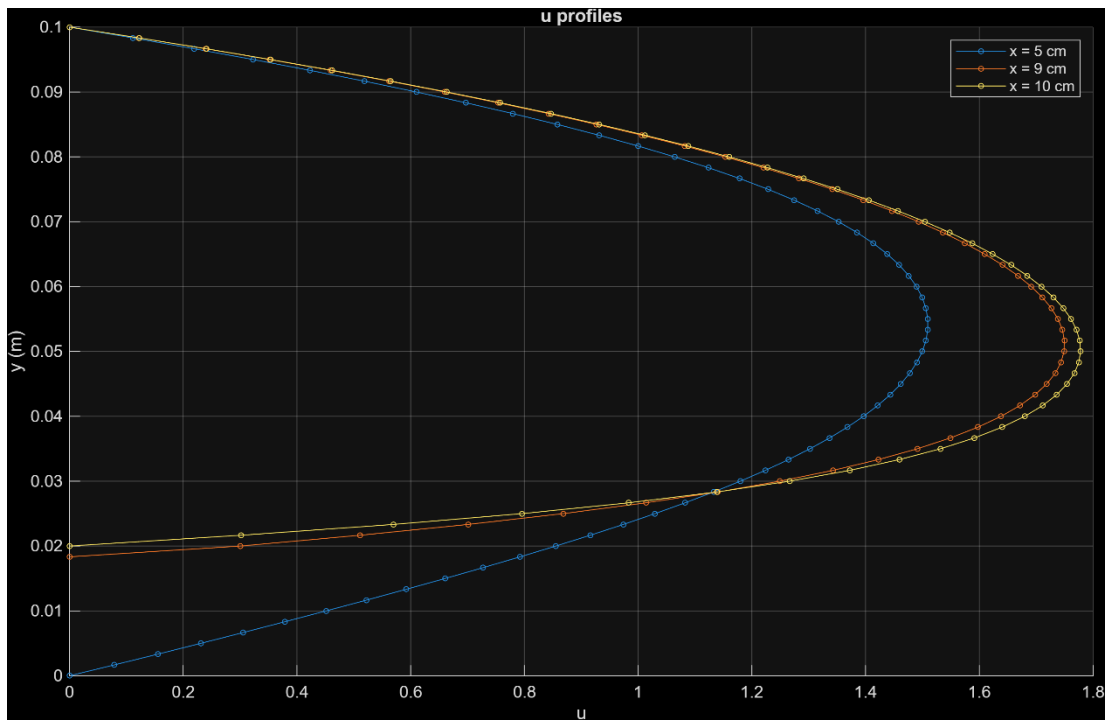


Figure 112. Horizontal velocity profiles at $x = 5, 9$ and 10 cm on the coarse mesh.

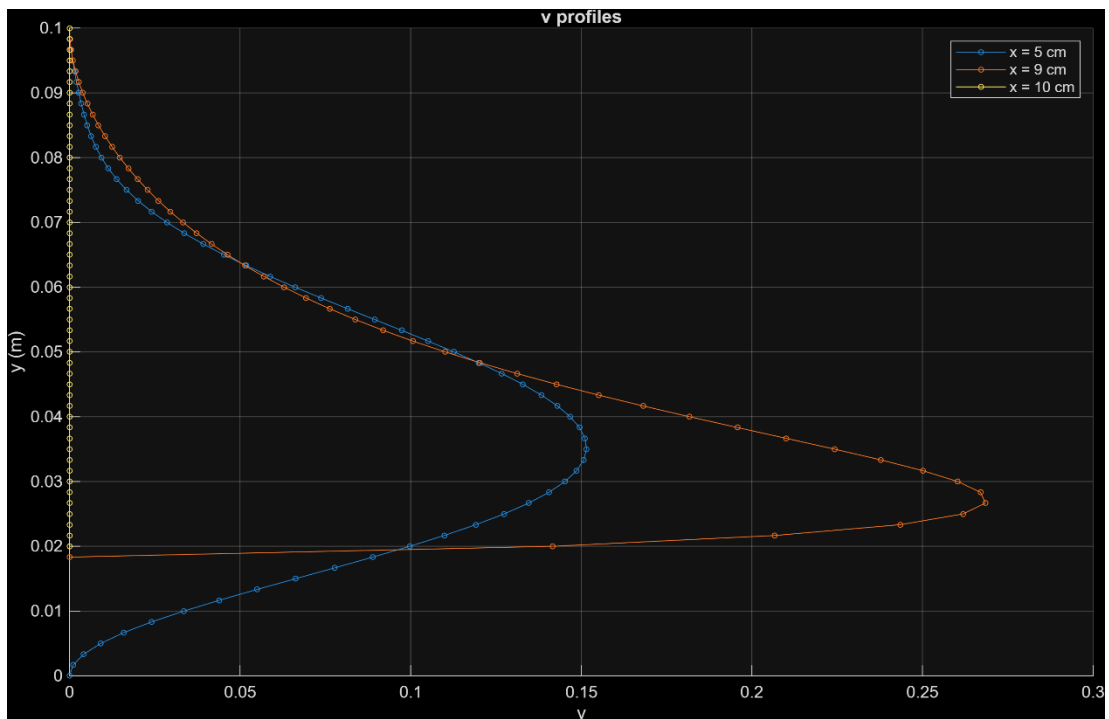


Figure 113. Vertical velocity profiles at $x = 5, 9$ and 10 cm on the coarse mesh.

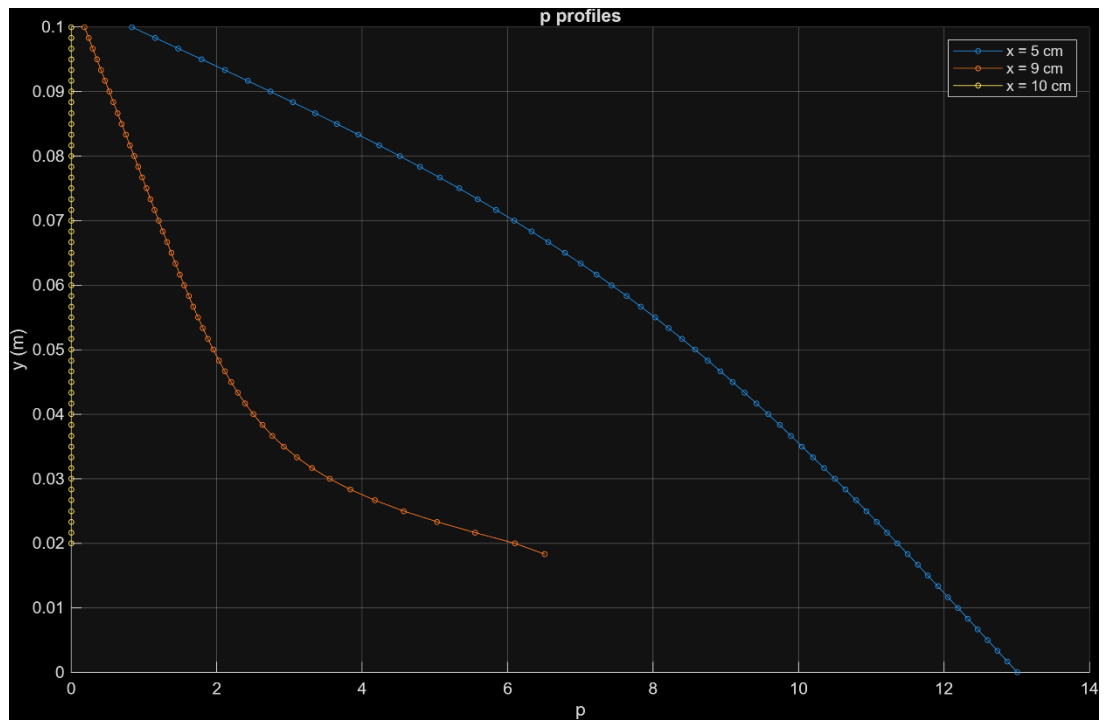


Figure 114. Pressure profiles at $x = 5, 9$ and 10 cm on the coarse mesh.

9.3 contours (intermediate mesh)

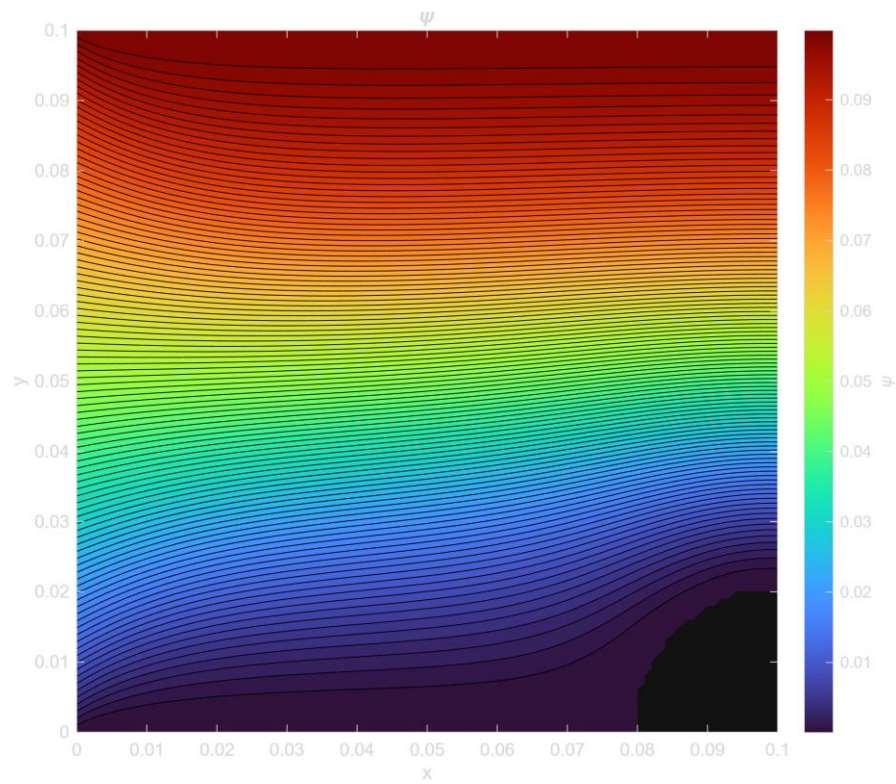


Figure 115. Stream function contour on the intermediate mesh.

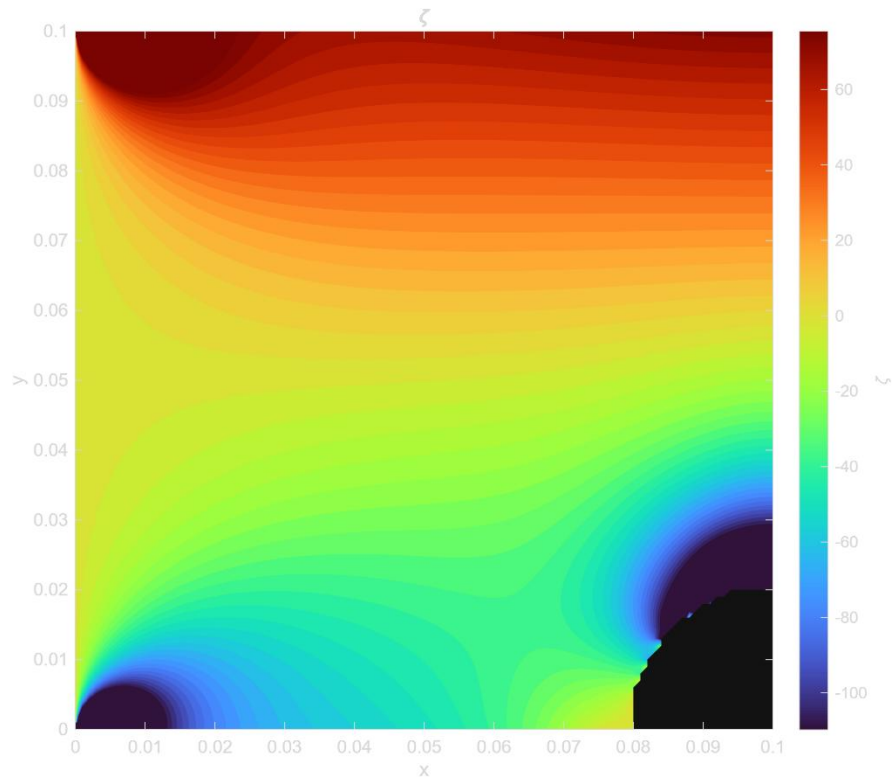


Figure 116. Vorticity contour on the intermediate mesh.

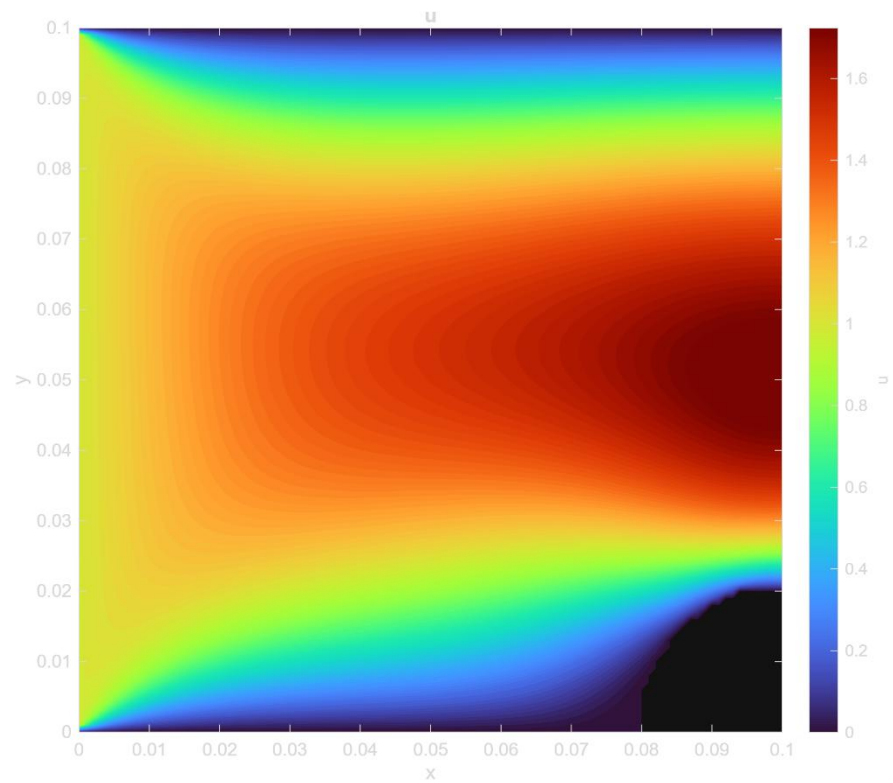


Figure 117. Horizontal velocity contour on the intermediate mesh.

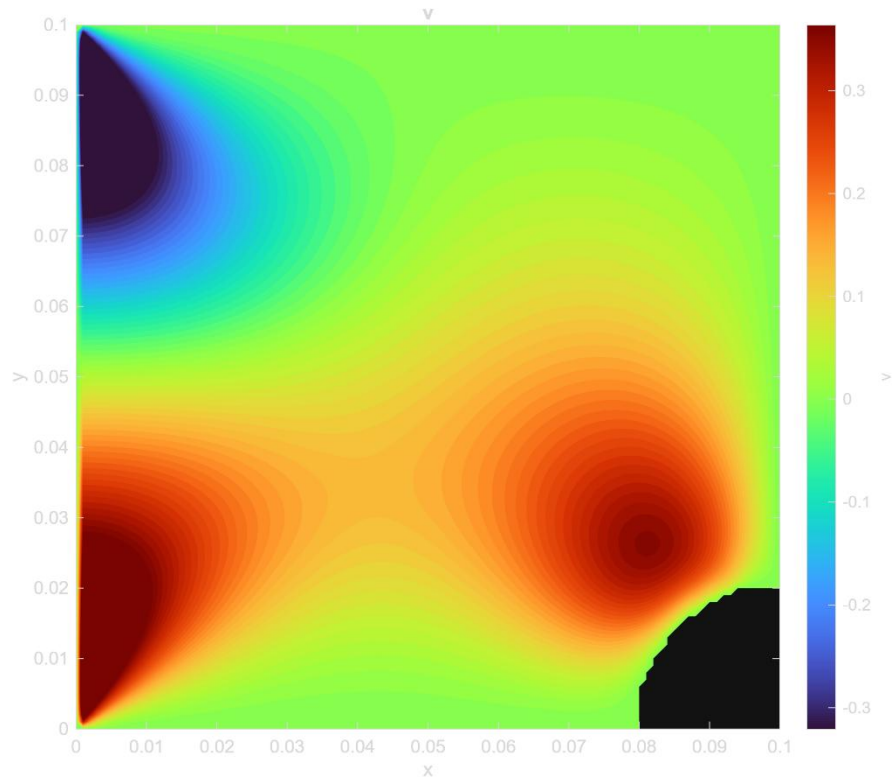


Figure 118. Vertical velocity contour on the intermediate mesh.

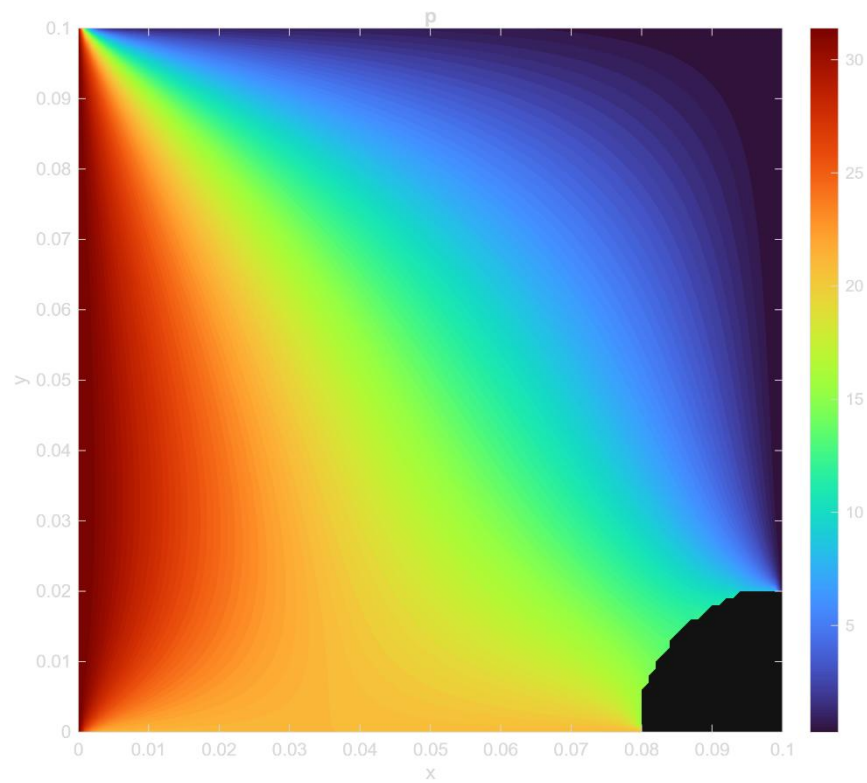


Figure 119. Pressure contour on the intermediate mesh.

9.4 line plots (intermediate mesh)

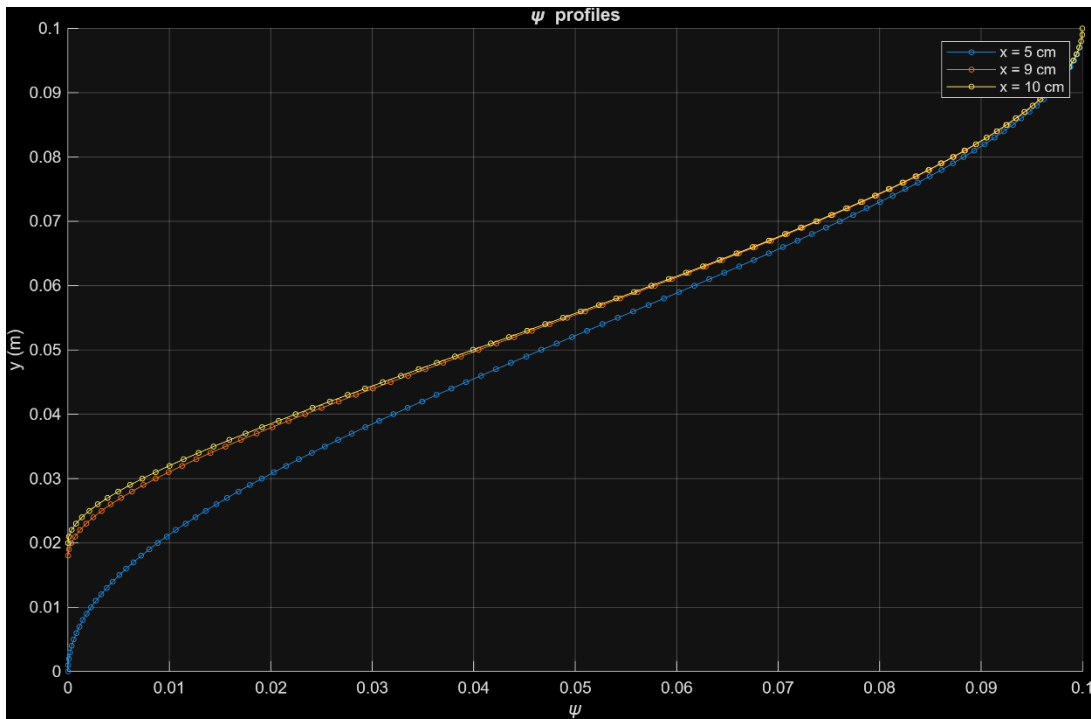


Figure 120. Stream function profiles at $x = 5, 9$ and 10 cm on the intermediate mesh.

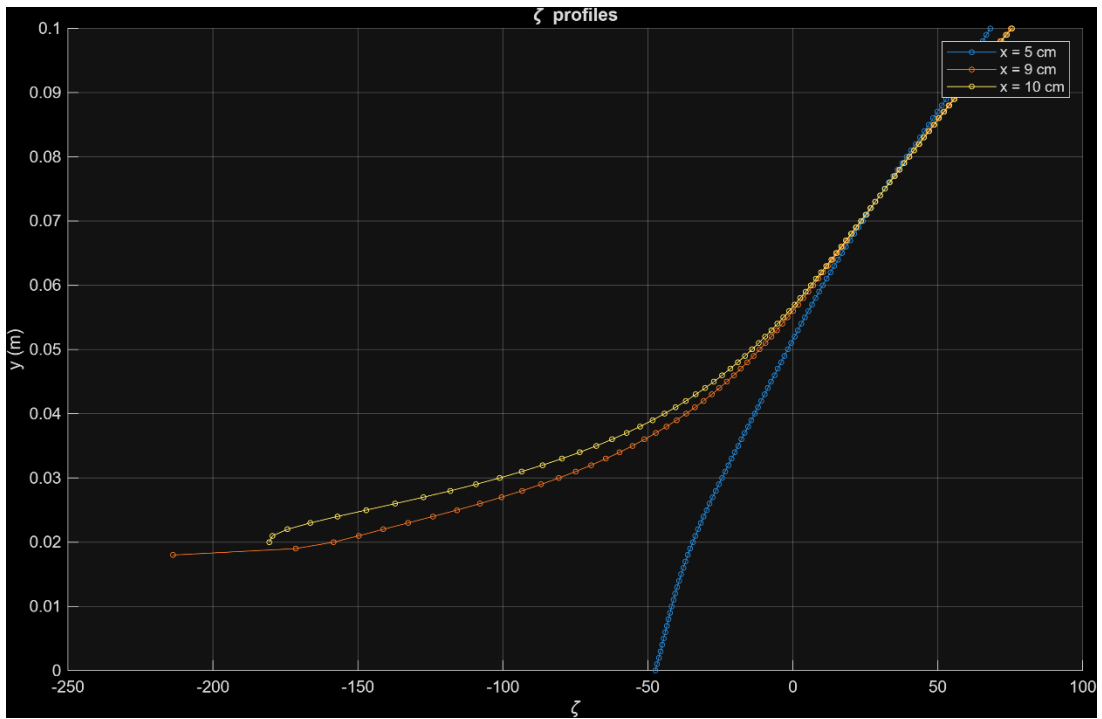


Figure 121. Vorticity profiles at $x = 5, 9$ and 10 cm on the intermediate mesh.

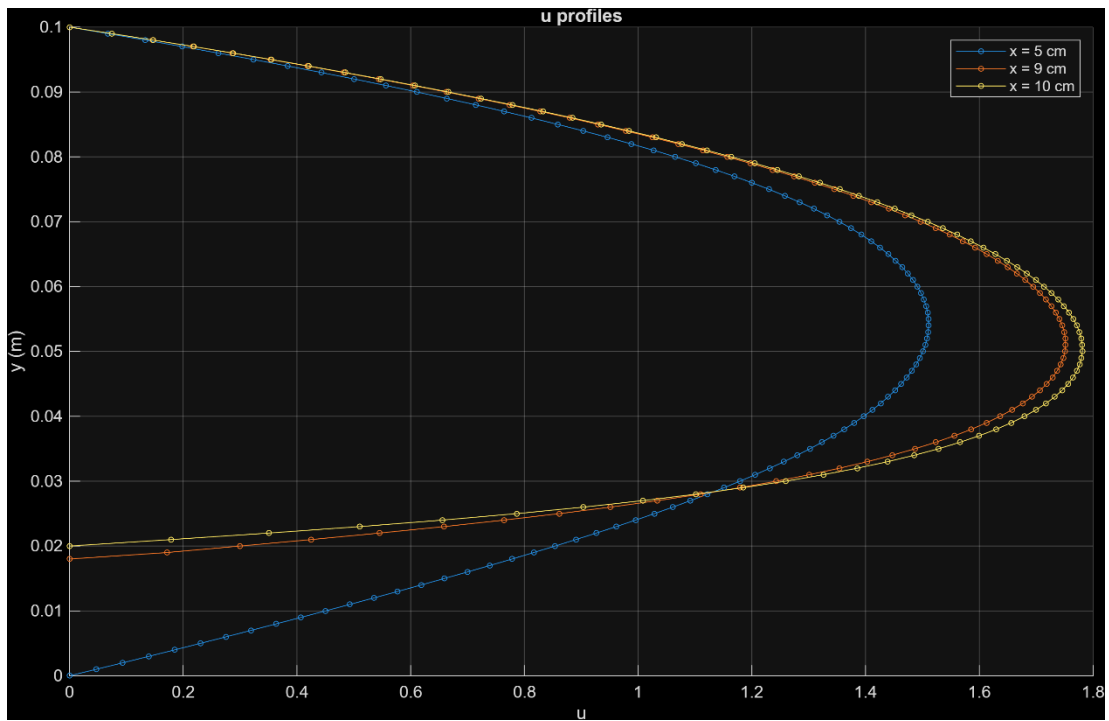


Figure 122. Horizontal velocity profiles at $x = 5, 9$ and 10 cm on the intermediate mesh.

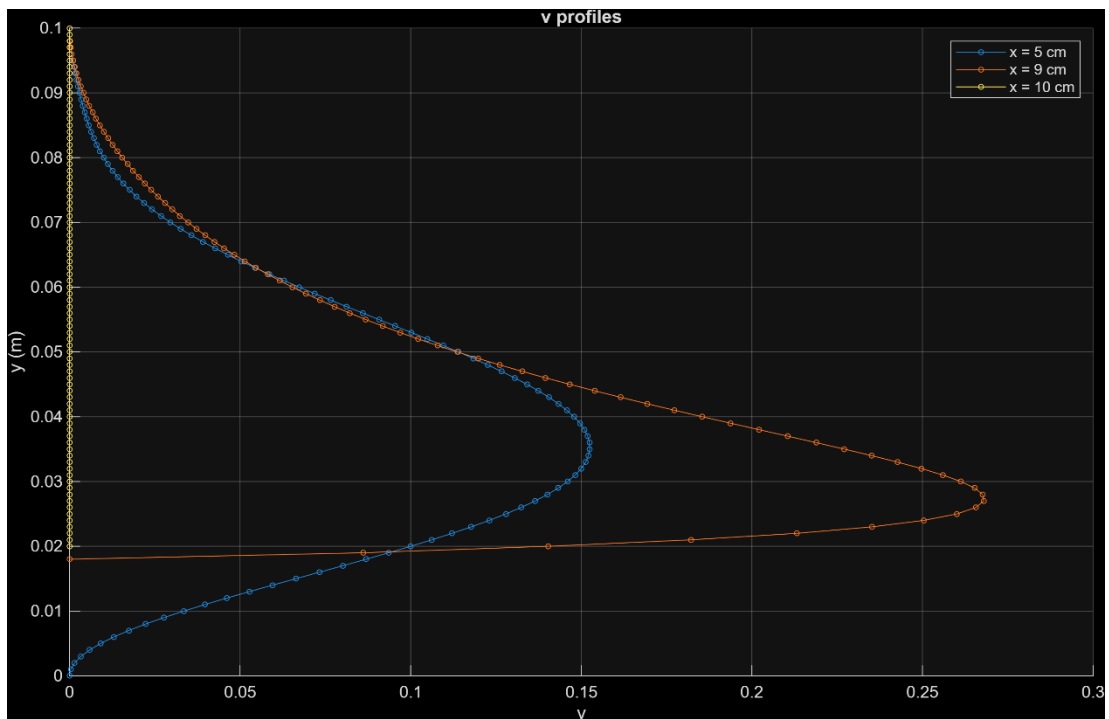


Figure 123. Vertical velocity profiles at $x = 5, 9$ and 10 cm on the intermediate mesh.

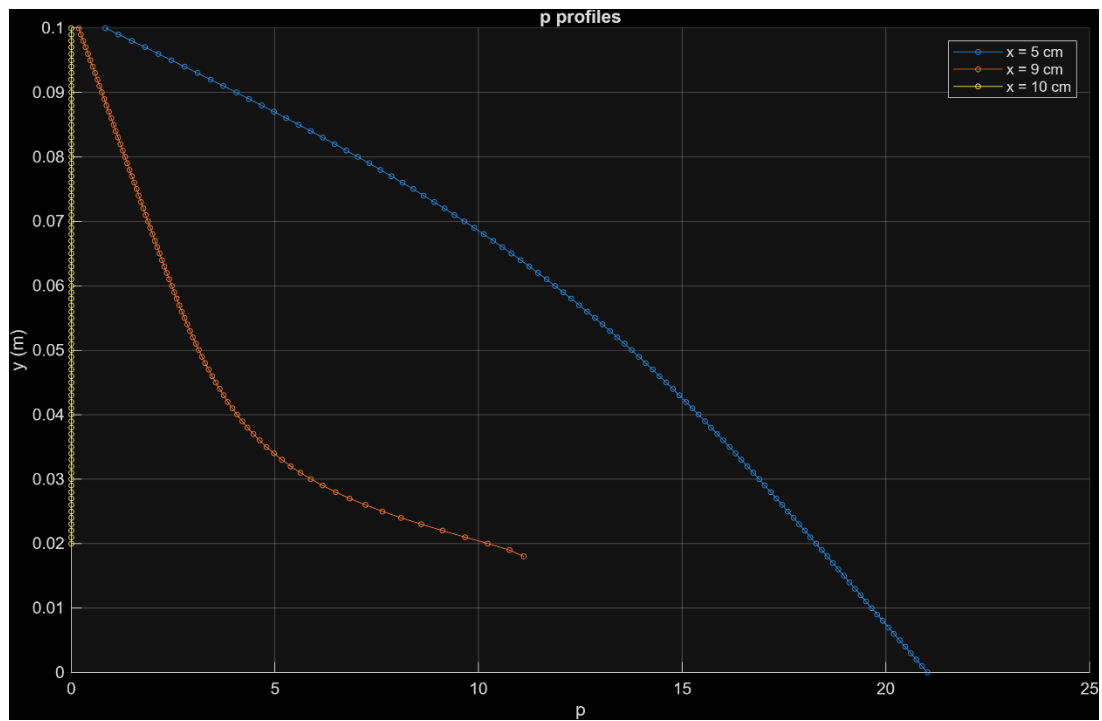


Figure 124. Pressure profiles at $x = 5, 9$ and 10 cm obtained with the Gauss–Seidel scheme on the intermediate mesh.

9.5 contours (fine mesh)

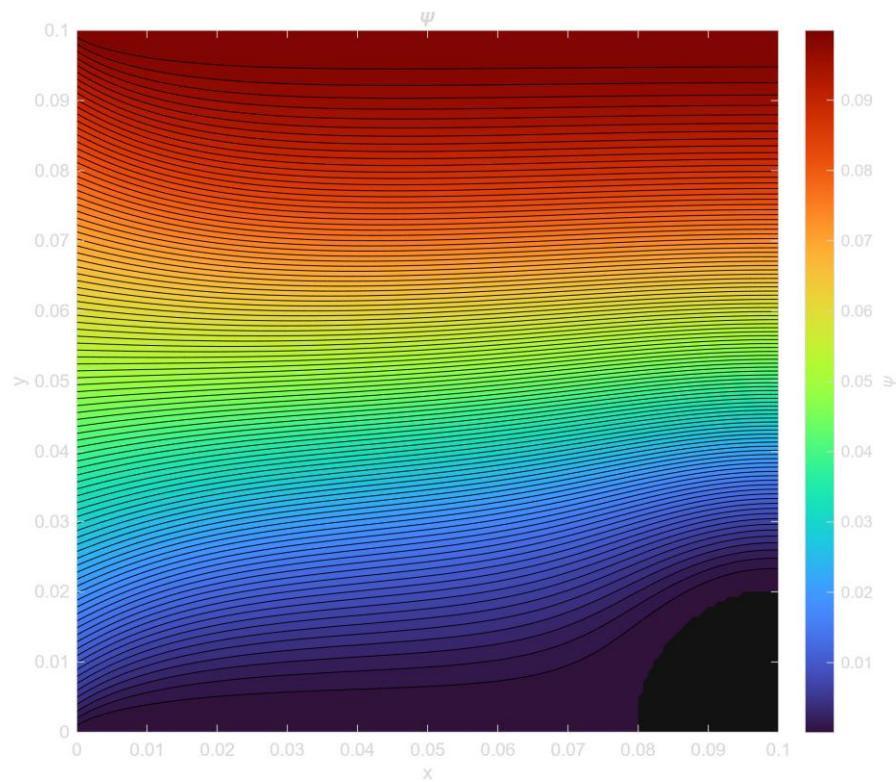


Figure 125. Stream function contour on the fine mesh.

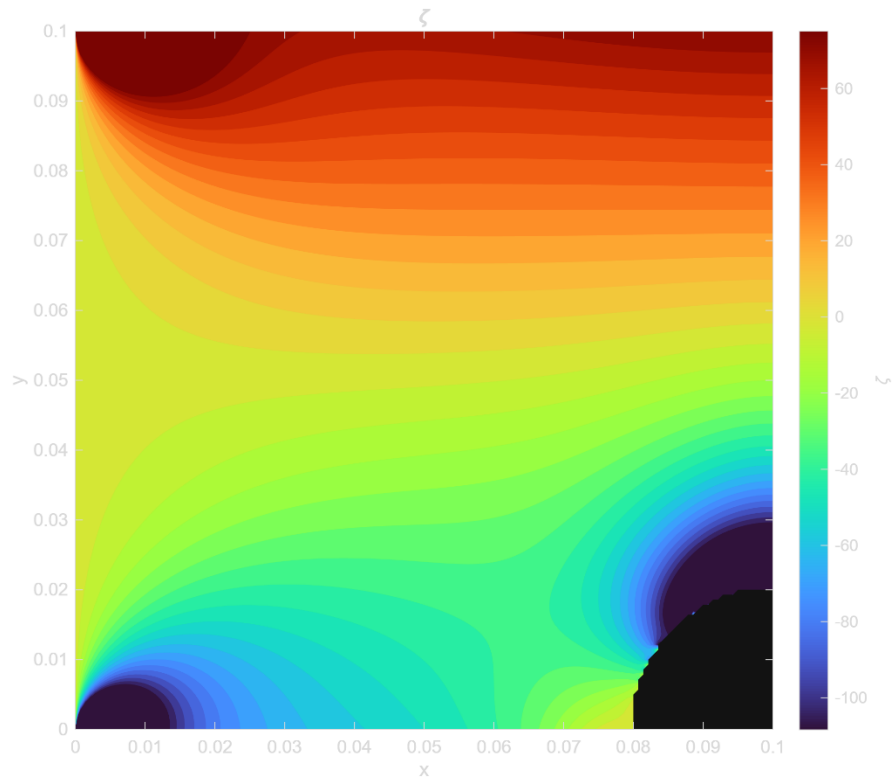


Figure 126. Vorticity contour on the fine mesh.

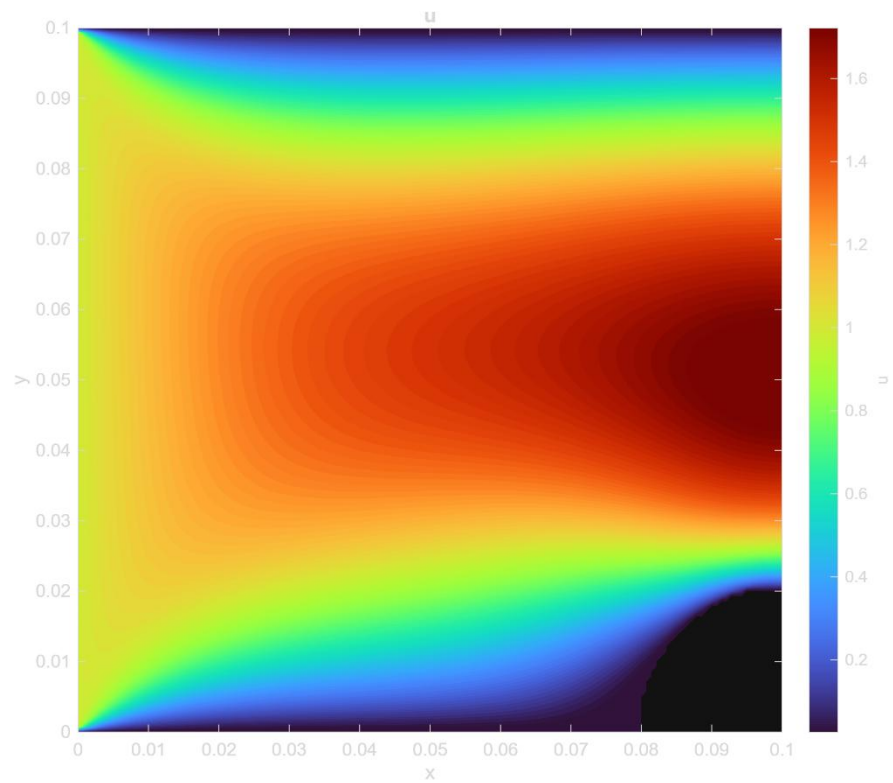


Figure 127. Horizontal velocity contour on the fine mesh.

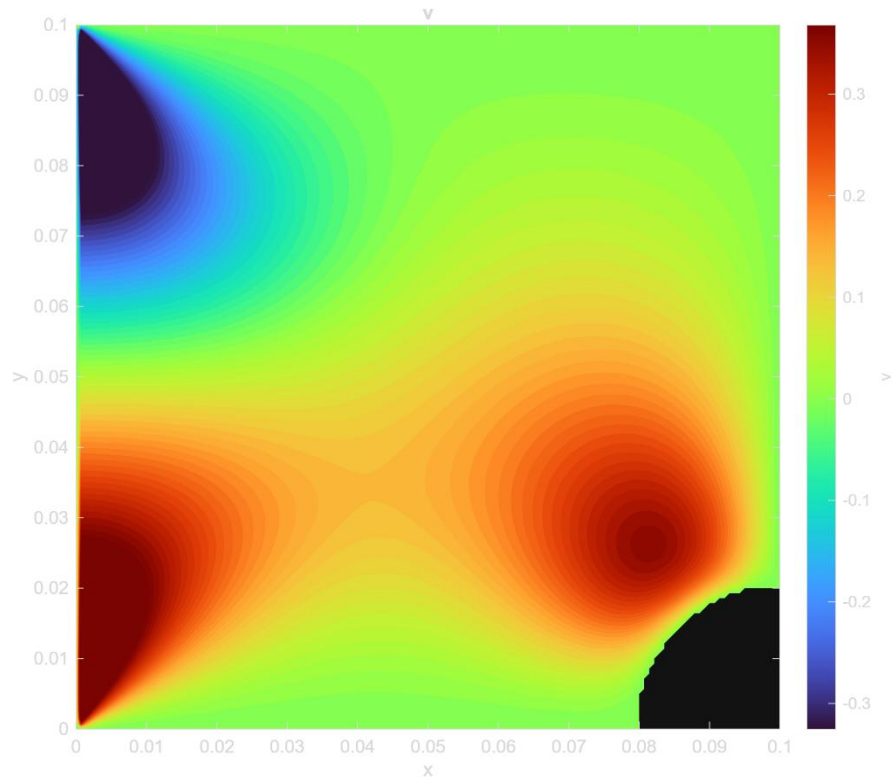


Figure 128. Vertical velocity contour on the fine mesh.

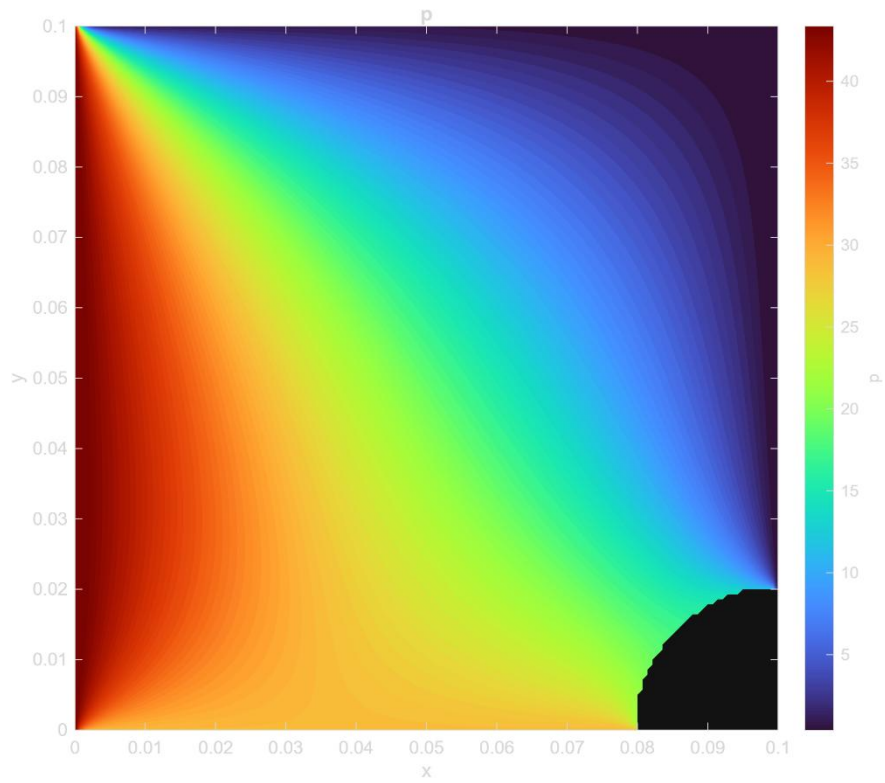


Figure 129. Pressure contour on the fine mesh.

9.6 line plots (fine mesh)

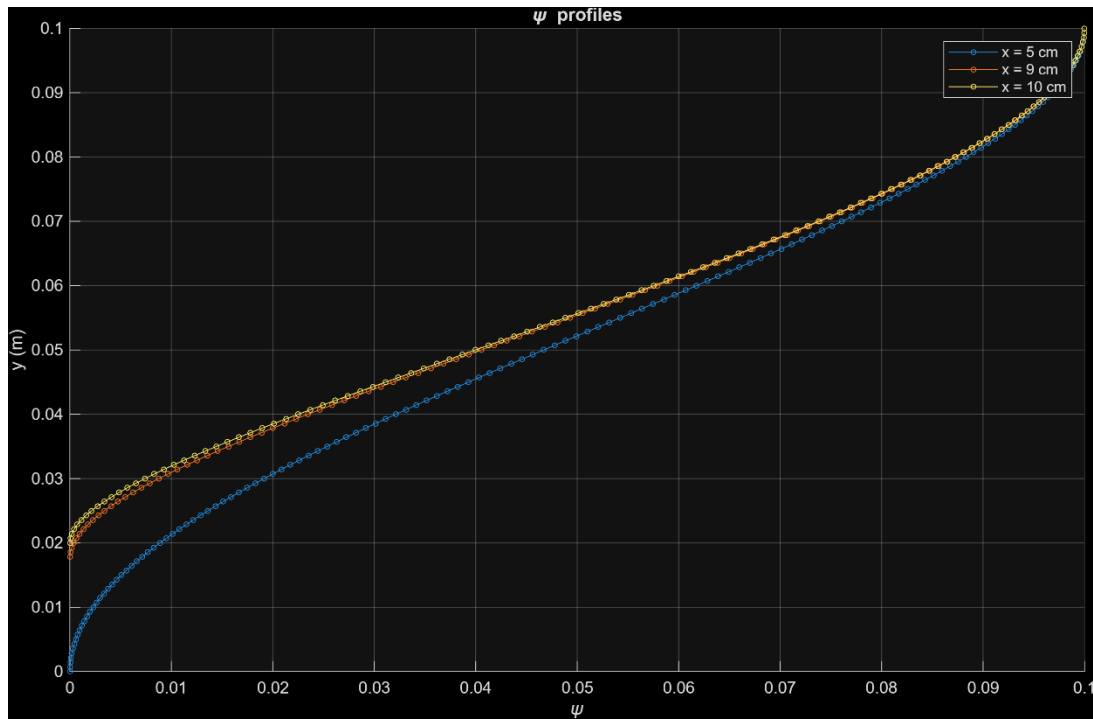


Figure 130. Stream function profiles at $x = 5, 9$ and 10 cm on the fine mesh.

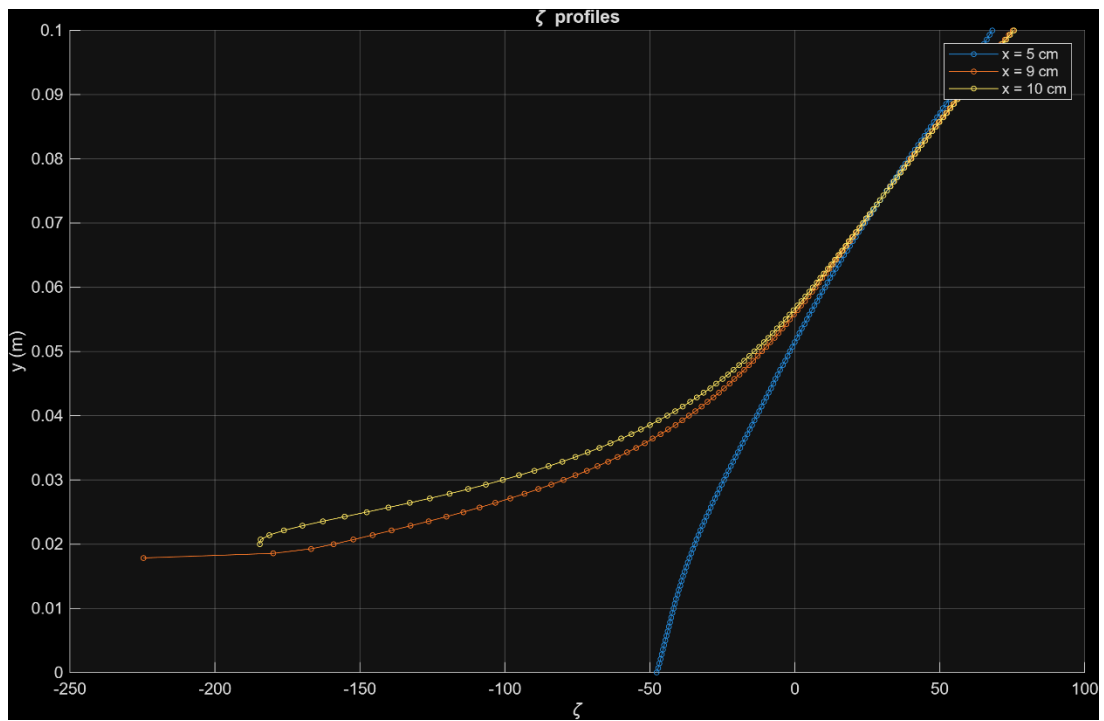


Figure 131. Vorticity profiles at $x = 5, 9$ and 10 cm on the fine mesh.

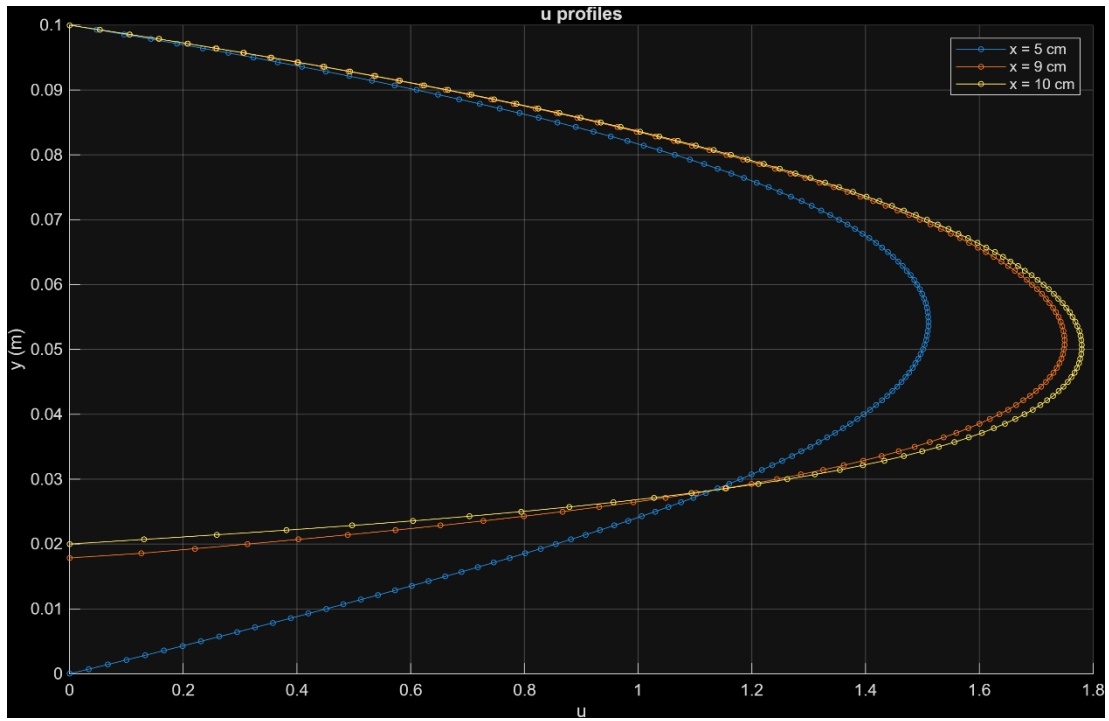


Figure 132. Horizontal velocity profiles at $x = 5, 9$ and 10 cm on the fine mesh.

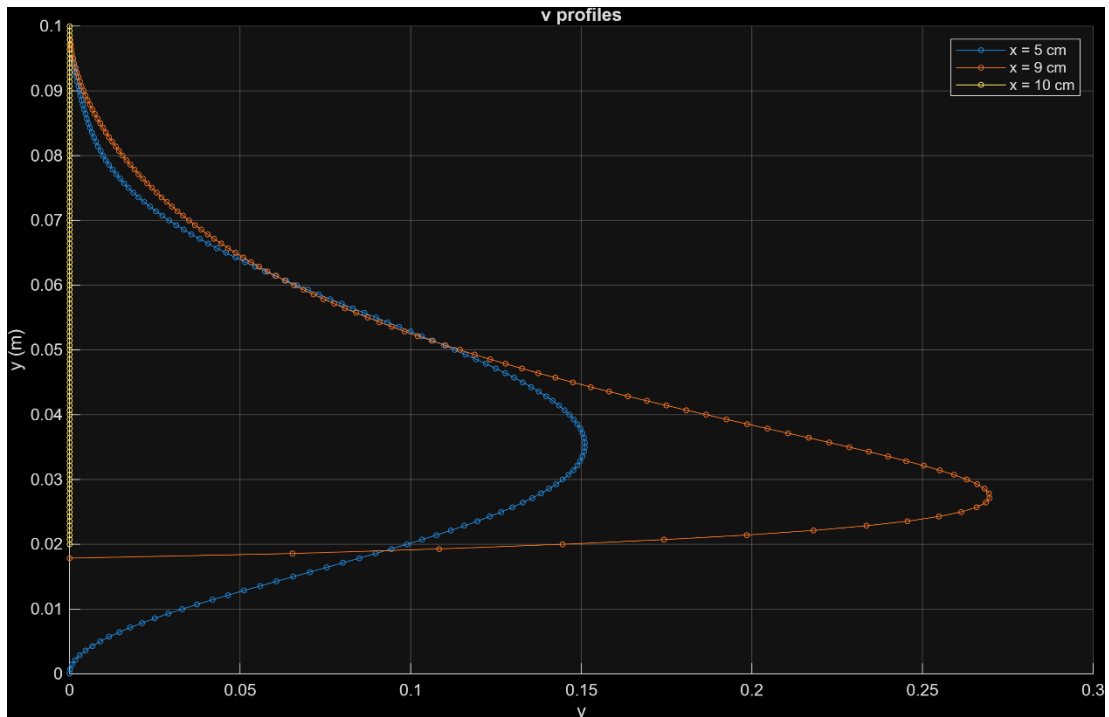


Figure 133. Vertical velocity profiles at $x = 5, 9$ and 10 cm on the fine mesh.

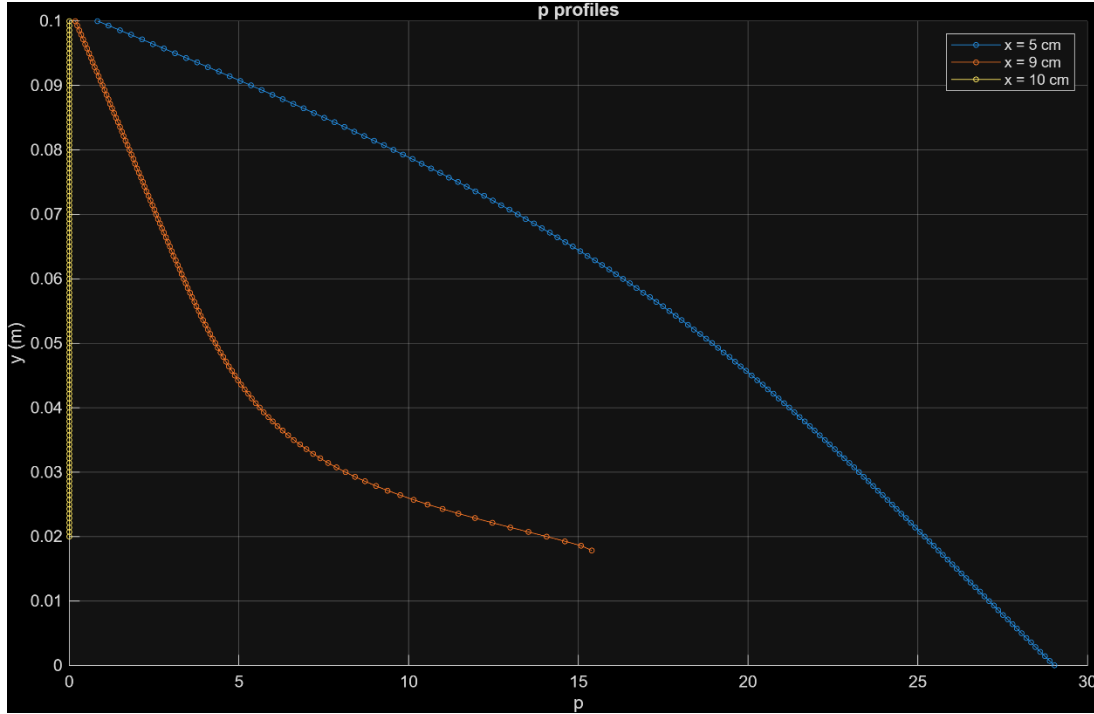


Figure 134. Pressure profiles at $x = 5, 9$ and 10 cm the fine mesh.

10 Results and Discussion

10.1 Mesh Details

Three uniform Cartesian meshes of increasing resolution were used to assess grid dependence of the Gauss-Seidel, SOR, SLOR, and SLOR+ADI schemes. Each mesh spans the full $10 \text{ cm} \times 10 \text{ cm}$ channel with node spacing $\Delta x = \Delta y = L/(N-1)$, where N is the number of nodes along each direction and $L = 0.1 \text{ m}$. Non-uniform spacing arises only in the row/column of nodes adjacent to the curved D–E boundary (radius $R = 2 \text{ cm}$), where the presence of the circular cutout forces the local node spacing to deviate from the uniform value.

Table 9. Mesh resolutions for the Gauss-Seidel, SOR, SLOR, and SLOR+ADI schemes.

Mesh	$N_x \times N_y$	$\Delta x = \Delta y$	Total nodes
Coarse	61×61	$0.1/60 \approx 1.667 \text{ mm}$	3,721

Intermediate	101×101	$0.1/100 = 1.000 \text{ mm}$	10,201
Fine	141×141	$0.1/140 \approx 0.714 \text{ mm}$	19,881

For the two-level Multigrid scheme, the fine mesh (141×141) was used as the fine grid, with a coarser grid of 71×71 nodes ($\Delta x = \Delta y = 0.1/70 \approx 1.429 \text{ mm}$) used for the correction step, giving a coarsening ratio of approximately 2:1 in each direction. consistent with the injection-based method used to go from the fine mesh to the coarse mesh in the V-cycle.

All simulations used the same physical parameters: channel height and length $L = 0.1 \text{ m}$, cylinder radius $R = 2 \text{ cm}$, inlet velocity $U_\infty = 1 \text{ m/s}$, and kinematic viscosity $\nu = 10^{-2} \text{ m}^2/\text{s}$, with a convergence tolerance of 10^{-7} applied simultaneously to ψ , ζ , u , and v .

10.2 Convergence Behavior

Convergence was defined by the simultaneous criterion $\max|\Delta\psi|$, $\max|\Delta\zeta|$, $\max|\Delta u|$, $\max|\Delta v| < 10^{-7}$, applied at every iteration of the flow-field solve. The reported wall-clock time is the total time to reach this criterion, measured with tic/toc, and includes the subsequent pressure-field solve performed as a post-processing step once ψ , ζ , u , and v have converged.

Table 10. Iterations and wall-clock time required for convergence of each scheme across the three mesh resolutions

Scheme	Coarse (61×61) Iter.	Coarse Time (s)	Intermediate (101×101) Iter.	Intermediate Time (s)	Fine (141×141) Iter.	Fine Time (s)
Gauss-Seidel	6,504	0.91	23,186	7.61	54,149	33.47
SOR	1,429	0.49	4,173	2.73	8,041	9.85
SLOR	2,131	1.18	6,682	11.78	17,048	68.79
SLOR+ADI	2,270	2.49	8,411	27.30	20,382	135.40
Multigrid (2-level)	—	—	—	—	25,472	56.60

For the two-level Multigrid scheme, only the fine-mesh (141×141 base / 71×71 coarse) result is reported in Table 2, consistent with the mesh used throughout the rest of Section 4. Runs were also carried out at the coarse and intermediate resolutions for reference; the iteration count rose from 2,557 (61×61) to 10,292 (101×101) to 25,472 (141×141).

Table 11. Relaxation parameters (ω_{opt}) applied to the ψ and ζ equations for SOR, SLOR, and SLOR+ADI.

Scheme	Mesh	$\omega_{opt}(\psi)$	$\omega_{opt}(\zeta)$
SOR	Coarse	1.70	1.40
	Intermediate	1.74	1.44
	Fine	1.76	1.46
SLOR	Coarse	1.7	1
	Intermediate	1.74	1.04
	Fine	1.76	1.06
SLOR+ADI	All meshes	1.3	1.00

Gauss-Seidel, having no relaxation acceleration, required the largest number of iterations at every mesh resolution, roughly 4.5–6.7 times more than SOR. SOR converged in both the fewest iterations and the least wall-clock time across all three meshes. Although SLOR solves each grid line implicitly via the Thomas algorithm, it required more iterations than point SOR here; this is consistent with the much lower ω_{opt} applied to the ζ equation in SLOR (1 - 1.06), compared to SOR (1.40 - 1.46), needed to keep the line-implicit update stable. SLOR+ADI required a comparable number of iterations to SLOR but roughly double the wall-clock time, reflecting the additional cost of performing both a horizontal and a vertical implicit sweep per iteration for each of ψ and ζ . Multigrid required fewer iterations than Gauss-Seidel on the fine mesh (25,472 vs. 54,149) but took longer in wall-clock time (56.60 s vs. 33.47 s), since each outer iteration includes six Gauss-Seidel sweeps on the fine mesh plus a full Gauss-Seidel solve of the delta equation on the coarse mesh, making its per-iteration cost far higher than a single Gauss-Seidel sweep.

11 Appendix

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central non-uniform discretization (first and second order)

$$\frac{\partial u}{\partial x} \Big|_{i,j} = \frac{u_{i,j+1} - u_{i,j-1}}{\Delta x} + \frac{u_{i,j+1} - u_{i,j}}{\Delta x} + \frac{u_{i,j} - u_{i,j-1}}{\Delta x} + O(\Delta x)^2$$

$$\begin{aligned} u_{i,j+1} &= u_{i,j} + (\Delta x) \frac{\partial u}{\partial x} \Big|_{i,j} + \frac{(\Delta x)^2}{2} \frac{\partial^2 u}{\partial x^2} \Big|_{i,j} + O(\Delta x)^3 \\ u_{i,j-1} &= u_{i,j} - (\Delta x) \frac{\partial u}{\partial x} \Big|_{i,j} + \frac{(\Delta x)^2}{2} \frac{\partial^2 u}{\partial x^2} \Big|_{i,j} + O(\Delta x)^3 \end{aligned} \rightarrow \frac{a \cdot b = 1}{\frac{a^2}{2} + \frac{b^2}{2} = 0} \rightarrow a + ab = 1 \rightarrow a = \frac{1}{a^2 + a}, b = \frac{-a^2}{a^2 + a}$$

$$\rightarrow \frac{u_{i,j+1} - u_{i,j-1}}{a^2 + a} = \frac{u_{i,j}(1-a^2)}{a^2 + a} + \Delta x \frac{\partial u}{\partial x} \Big|_{i,j} + O(\Delta x)^3 \rightarrow \frac{\partial u}{\partial x} \Big|_{i,j} = \frac{u_{i,j+1} - u_{i,j-1}}{(a^2 + a)\Delta x} + O(\Delta x)^2 \quad (1)$$

$$\frac{\partial u}{\partial x} \Big|_{i,j} \rightarrow \frac{a^2}{2} + \frac{b^2}{2} = 1 \rightarrow a + ab = 2 \rightarrow a = \frac{2}{a^2 + a}, b = \frac{2a}{a^2 + a}$$

$$\rightarrow \frac{\partial u}{\partial x} \Big|_{i,j} = \frac{2[u_{i,j+1} - (1+a)u_{i,j} + au_{i,j+1}]}{(a^2 + a)(\Delta x)^2} + O(\Delta x)^2 \quad (2)$$

viscosity equation, central discretization: $(\beta = \frac{\Delta x^2}{\Delta x^2}, a = \frac{\Delta x^2}{\Delta x^2})$

$$u \frac{\partial \tau}{\partial x} + v \frac{\partial \tau}{\partial y} = \nu \left(\frac{\partial^2 \tau}{\partial x^2} + \frac{\partial^2 \tau}{\partial y^2} \right)$$

$$u_{i,j} \left[\frac{j_{i,j+1} + (\beta^2)j_{i,j} - \beta^2 j_{i,j-1}}{\beta(\beta+1)\Delta x^2} \right] + v_{i,j} \left[\frac{j_{i,j+1} + (\beta^2)j_{i,j} - \beta^2 j_{i,j-1}}{\beta(\beta+1)\Delta y^2} \right] = \nu \left(\frac{2[j_{i,j+1} - (1+\beta)j_{i,j} + \beta j_{i,j+1}]}{\beta(\beta+1)(\Delta x)^2} + \frac{2[j_{i,j+1} - (1+\beta)j_{i,j} + \beta j_{i,j+1}]}{\beta(\beta+1)(\Delta y)^2} \right)$$

stream function equation, central discretization

$$\frac{\partial^2 \psi}{\partial x^2} + \frac{\partial^2 \psi}{\partial y^2} = -\zeta \rightarrow \frac{2[\psi_{i,j+1} - (1+\beta)\psi_{i,j} + \beta\psi_{i,j+1}]}{\beta(\beta+1)(\Delta x)^2} + \frac{2[\psi_{i,j+1} - (1+\beta)\psi_{i,j} + \beta\psi_{i,j+1}]}{\beta(\beta+1)(\Delta y)^2} = -\zeta_{i,j}$$

velocity equations, central discretization

$$u = \frac{\partial \psi}{\partial y} \Big|_{i,j} = \frac{\psi_{i,j+1} + (\beta^2)\psi_{i,j} - \beta^2\psi_{i,j-1}}{\beta(\beta+1)\Delta y}, \quad v = -\frac{\partial \psi}{\partial x} \Big|_{i,j} = \frac{-\psi_{i,j+1} + (\beta^2)\psi_{i,j} - \beta^2\psi_{i,j+1}}{\beta(\beta+1)\Delta x}$$

Boundary condition ψ

$$AB: \psi = U_a \bar{y}$$

$$AB: u = U_a \frac{x=0}{y=0 \text{ to } 1} \rightarrow U_a = \frac{\partial \psi}{\partial y} \rightarrow \psi = U_a y, c \rightarrow \psi = U_a \bar{y} \text{ (AB)} \rightarrow \psi_{i,j} = U_a \bar{y}_{i,j}$$

$$BC: \psi = U_a \bar{h} : (\partial \psi / \partial n) \Big|_{BC}$$

$$\frac{\partial \psi}{\partial n} \Big|_{BC} = U_a \bar{h} \rightarrow \psi = U_a \bar{h} \text{ (BC)} \rightarrow \psi_{i,j} = U_a \bar{h}_{i,j}$$

$$CD: \frac{\partial \psi}{\partial x} = 0 \rightarrow \frac{\psi_{i,j} - \psi_{i-1,j}}{\Delta x} = 0 \rightarrow \psi_{i,j} = \psi_{i-1,j} \rightarrow \psi_{i,j} = \psi_{i-1,j}$$

$$DE, EA: \psi_{i,j} = 0$$

Boundary conditions u, v

$$AB: u = u_m \rightarrow u_{i,j} = u_m \\ v = 0 \rightarrow v_{i,j} = 0$$

$$BC: u = 0 \rightarrow u_{i,\bar{j}} = 0 \\ v = 0 \rightarrow v_{i,\bar{j}} = 0$$

$$CD: u = \frac{\partial \psi}{\partial x} \xrightarrow{\text{central diff}} u_{i,j} = \frac{\psi_{i,j+1} + (1+\alpha)\psi_{i,j} - \alpha\psi_{i,j-1}}{\alpha(1+\alpha)\Delta x^-}$$

$$v = -\frac{\partial \psi}{\partial x} \xrightarrow{\text{backward diff}} v_{i,j} = -\frac{\psi_{i,j} - \psi_{i+1,j}}{\Delta x^-}$$

$$DE, EA: u = 0, v = 0$$

Boundary conditions f :

$$AB: f = 0 \text{ (uniform)}$$

$$BC: \psi_x + \psi_{\bar{x}\bar{x}} = f \xrightarrow{f_{m=0}} \psi_{\bar{x}\bar{x}} = f - (\psi_{\bar{x}})_{i,\bar{j}} = -f_{i,\bar{j}}$$

$$\psi_{i,\bar{j}+1} = \psi_{i,\bar{j}} - (\Delta \bar{x}) \frac{\partial \psi}{\partial \bar{x}} \bigg|_{i,\bar{j}} + \frac{(\Delta \bar{x})^2}{2} \frac{\partial^2 \psi}{\partial \bar{x}^2} \bigg|_{i,\bar{j}} \rightarrow f_{i,\bar{j}} = \frac{2(\psi_{i,\bar{j}} - \psi_{i,\bar{j}+1})}{(\Delta \bar{x})^2}$$

$$CD: f = -\psi_{xx} - \psi_{\bar{x}\bar{x}} \xrightarrow{\text{backward diff}} -\left[\frac{2[\psi_{i+2,\bar{j}} - (1+\beta_0)\psi_{i+1,\bar{j}} + \beta_0\psi_{i,\bar{j}}]}{\beta_0(1+\beta_0)(\Delta x^-)^2} + \frac{2[\psi_{i,\bar{j}+1} - (1+\alpha)\psi_{i,j} + \alpha\psi_{i,j-1}]}{\alpha(1+\alpha)(\Delta \bar{x}^-)^2} \right]$$

backward x: $\frac{\partial^2 u}{\partial x^2} = ?$, $\frac{\Delta x^-}{x} \frac{\Delta x^-}{x} \frac{\Delta x^-}{x} \rightarrow \beta_0 = \frac{\Delta x^-}{\Delta x^-} \rightarrow \Delta x^- + \Delta x^- = \Delta x^- (1 + \beta_0)$

$$\left. \begin{aligned} u_{i-1} &= u_i - \Delta x^- \frac{\partial u}{\partial x} \bigg|_i + \frac{(\Delta x^-)^2}{2} \frac{\partial^2 u}{\partial x^2} \bigg|_i + O(\Delta x^-)^3 \\ u_{i-2} &= u_i - \Delta x^- (1+\beta_0) \frac{\partial u}{\partial x} \bigg|_i + \frac{(\Delta x^-)^2 (1+\beta_0)^2}{2} \frac{\partial^2 u}{\partial x^2} \bigg|_i + O(\Delta x^-)^3 \end{aligned} \right\} \rightarrow \left. \begin{aligned} a + b(1+\beta_0) &= 0 \\ \frac{a}{2} + \frac{b(1+\beta_0)^2}{2} &= 1 \end{aligned} \right\} \rightarrow b[(1+\beta_0)^2 - (1+\beta_0)] = 2 \rightarrow b = \frac{2}{\beta_0^2 + \beta_0}$$

$$a = \frac{-2(1+\beta_0)}{\beta_0^2 + \beta_0}$$

$$\rightarrow \frac{-2(1+\beta_0)u_{i-1} + 2u_{i-2}}{\beta_0^2 + \beta_0} = \frac{-2\beta_0 u_{i,j}}{\beta_0^2 + \beta_0} + (\Delta x^-)^2 \frac{\partial^2 u}{\partial x^2} \bigg|_i \rightarrow \frac{\partial^2 u}{\partial x^2} \bigg|_i = \frac{2[u_{i-2} - (1+\beta_0)u_{i-1} + \beta_0 u_{i,j}]}{\beta_0(1+\beta_0)(\Delta x^-)^2}$$

$$DE: \frac{\partial^2 \psi}{\partial x^2} \xrightarrow{CD} \frac{\partial^2 \psi}{\partial x^2} \xrightarrow{CD} \psi_{\bar{x}\bar{x}} = f_{i,j} = \frac{2(0 - \psi_{i,j+1})}{(\Delta \bar{x}^-)^2}$$

$$\tilde{\psi}_{i,j} = \frac{2(0 - \psi_{i,j+1})}{(\Delta \bar{x}^-)^2}$$

$$\psi_{i,j} = \frac{2(0 - \psi_{i,j+1})}{(\Delta x^-)^2 (\Delta \bar{x}^-)^2}$$

$$EA: f_{i,1} = \frac{2(0 - \psi_{i,2})}{(\Delta \bar{x}^-)^2}$$